Project	**Mansourah Massarah Gold Project**
Customer:	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Title:	**DESIGN CALCULATION 3510 COUNTER CURRENT DECANTATION GRADE SLAB & PIPE RACK FOUNDATION**
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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
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QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

The Purpose of this document is to analysis & design of CCD area (3510) - pipe rack & stair case foundation, drain, pit and miscellaneous foundations. The member sizes and reinforcements are to be calculated. The report describes the methodology and the design parameters used in the design calculation. The analysis & design is carried out in accordance with project specifications,codes & standards.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD Pro. The STAAD modelling has been done with member & plate elements. Plate mat with subgrade modulus has been considered as per L&T approved Soil Report. Base Pressure check has been done from STAAD Pro output. Seismic load combinations are not shown as seismic loads are much less compared to wind loads.
For strength calculation Bending moment & Shear Force values are taken from STAAD output & designed in the excel sheet.

4. DESIGN REQUIREMENTS

4.1. Design References

Reference is made in this standard as indicated in Civil,Structural and Architectural Design Criteria Doc No. : MD-216-1000-EG-ST-CRT-1001-A02 , Structural Design Criteria - Process Building & Structures ITB Doc No. : MD-101-SMEM-EG-ST-CRT-1001 & Civil Design Criteria - General ITB Doc. No. MD-101-SMEM-EG-CV-CRT-1001.

The below mention codes & books are also used for this design calculation.

(i) SBC - 301
(ii) SBC - 304 CR - 2018
(iii) ACI 318M.
(iv) ACI SP-17M(09)

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4.2. Materials Property

Density of Material
- Density of Dry Soil : 1.9 T/m^3
- Density of Concrete : 2.5 T/m^3
- Density of Water : 1.0 T/m^3

Concrete Strength of 28 days
- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade
- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

Grout Grade
- Grout shall be cementitious under steel column bases of Strength 40 MPa

4.3. Minimum Reinforcing Steel (Rebar) Size

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.4 Concrete Clear Cover

Concrete Exposure	Member	Minimum Cove (mm)
Exposed to Soil	Foundation Bottom & Side	75
	Foundation Top	50
	Column & Beam	50
Exposed to Weather	Column	50
	Beam	50
	Slab	40

4.5. Allowable Soil Bearing Capacity

Recommended net allowable bearing capacity as per L&T soil report MD-216-3000-EG-CV-RPT-1008-A02, bore hole BH-12.

BH-no	Proposed Structures	Depth of foundation below FGL(m)	Recommended net allowable bearing capacity(KN/m2)	Vertical modulus of subgrage reaction(KN/m2/m)
BH-12	Neutralization and thickening	1.5	980	117600

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5. SAFETY FACTOR AGAINST STABILITY

5.1. Overturning : For foundation design a minimum factor of safety against overturning is consider 2.0
5.2. Sliding : For foundation design a minimum factor of safety against sliding is consider 1.5

6. LOAD CALCULATIONS

6.1. Primary Load Cases

For pipe rack foundation

1 FOUNDATION SELF WEIGHT(D1)
2 LIVE LOAD_5KN/SQM (L1)
3 WIND LOAD +X DIRN(W+X)
4 WIND LOAD -X DIRN(W-X)
5 WIND LOAD -Z DIRN(W-Z)
6 WIND LOAD +Z DIRN(W+Z)
7 TANK(D1)
8 TEMPERATURE +20(T+VE)
10 LIVE LOAD_10KN/SQM (L2)
11 LIVE LOAD_15KN/SQM (L3)
13 SELF WEIGHT+SECONDARY STEEL STRUCTURE(D2)
14 TEMPERATURE -20(T-VE)
15 MAXIMUM SEISMIC LOAD (E_MAX)
16 MINIMUM SEISMIC LOAD (E_MIN)

For pipe rack , primary loads are taken from OTE input drawing no. MD-216-3510-EG-ME-GAD-1025-A01

For stair case foundation

1 FOUNDATION SELF WEIGHT(D1)
2 LIVE LOAD_4KN/SQM (L1)
3 WIND LOAD +X DIRN(W+X)
4 WIND LOAD -X DIRN(W-X)
5 WIND LOAD -Z DIRN(W-Z)
6 WIND LOAD +Z DIRN(W+Z)
7 TEMPERATURE +20(T+VE)
8 TEMPERATURE -20(T-VE)
9 SELF WEIGHT+SECONDARY STEEL STRUCTURE(D2)
10 MAXIMUM SEISMIC LOAD (E_MAX)
11 MINIMUM SEISMIC LOAD (E_MIN)

For stair case, primary loads are taken from OTE input drawing no. MD-216-3510-EG-ME-GAD-1025-A01

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6.2. Load Combinations for Stress Calculation

For pipe rack foundation

LOAD CASE	LOAD COMBINATION
101	(D+D1+D2)
102	(D+D1+D2)+T(+VE)+L1
103	(D+D1+D2)+T(+VE)+L2
104	(D+D1+D2)+T(+VE)+L3
105	(D+D1+D2)+T(-VE)+L1
106	(D+D1+D2)+T(-VE)+L2
107	(D+D1+D2)+T(-VE)+L3
108	D+D1+D2+W(+X)
109	D+D1+D2+W(-X)
110	D+D1+D2+W(-Z)
111	D+D1+D2+W(+Z)
112	D+D1+D2+0.75L1+0.75W(+X)
113	D+D1+D2+0.75L1+0.75W(-X)
114	D+D1+D2+0.75L1+0.75W(-Z)
115	D+D1+D2+0.75L1+0.75W(+Z)
116	D+D1+D2+0.75L2+0.75W(+X)
117	D+D1+D2+0.75L2+0.75W(-X)
118	D+D1+D2+0.75L2+0.75W(-Z)
119	D+D1+D2+0.75L2+0.75W(+Z)
120	D+D1+D2+0.75L3+0.75W(+X)
121	D+D1+D2+0.75L3+0.75W(-X)
122	D+D1+D2+0.75L3+0.75W(-Z)
123	D+D1+D2+0.75L3+0.75W(+Z)
124	0.6(D1+D2)+W(+X)+D
125	0.6(D1+D2)+W(-X)+D
126	0.6(D1+D2)+W(-Z)+D
127	0.6(D1+D2)+W(+Z)+D
128	D+D1+D2+0.7E_MAX
129	D+D1+D2+0.7E_MIN
130	D+D1+D2+0.75L1+0.75*0.7E_MAX
131	D+D1+D2+0.75L1+0.75*0.7E_MIN
132	D+D1+D2+0.75L2+0.75*0.7E_MAX
133	D+D1+D2+0.75L2+0.75*0.7E_MIN
134	D+D1+D2+0.75L3+0.75*0.7E_MAX
135	D+D1+D2+0.75L3+0.75*0.7E_MIN
136	0.6(D1+D2)+0.7E_MAX+D
137	0.6(D1+D2)+0.7E_MIN+D

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For stair case foundation

LOAD CASE	LOAD COMBINATION
101	(D1+D2)
102	(D1+D2)+T(+VE)+L
103	(D1+D2)+T(-VE)+L1
104	D1+D2+W(+X)
105	D1+D2+W(-X)
106	D1+D2+W(-Z)
107	D1+D2+W(+Z)
108	D1+D2+0.75L1+0.75W(+X)
109	D1+D2+0.75L1+0.75W(-X)
110	D1+D2+0.75L1+0.75W(-Z)
111	D1+D2+0.75L1+0.75W(+Z)
112	0.6(D2)+W(+X)+D1
113	0.6(D2)+W(-X)+D1
114	0.6(D2)+W(-Z)+D1
115	0.6(D2)+W(+Z)+D1
116	D1+D2+0.7E_MAX
117	D1+D2+0.7E_MIN
118	D1+D2+0.75L+0.75*0.7E_MAX
119	D1+D2+0.75L+0.75*0.7E_MIN
120	0.6*D2+0.7E_MAX+D1
121	0.6*D2+0.7E_MIN+D1

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6.3 Load Combinations for Strength Calculation

For pipe rack foundation

LOAD CASE	LOAD COMBINATION
201	1.4D+1.4(D1+D2)
202	1.4(D+D1+D2)+1.4T(+VE)+1.7L1
203	1.4(D+D1+D2))+1.4T(+VE)+1.7L2
204	1.4(D+D1+D2)+1.4T(+VE)+1.7L3
205	1.4(D+D1+D2)+1.4T(-VE)+1.7L1
206	1.4(D+D1+D2)+1.4T(-VE)+1.7L2
207	1.4(D+D1+D2)+1.4T(-VE)+1.7L3
208	1.2(D+D1+D2)+0.8W(+X)
209	1.2(D+D1+D2)+0.8W(-X)
210	1.2(D+D1+D2)+0.8W(-Z)
211	1.2(D+D1+D2)+0.8W(+Z)
212	1.2(D+D1+D2)+1.0L1+1.6W(+X)
213	1.2(D+D1+D2)+1.0L1+1.6W(-X)
214	1.2(D+D1+D2)+1.0L1+1.6W(-Z)
215	1.2(D+D1+D2)+1.0L1+1.6W(+Z)
216	1.2(D+D1+D2)+1.0L2+1.6W(+X)
217	1.2(D+D1+D2)+1.0L2+1.6W(-X)
218	1.2(D+D1+D2)+1.0L2+1.6W(-Z)
219	1.2(D+D1+D2)+1.0L2+1.6W(+Z)
220	1.2(D+D1+D2)+1.0L3+1.6W(+X)
221	1.2(D+D1+D2)+1.0L3+1.6W(-X)
222	1.2(D+D1+D2)+1.0L3+1.6W(-Z)
223	1.2(D+D1+D2)+1.0L3+1.6W(+Z)
224	0.9(D1+D2)+W(+X)+D
225	0.9(D1+D2)+W(-X)+D
226	0.9(D1+D2)+W(-Z)+D
227	0.9(D1+D2)+W(+Z)+D
228	1.2(D+D1+D2)+1.0L1+1.0E_MAX
229	1.2(D+D1+D2)+1.0L1+1.0E_MIN
230	1.2(D+D1+D2)+1.0L2+1.0E_MAX
231	1.2(D+D1+D2)+1.0L2+1.0E_MIN
232	1.2(D+D1+D2)+1.0L3+1.0E_MAX
233	1.2(D+D1+D2)+1.0L3+1.0E_MIN
234	0.9(D1+D2)+1.0E_MAX(+X)+D
235	0.9(D1+D2)+1.0E_MIN(+X)+D

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For stair case foundation

LOAD CASE	LOAD COMBINATION
201	1.4(D1+D2)
202	1.4(D1+D2)+1.4T(+VE)+1.7L1
203	1.4(D1+D2)+1.4T(-VE)+1.7L1
204	1.2(D1+D2)+0.8W(+X)
205	1.2(D1+D2)+0.8W(-X)
206	1.2(D1+D2)+0.8W(-Z)
207	1.2(D1+D2)+0.8W(+Z)
208	1.2(D1+D2)+1.0L1+1.6W(+X)
209	1.2(D1+D2)+1.0L1+1.6W(-X)
210	1.2(D1+D2)+1.0L1+1.6W(-Z)
211	1.2(D1+D2)+1.0L1+1.6W(+Z)
212	0.9(D2)+W(+X)+D1
213	0.9(D2)+W(-X)+D1
214	0.9(D2)+W(-Z)+D1
215	0.9(D2)+W(+Z)+D1
216	1.2(D1+D2)+1.0L+1.0E_MAX
217	1.2(D1+D2)+1.0L+1.0E_MIN
218	0.9*D2+1.0E_MAX(+X)+D1
219	0.9*D2+1.0E_MIN(+X)+D1

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6.4. Primary Load Cases for pit design

1 DEAD LOAD(D)
2 SOIL PRESSURE(DRY CONDITION) (H)
3 SURCHARGE PRESSURE (L)

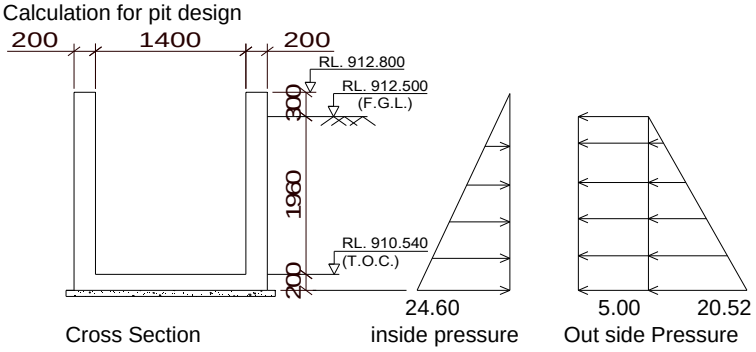
6.5. Load Combinations for Stress Calculation(pit design)

LOAD CASE	LOAD COMBINATION
101	D
102	D+H+L

6.6. Load Combinations for Strength Calculation(pit design)

LOAD CASE	LOAD COMBINATION
201	1.4D
202	1.4D+1.7(H+L)

6.7. Primary Load Calculation for pit design



Cross Section | inside pressure | Out side Pressure

Input Data :

Clear height of pit from FGL =	1960	mm	C/C distance along width of Pit =	1600	mm
Clear width of Pit =	1400	mm	C/C distance along length of Pit =	1600	mm
Clear length of Pit =	1400	mm	Bottom of pit from FGL,h =	2160	mm
Vertical wall thickness =	200	mm			
Bottom slab thickness =	200	mm			
Pressure coefficient, C_p =	0.5				
Surcharge load at side of pit, P =	10	kN/m^2			
Dry unit weight of soil, $\gamma_{dry\ soil}$ =	19	kN/m^3			

1 DEAD LOAD(D)
Selfweight of structure

2 SOIL PRESSURE(DRY CONDITION) (H)
Lateral earth pressure at side wall due to dry soil = $C_p \times \gamma_{dry\ soil} \times h$ 20.52 kN/m^2

3 SURCHARGE PRESSURE (L)
Lateral earth pressure at side wall due to surcharged load = $C_p \times P$ 5.00 kN/m^2

4 WATER PRESSURE (F)
Water pressure at side wall = 24.60 kN/m^2

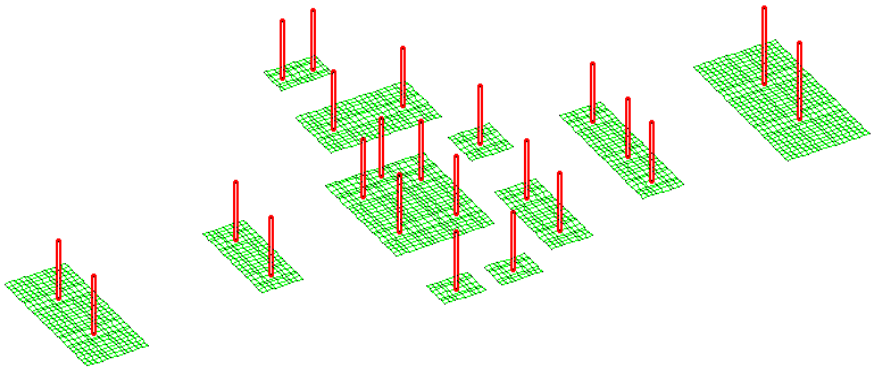
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7. Design Calculations

7.1. Design of Columns



Column group as per STAAD_pro

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Design of column C1

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.90

% of reinforcement, p_t = 1 % i.e. ρ_g = 0.01

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 700 mm

Depth parallel to Z-Z axis, b = 700 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.79

Y_b = (b-d')/b = 0.79

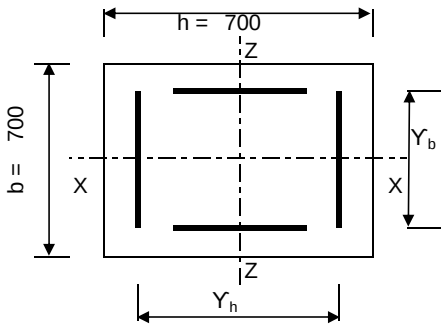
Unsupported length of column parallel to X-X axis, l_{ux} = 1.9 m kl_{ux}/r_b = 9.048 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.9 m kl_{uz}/r_h = 9.048

$P_{n,\,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 13501.8 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity						Check for Interaction Ratio		
								about X-X axis			about ZZ axis					
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
Max Fx	849	3.0	32.2	943	3.3	35.8	0.055	0.070	840.4	0.00	0.070	840.4	0.04		0.01	safe
Min Fx	-476.2	0.0	0.0	-529	0.0	0.0	-0.03	0.070	840.4	0.00	0.070	840.4	0.00		0	safe
Max Fy	45.9	13.9	200.6	51	15.5	222.9	0.003	0.070	840.4	0.02	0.070	840.4	0.27		0.12	safe
Min Fy	616.3	100.4	206.3	685	111.6	229.3	0.04	0.070	840.4	0.13	0.070	840.4	0.27		0.16	safe
Max Fz	425.2	399.7	3.8	472	444.1	4.2	0.028	0.070	840.4	0.53	0.070	840.4	0.00		0.36	safe
Min Fz	-445.6	736.5	0.4	-495	818.4	0.5	-0.03	0.070	840.4	0.97	0.070	840.4	0.00		0.96	safe
Max Mx	196.4	201.5	9.8	218	223.9	10.9	0.013	0.070	840.4	0.27	0.070	840.4	0.01	1.61	0.12	safe
Min Mx	-11.2	133.4	74.0	-12	148.2	82.2	-0	0.070	840.4	0.18	0.070	840.4	0.10		0.09	safe
Max My	-445.6	736.5	0.4	-495	818.4	0.5	-0.03	0.070	840.4	0.97	0.070	840.4	0.00		0.96	safe
Min My	425.2	399.7	3.8	472	444.1	4.2	0.028	0.070	840.4	0.53	0.070	840.4	0.00		0.36	safe
Max Mz	676.6	56.1	202.9	752	62.3	225.5	0.044	0.070	840.4	0.07	0.070	840.4	0.27		0.14	safe
Min Mz	616.3	100.4	206.3	685	111.6	229.3	0.04	0.070	840.4	0.13	0.070	840.4	0.27		0.16	safe

Reinforcement Calculation							Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
4900	12	25				5893	1.2	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

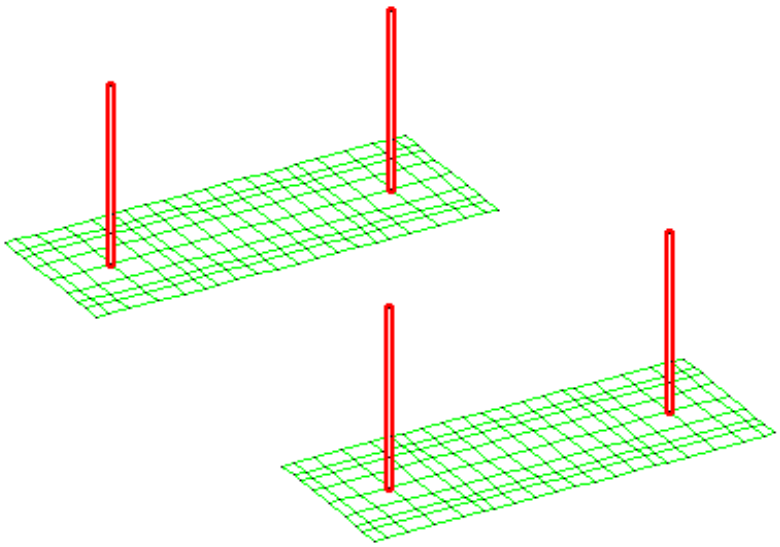
ACI DESIGN HANDBOOK-SP-17M(09)

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Design of Column C1



Column group as per STAAD_pro

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Design of column C1

Materials :

Concrete Grade, f_c' = 35 Mpa

Reinforcement Grade, f_y = 420 Mpa

Assumptions :

φ = 0.90

% of reinforcement, p_t = 1.2 % i.e. ρ_g = 0.012

β = 0.65

k = 1

Column Dimensions :

Depth parallel to X-X axis, h = 700 mm

Depth parallel to Z-Z axis, b = 700 mm

Clear Cover = 50 mm

d' = 72.5 mm

Y_h = (h-d')/h = 0.79

Y_b = (b-d')/b = 0.79

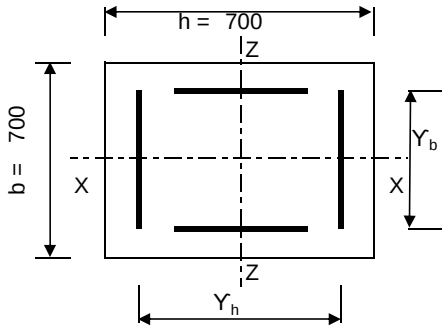
Unsupported length of column parallel to X-X axis, l_{ux} = 1.9 m kl_{ux}/r_b = 9.048 < 22 neglect slenderness effect

Unsupported length of column parallel to Z-Z axis, l_{uz} = 1.9 m kl_{uz}/r_h = 9.048

$P_{n,Max} = 0.8[0.85*f_c'(A_g-A_{st})+f_yA_{st}]$ = 13502 KN safe

Factored Force Value				Design force Value				Uniaxial moment capacity about X-X axis			Uniaxial moment capacity about ZZ axis			Check for Interaction Ratio		
L/C	P_u	M_{ux}	M_{uz}	$P_n = P_u/\varphi$	$M_{nx} = M_{ux}/\varphi$	$M_{nz} = M_{uz}/\varphi$	$K_n = P_n/f_c'bh$	$R_{nx} = M_{nx1}/f_c'b^2h$	M_{nx1}	M_{nx}/M_{nx1}	$R_{nz} = M_{nz1}/f_c'h^2b$	M_{nz1}	M_{nz}/M_{nz1}	α=log0.5/logβ	$(M_{nx}/M_{nx1})^\alpha + (M_{nz}/M_{nz1})^\alpha$	Remarks
	KN	KN-m		KN	KN-m				KN-m			KN-m				
Max Fx	197	44.7	4.6	219	49.6	5.1	0.013	0.100	1201	0.04	0.100	1201	0.00		0.01	safe
Max Fy	100.8	0.3	984.6	112	0.4	1094.0	0.007	0.100	1201	0.00	0.100	1201	0.91		0.86	safe
Min Fy	134.2	3.0	989.5	149	3.3	1099.5	0.009	0.100	1201	0.00	0.100	1201	0.92		0.87	safe
Max Fz	63.6	35.2	13.3	71	39.1	14.8	0.004	0.100	1201	0.03	0.100	1201	0.01		0.01	safe
Min Fz	146.5	56.6	17.5	163	62.9	19.5	0.009	0.100	1201	0.05	0.100	1201	0.02		0.01	safe
Max Mx	80.8	0.3	8.3	90	0.3	9.2	0.005	0.100	1201	0.00	0.100	1201	0.01	1.61	0	safe
Min Mx	80.8	0.3	8.3	90	0.3	9.2	0.005	0.100	1201	0.00	0.100	1201	0.01		0	safe
Max My	146.5	56.6	17.5	163	62.9	19.5	0.009	0.100	1201	0.05	0.100	1201	0.02		0.01	safe
Min My	63.6	35.2	13.3	71	39.1	14.8	0.004	0.100	1201	0.03	0.100	1201	0.01		0.01	safe
Max Mz	100.8	0.3	984.6	112	0.4	1094.0	0.007	0.100	1201	0.00	0.100	1201	0.91		0.86	safe
Min Mz	134.2	3.0	989.5	149	3.3	1099.5	0.009	0.100	1201	0.00	0.100	1201	0.92		0.87	safe

Reinforcement Calculation								Calculation of Ties		
A_{st} (mm^2)	Combination of Reinforcement Bars,db					A_{st} provided(mm^2)	Reinforcement provided (%)	Dia, d_{tie} (cl : 25.7.2.2 SBC 304-CR-18)	Pitch (cl : 25.7.2.1 SBC 304-CR-18)	Spacing of ties provided
5880	12	25	+	0	0	5893	1.2	10	400	250



P_n : Nominal axial load P_u : Factored design axial load

M_n : Nominal moment M_u : Factored design moment

Use interaction diagram R35-420.8

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7.2. Design of pipe rack & stair case foundations

Foundation Base Pressure :

refer staad file : MD-216-3510-EG-CV-CAL-1281_pipe rack fdn in CCD area

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	100	101	0	66.549	0
Min Px	100	101	0	66.549	0
Max Py	1770	133	0	256.362	0
Min Py	635	102	0	0	0
Max Pz	100	101	0	66.549	0
Min Pz	100	101	0	66.549	0

Gross Maximum Base pressure, p_{max} = 256.4 kN/m^2

Net Allowable Safe Bearing Capacity = 980 kN/m^2 (considered)

Depth of footing from FGL = 2 m

Unit weight of soil = 19 kN/m^3

Gross Safe Bearing Capacity = 1018 kN/m^2

> 256.4 kN/m^2 safe

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Design of Foundation F1:

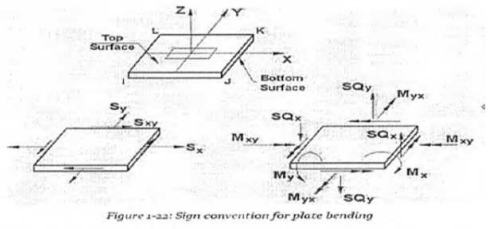
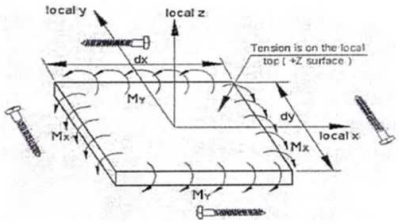


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G1 : F1_MOMENT At the face of the column
refer staad file : MD-216-3510-EG-CV-CAL-1281_pipe rack fdn in CCD area

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	730	1.036	-0.421	0	0	0	85.29	117.7	-26.66	112	144.4	227 0.9(D1+D2)+W(+Z)+D
Min Qx	250	-1.027	-0.417	0	0	0	85.83	118.5	0	85.83	118.5	227 0.9(D1+D2)+W(+Z)+D
Max Qy	148	0.052	0.921	0	0	0	7.609	-2.354	24.49	32.1	-26.84	227 0.9(D1+D2)+W(+Z)+D
Min Qy	610	0.315	-0.934	0	0	0	106.2	186.9	-8.541	114.8	195.4	215 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Max Sx	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Min Sx	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Max Sy	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Min Sy	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Max Sxy	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Min Sxy	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Max Mx	130	-0.314	-0.925	0	0	0	114.8	173.1	2.892	117.7	176	227 0.9(D1+D2)+W(+Z)+D
Min Mx	130	0.19	0.669	0	0	0	-75.35	-129.1	2.481	-77.83	-131.6	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Max My	130	-0.306	-0.933	0	0	0	106.5	187.2	4.63	111.1	191.8	215 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min My	514	-0.231	-0.448	0	0	0	-10.71	-214.2	-23.1	-33.81	-237.3	227 0.9(D1+D2)+W(+Z)+D
Max Mxy	154	0.698	-0.102	0	0	0	-24.28	-160	55.24	-79.52	-215.2	227 0.9(D1+D2)+W(+Z)+D
Min Mxy	634	-0.692	-0.102	0	0	0	-24.54	-160.2	-59.93	-84.47	-220.1	227 0.9(D1+D2)+W(+Z)+D

PLATE CENTER STRESS ; G2 : F1_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	970	0.237	0.13	0	0	0	6.02	33.24	-10.87	16.89	44.11	227 0.9(D1+D2)+W(+Z)+D
Min Qx	490	-0.235	0.132	0	0	0	7.564	35.11	0	7.564	35.11	227 0.9(D1+D2)+W(+Z)+D
Max Qy	388	-0.036	0.28	0	0	0	7.565	-50.07	30.1	37.67	-80.17	227 0.9(D1+D2)+W(+Z)+D
Min Qy	622	0.111	-0.361	0	0	0	76.76	59.05	4.159	80.92	63.21	215 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Max Sx	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Min Sx	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Max Sy	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Min Sy	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Max Sxy	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Min Sxy	2639	0.004	-0.001	0	0	0	0.058	-0.129	0.194	0.252	-0.323	201 1.4D+1.4(D1+D2)
Max Mx	142	-0.106	-0.347	0	0	0	82.64	51.24	-8.437	91.08	59.67	227 0.9(D1+D2)+W(+Z)+D
Min Mx	622	-0.058	0.28	0	0	0	-45.06	-43.04	-5.313	-50.37	-48.35	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Max My	142	-0.099	-0.351	0	0	0	71.46	59.69	-6.501	77.96	66.19	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Min My	46	0.062	-0.151	0	0	0	-20.77	-134.9	-0.529	-21.3	-135.5	227 0.9(D1+D2)+W(+Z)+D
Max Mxy	388	-0.008	0.203	0	0	0	7.166	-16.34	31.48	38.65	-47.82	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Min Mxy	868	0.02	0.21	0	0	0	6.524	-18.22	-34.96	41.48	-53.18	215 1.2(D+D1+D2)+1.0L1+1.6W(+Z)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

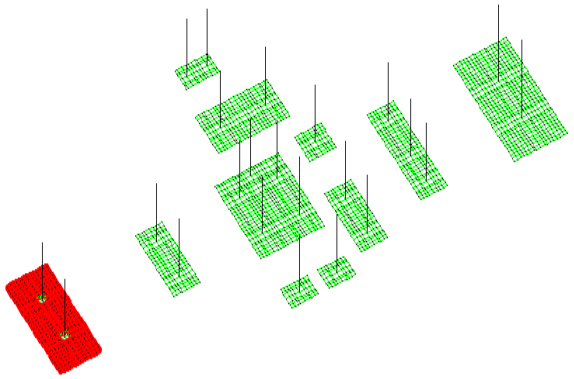
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	117.7	top face	50	442	117.7	0.014	712.6		16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min Mxy	-84.5	bottom face	75	417	84.47	*As	540.8	900	16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.237		118.5	16.89	0.004	0.442	501.2	445.1	758.3	445.1	1628	> 118.5	ok	333.8	φVc>Vu	not required
	Min Qx	0.235		117.5	7.564			601.8			445.1	1628	> 117.5	ok	333.8	φVc>Vu	not required

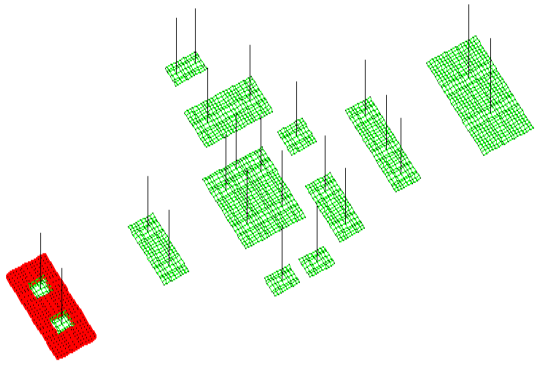
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	191.8	top face	50	440	191.8	0.014	1175.3	900	20	267.4	200	1571.5	22.16	27.7	0.045	tension control
Min My	-237.3	bottom face	75	415	237.3		1553.5		20	202.3	200	1571.5	22.16	27.7	0.042	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.28		140	80.17	0.004	0.44	439	445.1	758.3	439.01	459	> 140	ok	329.3	φVc>Vu	not required
	Min Qy	0.361		180.5	63.21			452.1			445.1	463	> 180.5	ok	333.8	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F2:

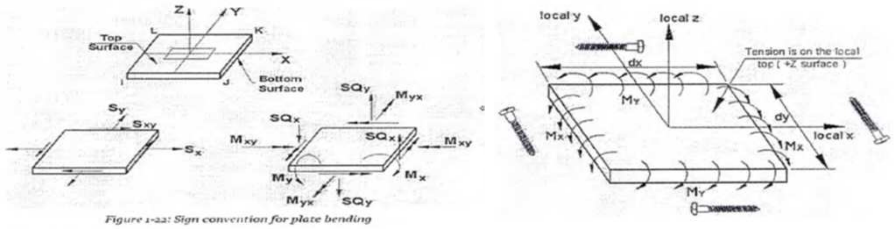


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G3 : F2_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	215	0.609	0.117	0	0	0	-65.11	-74.41	1.38	-66.49	-75.79	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3
Min Qx	725	-0.604	-0.04	0	0	0	-67.34	-51.86	0	-67.34	-51.86	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Max Qy	209	0.415	0.571	0	0	0	-51.56	-42.12	2.202	-53.76	-44.32	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3
Min Qy	35	0.108	-0.598	0	0	0	-54	-109	1.129	-55.13	-110.1	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Sx	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Min Sx	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Max Sy	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Min Sy	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Max Sxy	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Min Sxy	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Max Mx	149	-0.128	-0.09	0	0	0	29.77	8.778	-16.27	46.04	25.05	202 1.4(D+D1+D2)+1.4T(+VE)+1.7L1
Min Mx	131	0.265	0.3	0	0	0	-69.78	-28.77	9.305	-79.09	-38.08	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Max My	89	-0.068	-0.267	0	0	0	11.41	96.82	9.409	20.82	106.2	227 0.9(D1+D2)+W(+Z)+D
Min My	35	0.088	-0.411	0	0	0	-28.99	-111.2	6.866	-35.85	-118	227 0.9(D1+D2)+W(+Z)+D
Max Mxy	209	0.043	0.043	0	0	0	1.112	15.48	40.13	41.24	55.61	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min Mxy	155	-0.065	0.028	0	0	0	8.787	42.28	-43.43	52.22	85.71	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2

PLATE CENTER STRESS ; G4 : F2_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	455	0.106	0.122	0	0	0	-4.08	-52.23	-0.831	-4.911	-53.06	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3
Min Qx	965	-0.107	-0.045	0	0	0	-4.155	-34.41	0	-4.155	-34.41	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Max Qy	77	0.033	0.262	0	0	0	-27.26	-17.78	0.588	-27.85	-18.37	222 1.2(D+D1+D2)+1.0L3+1.6W(-Z)
Min Qy	47	0.037	-0.3	0	0	0	-28.66	-23.77	0.686	-29.35	-24.46	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Sx	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Min Sx	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Max Sy	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Min Sy	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Max Sxy	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Min Sxy	983	-0.005	0.004	0	0	0	-0.79	-0.728	-0.43	-1.22	-1.158	201 1.4D+1.4(D1+D2)
Max Mx	623	0.026	-0.036	0	0	0	16.23	1.872	-0.049	16.28	1.921	227 0.9(D1+D2)+W(+Z)+D
Min Mx	623	-0.047	0.07	0	0	0	-47.04	-1.904	1.001	-48.04	-2.905	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Max My	899	-0.015	-0.023	0	0	0	-1.65	88.99	-0.877	-2.527	89.87	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min My	389	0.049	-0.023	0	0	0	-0.687	-64.97	4.813	-5.5	-69.78	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Mxy	197	0.02	0.123	0	0	0	-12.49	43.86	16.09	-28.58	59.95	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Min Mxy	677	-0.037	0.126	0	0	0	-15.47	13.54	-18.39	-33.86	31.93	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

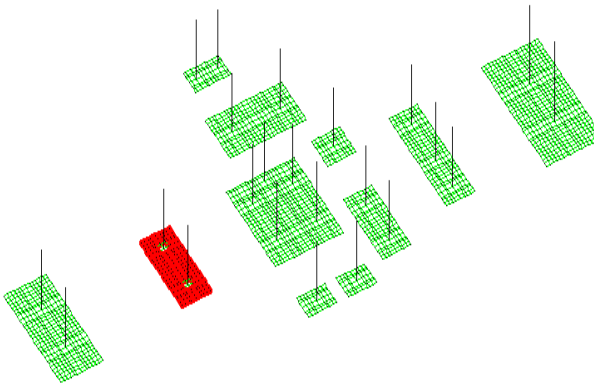
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	46.0	top face	50	442	46.04	0.014	276.8	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min Mx	-79.1	bottom face	75	417	79.09		506.1		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.106		53	4.911	0.002	0.442	499.9	435.5	758.3	435.48	1621	> 53	ok	326.6	φVc>Vu	not required
	Min Qx	0.107		53.5	4.155			515.7			435.48	1621	> 53.5	ok	326.6	φVc>Vu	not required

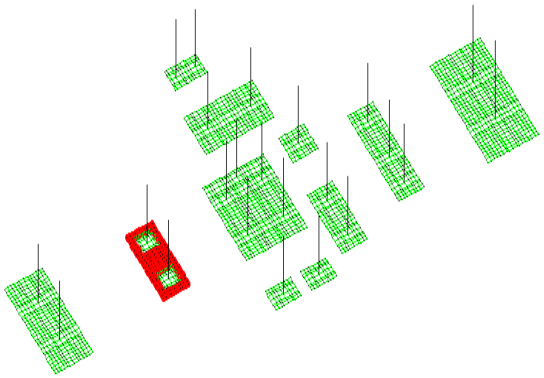
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	106.2	top face	50	442	106.2	0.014	642.4	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min My	-118.0	bottom face	75	417	118		758.6		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.262		131	18.37	0.002	0.442	472.3	435.5	758.3	435.48	456	> 131	ok	326.6	φVc>Vu	not required
	Min Qy	0.3		150	24.46			464.7			435.48	456	> 150	ok	326.6	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F3 :

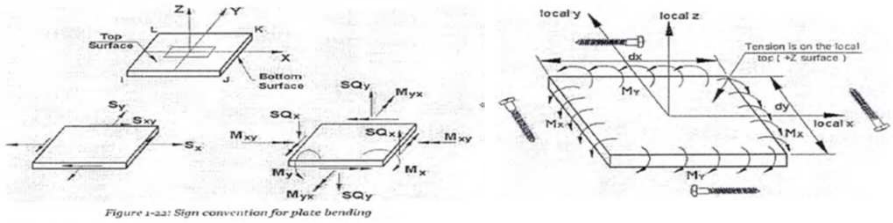


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G5 : F3_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	228	0.861	-0.452	0	0	0	-33.65	-103.2	8.718	-42.37	-111.9	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Min Qx	709	-0.957	-0.535	0	0	0	-48.91	-116.8	0	-48.91	-116.8	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max Qy	151	0.165	0.73	0	0	0	-13.58	-64.12	0.31	-13.89	-64.43	227 0.9(D1+D2)+W(+Z)+D
Min Qy	589	-0.303	-0.99	0	0	0	-93.33	-159.7	-0.983	-94.31	-160.6	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max Sx	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Min Sx	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Max Sy	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Min Sy	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Max Sxy	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Min Sxy	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Max Mx	133	-0.205	-0.591	0	0	0	54.68	96.99	-5.072	59.75	102.1	227 0.9(D1+D2)+W(+Z)+D
Min Mx	109	0.22	-0.984	0	0	0	-98.68	-163.7	-9.254	-107.9	-173	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max My	90	-0.115	-0.263	0	0	0	2.689	114	22.04	24.72	136	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Min My	37	0.2	-0.481	0	0	0	-35.15	-224.2	-0.11	-35.26	-224.3	227 0.9(D1+D2)+W(+Z)+D
Max Mxy	210	-0.389	-0.065	0	0	0	29.95	72.99	50.31	80.26	123.3	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Min Mxy	691	0.363	-0.009	0	0	0	19.44	68.45	-61.3	80.74	129.8	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)

PLATE CENTER STRESS ; G6 : F3_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	468	0.241	0.007	0	0	0	37.73	-45.67	0.667	38.4	-46.34	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Min Qx	949	-0.288	-0.006	0	0	0	32.02	-52.83	0	32.02	-52.83	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max Qy	79	0.038	0.333	0	0	0	-33.89	-37.11	0.6	-34.49	-37.71	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Min Qy	121	0.053	-0.377	0	0	0	-54.42	-40.14	-9.75	-64.17	-49.89	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max Sx	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Min Sx	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Max Sy	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Min Sy	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Max Sxy	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Min Sxy	2207	-0.004	-0.005	0	0	0	4.671	-0.503	1.128	5.799	-1.631	201 1.4D+1.4(D1+D2)
Max Mx	1995	0.045	0.004	0	0	0	53.69	-29.6	0.595	54.28	-30.2	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Min Mx	121	0.053	-0.377	0	0	0	-54.42	-40.14	-9.75	-64.17	-49.89	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max My	49	-0.024	0.097	0	0	0	1.488	62.51	-5.843	7.331	68.35	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Min My	49	0.051	-0.221	0	0	0	-27.29	-148.6	-9.576	-36.87	-158.2	227 0.9(D1+D2)+W(+Z)+D
Max Mxy	456	0.088	0.149	0	0	0	28.68	-2.665	20.65	49.33	-23.32	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Min Mxy	967	0.062	0.103	0	0	0	-11.86	-8.547	-29.5	-41.36	-38.05	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

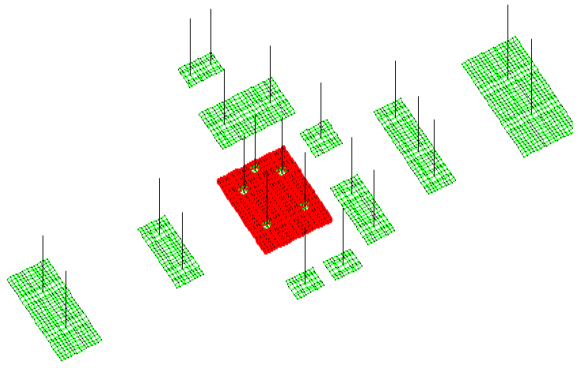
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	59.7	top face	50	442	59.75	0.014	359.7	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min Mx	-107.9	bottom face	75	417	107.9		692.9		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.241		120.5	38.4	0.002	0.442	442.1	435.5	758.3	435.48	1621	> 120.5	ok	326.6	φVc>Vu	not required
	Min Qx	0.288		144	32.02			452.4			435.48	1621	> 144	ok	326.6	φVc>Vu	not required

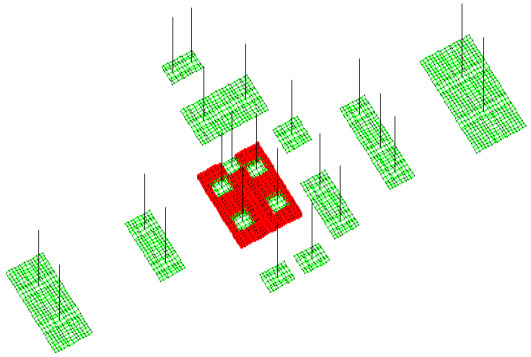
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	136.0	top face	50	442	136	0.014	824.9	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min My	-224.3	bottom face	75	415	224.3		1466.7		20	214.28	200	1571.5	22.16	27.7	0.042	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.333		166.5	37.71	0.002	0.442	451.8	435.5	758.3	435.48	456	> 166.5	ok	326.6	φVc>Vu	not required
	Min Qy	0.377		188.5	49.89			446.9			435.48	456	> 188.5	ok	326.6	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F4:

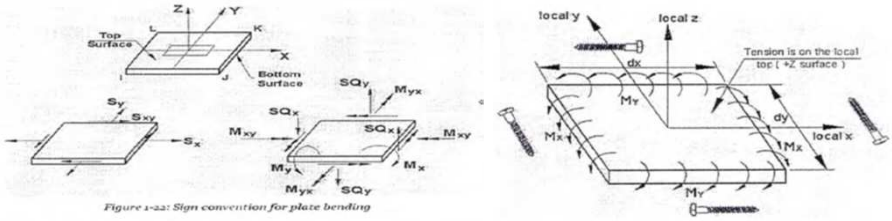


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G7 : F4_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1058	0.299	0.022	0	0	0	22.55	-2.616	3.009	25.56	-5.625	202 1.4(D+D1+D2)+1.4T(+VE)+1.7L1
Min Qx	1068	-0.302	0.106	0	0	0	10.3	-3.173	0	10.3	-3.173	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Max Qy	1069	0.008	0.255	0	0	0	-0.663	-6.103	-1.2	-1.863	-7.303	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min Qy	1013	0.008	-0.255	0	0	0	-0.664	-6.109	1.189	-1.853	-7.298	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Max Sx	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Min Sx	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Max Sy	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Min Sy	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Max Sxy	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Min Sxy	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Max Mx	1048	0.017	-0.013	0	0	0	42.4	1.646	-0.194	42.59	1.84	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Min Mx	989	0.066	-0.006	0	0	0	-5.663	-4.057	-0.562	1.775	-4.619	224 0.9(D1+D2)+W(+X)+D
Max My	991	0.002	0.016	0	0	0	41.44	1.88	0.041	41.48	1.921	202 1.4(D+D1+D2)+1.4T(+VE)+1.7L1
Min My	999	0.069	-0.056	0	0	0	-1.234	-19.47	-0.624	-1.858	-20.09	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Max Mxy	1021	-0.092	-0.01	0	0	0	25.58	-0.953	3.076	28.66	-4.029	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min Mxy	1077	-0.091	0.01	0	0	0	25.61	-0.952	-3.105	28.71	-4.057	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3

PLATE CENTER STRESS ; G8 : F4_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1072	0.093	0.01	0	0	0	25.42	-0.951	3.022	28.44	-3.973	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min Qx	1021	-0.092	-0.01	0	0	0	25.58	-0.951	0	25.58	-0.951	202 1.4(D+D1+D2)+1.4T(+VE)+1.7L1
Max Qy	1083	-0.04	0.069	0	0	0	0.746	-0.302	-1.037	1.783	-1.339	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min Qy	1027	-0.04	-0.069	0	0	0	0.744	-0.304	1.032	1.776	-1.336	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Max Sx	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Min Sx	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Max Sy	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Min Sy	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Max Sxy	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Min Sxy	1083	0.002	0.009	0	0	0	-0.229	-0.455	0.101	-0.33	-0.556	201 1.4D+1.4(D1+D2)
Max Mx	1047	0.003	-0.016	0	0	0	41.46	1.871	-0.109	41.57	1.98	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Min Mx	1072	0.005	0.01	0	0	0	-4.865	-0.044	-0.652	-5.517	-0.696	224 0.9(D1+D2)+W(+X)+D
Max My	991	0.002	0.016	0	0	0	41.44	1.88	0.041	41.48	1.921	202 1.4(D+D1+D2)+1.4T(+VE)+1.7L1
Min My	1046	0.043	0.01	0	0	0	-0.283	-4.069	0.217	-0.5	-4.286	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Max Mxy	1021	-0.092	-0.01	0	0	0	25.58	-0.953	3.076	28.66	-4.029	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min Mxy	1077	-0.091	0.01	0	0	0	25.61	-0.952	-3.105	28.71	-4.057	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		300	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

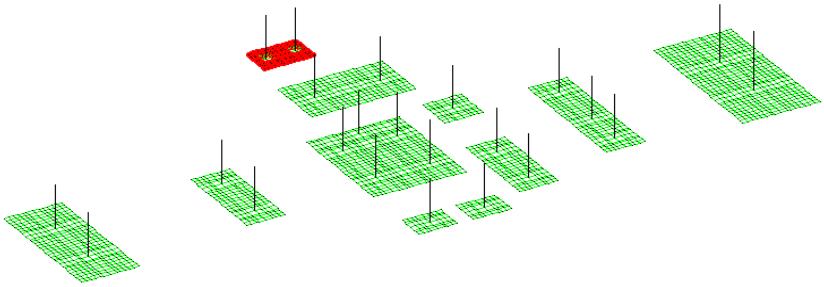
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	42.6	top face	50	244	42.59	0.014	468.1	540	12	209.52	200	565.8	7.978	9.972	0.07	tension control
Min Mx	1.8	top face	50	244	1.775		19.3		12	209.52	200	565.8	7.978	9.972	0.07	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.093		27.9	28.44	0.002	0.244	233.3	240.6	418.6	233.27	889	> 27.9	ok	174.9	φVc>Vu	not required
	Min Qx	0.092		27.6	25.58			233.5			233.5	890	> 27.6	ok	175.1	φVc>Vu	not required

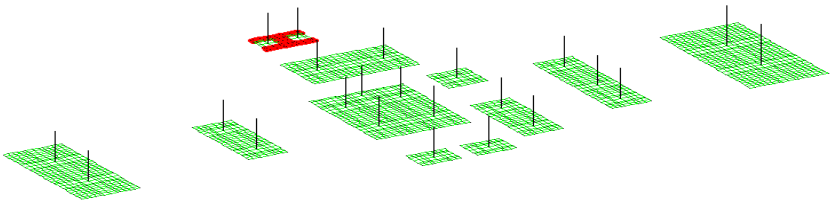
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	KN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	1.9	top face	50	244	1.921	0.014	20.8	540	12	209.52	200	565.8	7.978	9.972	0.07	tension control
Min My	-20.1	bottom face	75	219	20.09		244.6		12	209.52	200	565.8	7.978	9.972	0.063	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.069		20.7	1.339	0.004	0.244	267.2	240.6	418.6	240.58	252	> 20.7	ok	180.4	φVc>Vu	not required
	Min Qy	0.069		20.7	1.336			267.3			240.58	252	> 20.7	ok	180.4	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F5:

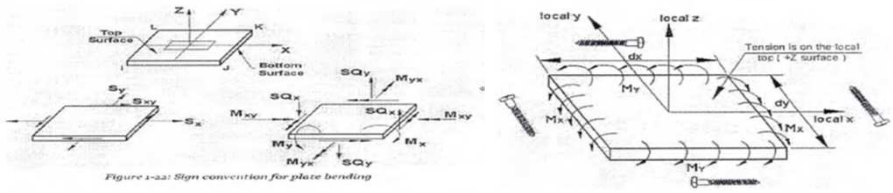


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G9 : F5_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1182	0.396	0.277	0	0	0	-34	-55.91	-14.16	-48.16	-70.07	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Min Qx	1111	-0.383	-0.084	0	0	0	-54.76	-60.42	0	-54.76	-60.42	233 1.2(D+D1+D2)+1.0L3+1.0E_MIN
Max Qy	1180	0.057	0.407	0	0	0	-40.5	-70.71	-8.384	-48.88	-79.09	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Min Qy	1127	-0.228	-0.306	0	0	0	-45.76	-37.57	10.36	-56.12	-47.93	233 1.2(D+D1+D2)+1.0L3+1.0E_MIN
Max Sx	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Min Sx	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Max Sy	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Min Sy	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Max Sxy	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Min Sxy	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Max Mx	1110	0.191	0.059	0	0	0	31.37	12.7	5.178	36.55	17.88	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Min Mx	1166	0.26	0.067	0	0	0	-64.68	-23.42	-4.254	-60.94	-27.68	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Max My	1188	0.002	-0.202	0	0	0	11.55	38.45	1.509	13.06	39.96	226 0.9(D1+D2)+W(-Z)+D
Min My	1180	0.057	0.407	0	0	0	-40.5	-70.71	-8.384	-48.88	-79.09	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Mxy	1183	0.088	0.025	0	0	0	-7.037	-23.46	21.37	-28.4	-44.82	233 1.2(D+D1+D2)+1.0L3+1.0E_MIN
Min Mxy	1118	-0.125	-0.053	0	0	0	-3.207	10.67	-27.33	-30.54	38	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)

PLATE CENTER STRESS ; G10 : F5_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1186	0.124	0.021	0	0	0	3.557	-27.07	-9.359	12.92	-36.43	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Min Qx	1115	-0.163	-0.033	0	0	0	-11.33	-35.98	0	-11.33	-35.98	233 1.2(D+D1+D2)+1.0L3+1.0E_MIN
Max Qy	1212	0.038	0.156	0	0	0	-27.49	-18.16	-5.943	-33.43	-24.1	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Min Qy	1157	0.004	-0.111	0	0	0	-30.91	-6.261	1.136	-32.04	-7.397	233 1.2(D+D1+D2)+1.0L3+1.0E_MIN
Max Sx	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Min Sx	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Max Sy	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Min Sy	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Max Sxy	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Min Sxy	2747	0.007	-0.003	0	0	0	1.108	-0.409	0.768	1.876	-1.177	201 1.4D+1.4(D1+D2)
Max Mx	2742	-0.014	-0.005	0	0	0	17.42	-13.82	2.777	20.2	-16.6	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min Mx	1157	0.004	-0.111	0	0	0	-30.91	-6.261	1.136	-32.04	-7.397	233 1.2(D+D1+D2)+1.0L3+1.0E_MIN
Max My	1194	0.022	0.009	0	0	0	0.328	16.73	0.813	1.141	17.55	226 0.9(D1+D2)+W(-Z)+D
Min My	1171	0.026	0.012	0	0	0	-0.354	-37.31	2.006	-2.36	-39.32	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Max Mxy	1147	-0.085	-0.07	0	0	0	-11.22	-17.15	12.7	-23.91	-29.85	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3
Min Mxy	1106	0.07	-0.055	0	0	0	1.963	-17.65	-19.77	21.74	-37.42	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

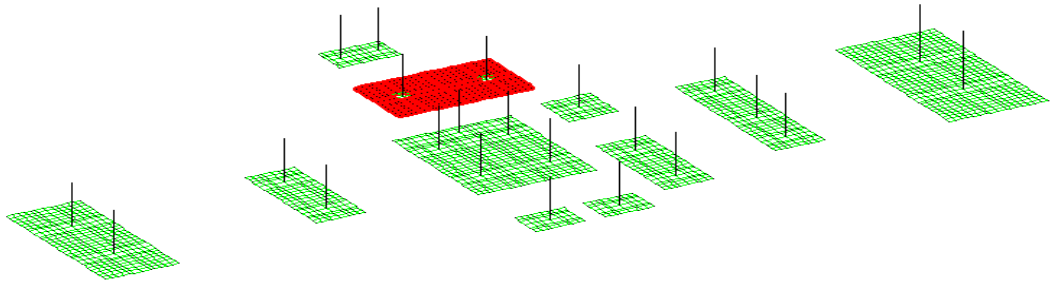
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	36.5	top face	50	442	36.55	0.014	219.5	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min Mx	-60.9	bottom face	75	417	60.94		389.2		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.124		62	12.92	0.002	0.442	454.7	435.5	758.3	435.48	1621	> 62	ok	326.6	φVc>Vu	not required
	Min Qx	0.163		81.5	11.33			472.7			435.48	1621	> 81.5	ok	326.6	φVc>Vu	not required

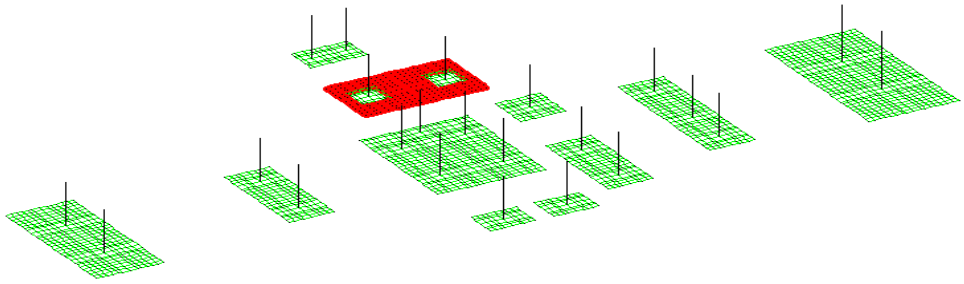
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	40.0	top face	50	442	39.96	0.014	240.1	900	16	223.49	200	1005.8	14.18	17.73	0.072	tension control
Min My	-79.1	bottom face	75	417	79.09		506.1		16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.156		78	24.1	0.002	0.442	442.8	435.5	758.3	435.48	456	> 78	ok	326.6	φVc>Vu	not required
	Min Qy	0.111		55.5	7.397			475.1			435.48	456	> 55.5	ok	326.6	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F6:

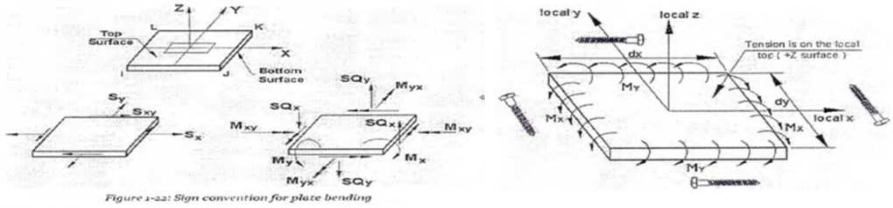


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G11 : F6_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1302	0.405	0.047	0	0	0	-23.29	-32.09	5.083	-28.37	-37.17	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Min Qx	1312	-0.287	0.033	0	0	0	-16.4	-22.94	0	-16.4	-22.94	221 1.2(D+D1+D2)+1.0L3+1.6W(-X)
Max Qy	1320	0.214	0.331	0	0	0	-21.04	-12.09	6.971	-28.01	-19.06	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Min Qy	1248	0.214	-0.331	0	0	0	-21.04	-12.09	-6.971	-28.01	-19.06	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Max Sx	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Min Sx	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Max Sy	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Min Sy	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Max Sxy	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Min Sxy	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Max Mx	1302	-0.102	-0.015	0	0	0	8.661	2.887	-0.025	8.686	2.912	225 0.9(D1+D2)+W(-X)+D
Min Mx	1246	-0.034	-0.269	0	0	0	-26.89	-11.23	-4.123	-23.01	-15.36	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3
Max My	1302	-0.102	-0.015	0	0	0	8.661	2.887	-0.025	8.686	2.912	225 0.9(D1+D2)+W(-X)+D
Min My	1302	0.405	0.047	0	0	0	-23.29	-32.09	5.083	-28.37	-37.17	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Max Mxy	1258	-0.124	-0.097	0	0	0	-9.296	-11.33	9.701	-19	-21.03	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3
Min Mxy	1330	-0.123	0.097	0	0	0	-9.292	-11.31	-9.702	-18.99	-21.01	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3

PLATE CENTER STRESS ; G12 : F6_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1284	0.067	-0.027	0	0	0	-12.33	-0.193	-0.173	-12.5	-0.366	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Min Qx	1294	-0.047	-0.02	0	0	0	-8.846	-0.156	0	-8.846	-0.156	221 1.2(D+D1+D2)+1.0L3+1.6W(-X)
Max Qy	1324	0.051	0.058	0	0	0	-1.644	-12.51	2.493	-4.137	-15	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Min Qy	1252	0.051	-0.058	0	0	0	-1.644	-12.51	-2.493	-4.137	-15	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Max Sx	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Min Sx	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Max Sy	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Min Sy	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Max Sxy	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Min Sxy	1370	-0.006	0.007	0	0	0	-0.733	-0.744	-0.501	-1.234	-1.245	201 1.4D+1.4(D1+D2)
Max Mx	1244	0.019	0.01	0	0	0	0.806	-1.069	0.287	1.093	-1.356	224 0.9(D1+D2)+W(+X)+D
Min Mx	1282	0.024	-0.037	0	0	0	-17.11	-0.407	-0.573	-17.69	-0.98	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3
Max My	1306	-0.021	-0.011	0	0	0	0.749	1.776	0.034	0.783	1.81	225 0.9(D1+D2)+W(-X)+D
Min My	1234	0.066	-0.029	0	0	0	-1.433	-20.12	-1.342	-2.775	-21.46	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Max Mxy	1292	0.028	-0.023	0	0	0	-11.97	-0.485	4.281	-16.25	-4.766	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
Min Mxy	1364	0.028	0.023	0	0	0	-11.97	-0.485	-4.281	-16.25	-4.766	220 1.2(D+D1+D2)+1.0L3+1.6W(+X)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		300	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

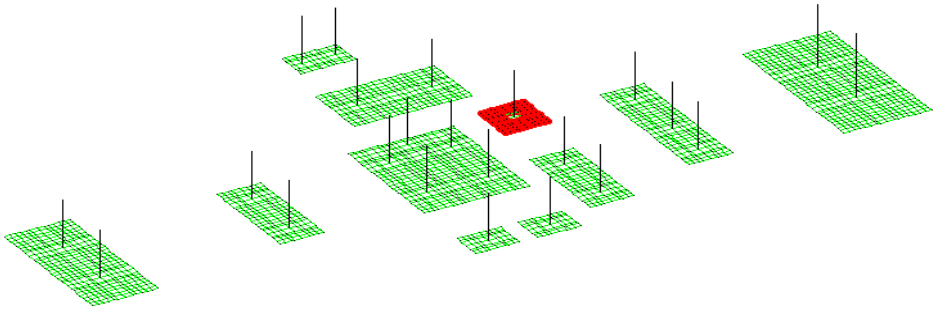
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	8.7	top face	50	244	8.686	0.014	94.4	540	12	209.52	200	565.8	7.978	9.972	0.07	tension control
Min Mx	-23.0	bottom face	75	219	23.01		280.5		12	209.52	200	565.8	7.978	9.972	0.063	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.067		20.1	12.5	0.002	0.244	234.7	240.6	418.6	234.74	891	> 20.1	ok	176.1	φVc>Vu	not required
	Min Qx	0.047		14.1	8.846			234.7			234.7	891	> 14.1	ok	176	φVc>Vu	not required

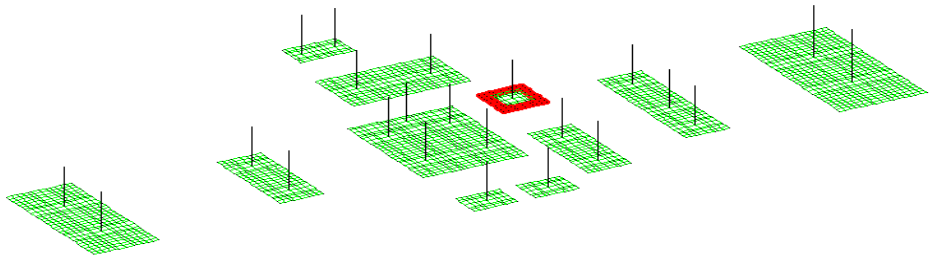
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	2.9	top face	50	244	2.912	0.014	31.6	540	12	209.52	200	565.8	7.978	9.972	0.07	tension control
Min My	-37.2	bottom face	75	219	37.17		455.7		12	209.52	200	565.8	7.978	9.972	0.063	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.058		17.4	15	0.004	0.244	233.7	240.6	418.6	233.69	247	> 17.4	ok	175.3	φVc>Vu	not required
	Min Qy	0.058		17.4	15			233.7			233.69	247	> 17.4	ok	175.3	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F7:

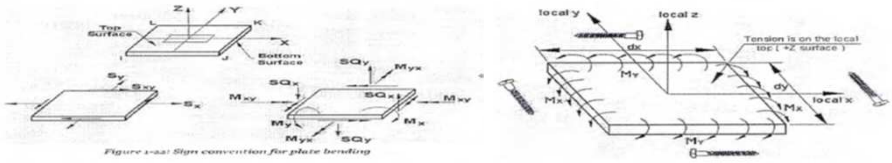


Figure 1-22: Sign convention for plate bending

Sign convention :

Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :

Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :

SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :

+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:

PLATE CENTER STRESS ; G13 : F7_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1509	0.201	0.032	0	0	0	-14.48	-37.4	7.053	-21.53	-44.45	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Min Qx	1513	-0.202	0.032	0	0	0	-14.52	-37.42	0	-14.52	-37.42	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max Qy	1516	0.031	0.244	0	0	0	-32.61	-17.89	6.188	-38.8	-24.07	206 1.4(D+D1+D2)+1.4T(-VE)+1.7L2
Min Qy	1544	-0.025	-0.191	0	0	0	-34.38	-15.4	6.203	-40.58	-21.6	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max Sx	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Min Sx	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Max Sy	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Min Sy	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Max Sxy	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Min Sxy	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Max Mx	1319	-0.043	-0.129	0	0	0	11.67	17.27	1.498	13.17	18.77	227 0.9(D1+D2)+W(+Z)+D
Min Mx	1540	0.025	-0.191	0	0	0	-34.38	-15.39	-6.198	-32.58	-21.59	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max My	1293	0.001	-0.008	0	0	0	-3.371	24.27	0.478	-3.849	24.75	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Min My	1513	-0.182	0.002	0	0	0	-12.9	-40.43	-6.521	-19.42	-46.95	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max Mxy	1517	0.138	0.165	0	0	0	-14.8	-17.97	13.48	-28.27	-31.45	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Min Mxy	1521	-0.139	0.165	0	0	0	-14.82	-17.99	-13.45	-28.27	-31.43	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN

PLATE CENTER STRESS ; G14 : F7_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1367	0.031	-0.014	0	0	0	5.905	0.821	0.127	6.032	0.948	227 0.9(D1+D2)+W(+Z)+D
Min Qx	1357	-0.031	-0.014	0	0	0	5.905	0.821	0	5.905	0.821	227 0.9(D1+D2)+W(+Z)+D
Max Qy	1532	0.014	0.104	0	0	0	-15.27	8.798	2.625	-17.89	11.42	206 1.4(D+D1+D2)+1.4T(-VE)+1.7L2
Min Qy	1283	0.012	-0.056	0	0	0	-7.673	-12.62	0.059	-7.732	-12.68	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max Sx	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Min Sx	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Max Sy	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Min Sy	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Max Sxy	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Min Sxy	1643	-0.002	-0.006	0	0	0	-0.397	8.464	-0.204	-0.601	8.668	201 1.4D+1.4(D1+D2)
Max Mx	1355	-0.017	-0.022	0	0	0	9.64	1.107	-0.057	9.697	1.164	227 0.9(D1+D2)+W(+Z)+D
Min Mx	1560	-0.013	-0.032	0	0	0	-21.62	-0.927	1.53	-23.15	-2.457	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max My	1293	0.001	-0.008	0	0	0	-3.371	24.27	0.478	-3.849	24.75	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Min My	1515	-0.027	-0.013	0	0	0	-0.405	-29.22	-2.178	-2.583	-31.4	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Max Mxy	1534	0.024	0.042	0	0	0	-5.212	-10.43	6.22	-11.43	-16.65	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Min Mxy	1538	-0.024	0.042	0	0	0	-5.213	-10.43	-6.214	-11.43	-16.65	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

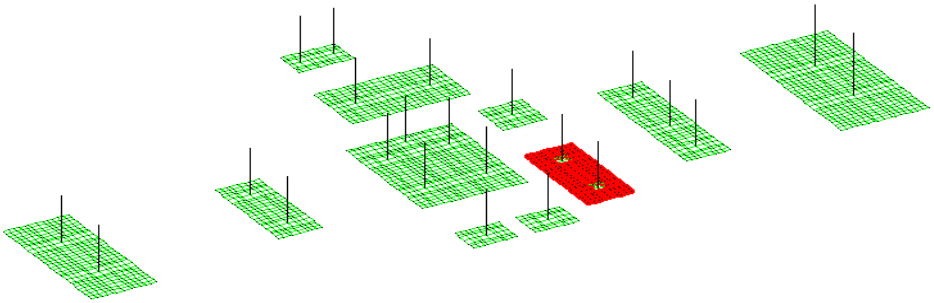
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	13.2	top face	50	444	13.17	0.014	78.6		12	251.43	200	565.8	7.978	9.972	0.131	tension control
Min Mx	-32.6	bottom face	75	417	32.58		207.4	900	16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.031		15.5	6.032	0.001	0.444	431.3	429.9	761.8	429.9	1623	> 15.5	ok	322.4	φVc>Vu	not required
	Min Qx	0.031		15.5	5.905			431.5			429.9	1623	> 15.5	ok	322.4	φVc>Vu	not required

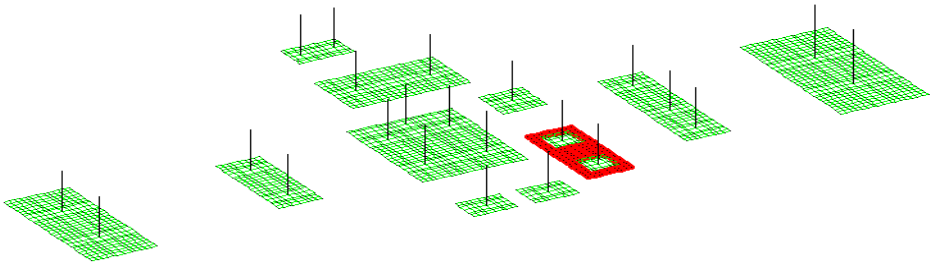
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	24.7	top face	50	444	24.75	0.014	147.8		12	251.43	200	565.8	7.978	9.972	0.131	tension control
Min My	-46.9	bottom face	75	417	46.95		299.4	900	16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.104		52	11.42	0.002	0.444	439.7	429.9	761.8	429.9	452	> 52	ok	322.4	φVc>Vu	not required
	Min Qy	0.056		28	12.68			429.7			429.71	452	> 28	ok	322.3	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F8:

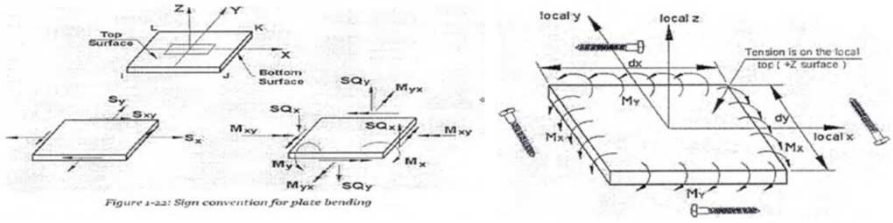


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G15 : F8_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1391	0.526	-0.293	0	0	0	-27.19	-18.28	-11.13	-38.31	-29.41	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Min Qx	1382	-0.613	-0.018	0	0	0	-35.54	-39.42	0	-35.54	-39.42	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max Qy	1460	0.128	0.384	0	0	0	-56.15	-7.805	-6.123	-62.28	-13.93	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Min Qy	1389	0.094	-0.561	0	0	0	-63.46	-20.39	-7.51	-70.97	-27.9	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max Sx	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Min Sx	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Max Sy	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Min Sy	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Max Sxy	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Min Sxy	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Max Mx	1382	0.139	0.022	0	0	0	10.39	4.646	-0.337	10.73	4.983	224 0.9(D1+D2)+W(+X)+D
Min Mx	1389	0.094	-0.561	0	0	0	-63.46	-20.39	-7.51	-62.97	-27.9	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max My	1453	0.01	-0.003	0	0	0	-23.03	6.99	4.246	-27.27	11.24	219 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
Min My	1446	-0.607	0.045	0	0	0	-35.61	-39.44	-8.751	-44.36	-48.19	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max Mxy	1455	0.016	0.051	0	0	0	-22.01	0.637	16.26	-38.27	16.9	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Min Mxy	1463	-0.015	0.05	0	0	0	-21.97	0.643	-16.26	-38.23	16.9	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN

PLATE CENTER STRESS ; G16 : F8_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	1377	0.278	-0.024	0	0	0	-8.493	-22.52	4.164	-12.66	-26.69	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Min Qx	1384	-0.351	-0.072	0	0	0	-14.89	-27.24	0	-14.89	-27.24	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max Qy	1464	-0.271	0.136	0	0	0	-16.71	-13.16	-9.023	-25.73	-22.19	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Min Qy	1405	0.119	-0.197	0	0	0	-54.4	-4.67	-2.759	-57.16	-7.429	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max Sx	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Min Sx	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Max Sy	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Min Sy	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Max Sxy	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Min Sxy	2771	0.012	0.001	0	0	0	-1.484	-1.175	1.279	-2.763	-2.454	201 1.4D+1.4(D1+D2)
Max Mx	1384	0.077	0.021	0	0	0	4.159	3.499	-0.234	4.393	3.733	224 0.9(D1+D2)+W(+X)+D
Min Mx	1405	0.119	-0.197	0	0	0	-54.4	-4.67	-2.759	-57.16	-7.429	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max My	1448	0.077	-0.02	0	0	0	4.129	3.571	0.177	4.306	3.748	224 0.9(D1+D2)+W(+X)+D
Min My	1384	-0.351	-0.072	0	0	0	-14.89	-27.24	4.052	-18.94	-31.29	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Max Mxy	1457	0.138	0.107	0	0	0	-13.08	-7.454	11.24	-24.33	-18.7	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
Min Mxy	1465	-0.138	0.107	0	0	0	-13.06	-7.445	-11.24	-24.3	-18.69	231 1.2(D+D1+D2)+1.0L2+1.0E_MIN

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		300	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

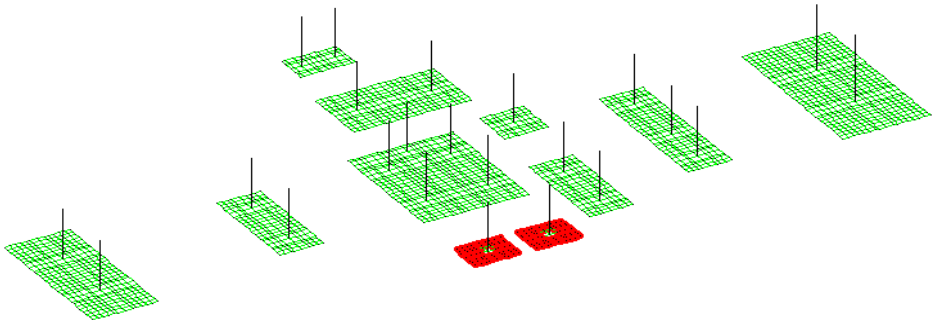
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	10.7	top face	50	244	10.73	0.014	116.7	540	12	209.52	200	565.8	7.978	9.972	0.07	tension control
Min Mx	-63.0	bottom face	75	217	62.97		787.9		16	255.3	200	1005.8	14.18	17.73	0.034	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φVc	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN			
Load condition	Max Qx	0.278		83.4	12.66	0.002	0.244	246.4	240.6	418.6	240.58	895	> 83.4	ok	180.4	φVc>Vu	not required
	Min Qx	0.351		105.3	14.89			247.6			240.58	895	> 105.3	ok	180.4	φVc>Vu	not required

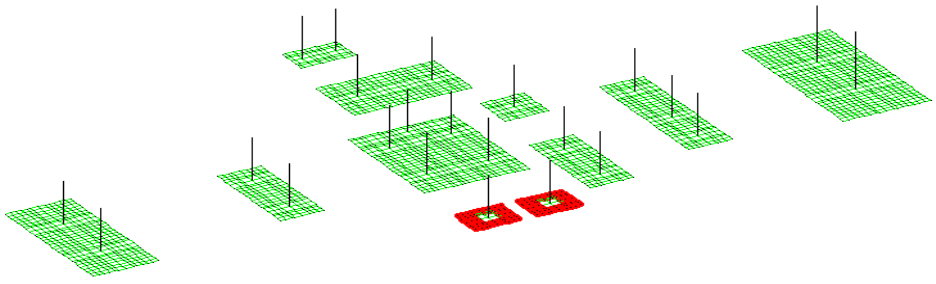
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	11.2	top face	50	244	11.24	0.014	122.3	540	12	209.52	200	565.8	7.978	9.972	0.07	tension control
Min My	-48.2	bottom face	75	217	48.19		599.2		16	335.7	200	1005.8	14.18	17.73	0.034	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φVc	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN			
Load condition	Max Qy	0.136		40.8	22.19	0.004	0.244	235.3	240.6	418.6	235.28	248	> 40.8	ok	176.5	φVc>Vu	not required
	Min Qy	0.197		59.1	7.429			249.6			240.58	252	> 59.1	ok	180.4	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F9:

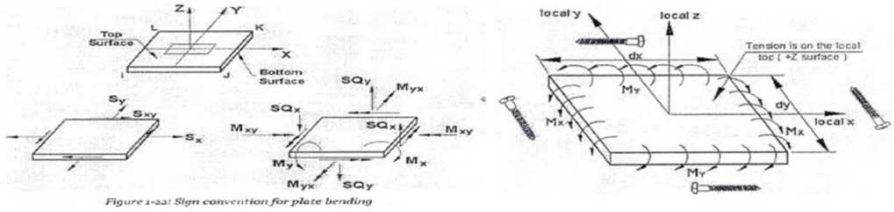


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G17 : F9_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	248	0.564	-0.028	0	0	0	-57.92	-59.41	9.101	-67.02	-68.51	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Min Qx	632	-0.547	-0.058	0	0	0	-52.63	-77.54	0	-52.63	-77.54	206 1.4(D+D1+D2)+1.4T(-VE)+1.7L2
Max Qy	572	-0.098	0.559	0	0	0	-54.04	-89.38	-1.904	-55.95	-91.28	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Min Qy	518	-0.138	-0.66	0	0	0	-61.94	-139.2	0.718	-62.66	-139.9	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Sx	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Min Sx	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Max Sy	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Min Sy	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Max Sxy	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Min Sxy	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Max Mx	254	-0.359	-0.094	0	0	0	21.12	14.12	21.18	42.3	35.29	227 0.9(D1+D2)+W(+Z)+D
Min Mx	614	-0.214	0.359	0	0	0	-73.7	-33.39	-9.941	-75.64	-43.33	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Max My	572	0.08	-0.295	0	0	0	15.12	73.76	-8.029	23.15	81.79	227 0.9(D1+D2)+W(+Z)+D
Min My	518	-0.138	-0.66	0	0	0	-61.94	-139.2	0.718	-62.66	-139.9	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Mxy	638	-0.02	-0.073	0	0	0	-1.684	24.38	41.63	-43.32	66.01	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Min Mxy	230	0.055	-0.042	0	0	0	-2.218	-38.37	-35.11	-37.33	-73.48	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3

PLATE CENTER STRESS ; G18 : F9_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	380	0.159	0.14	0	0	0	-21.78	-17.61	10.8	-32.57	-28.4	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Min Qx	380	-0.106	0.022	0	0	0	5.422	-1.447	0	5.422	-1.447	227 0.9(D1+D2)+W(+Z)+D
Max Qy	560	-0.034	0.268	0	0	0	-28.31	-13.14	-1.144	-29.46	-14.29	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Min Qy	530	-0.044	-0.314	0	0	0	-32.21	-48.76	1.869	-34.07	-50.63	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Sx	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Min Sx	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Max Sy	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Min Sy	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Max Sxy	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Min Sxy	2876	-0.002	-0.001	0	0	0	-0.308	-0.264	0.195	-0.503	-0.459	201 1.4D+1.4(D1+D2)
Max Mx	146	-0.035	-0.039	0	0	0	18.97	2.027	0.189	19.16	2.216	227 0.9(D1+D2)+W(+Z)+D
Min Mx	146	0.043	0.074	0	0	0	-49.12	-2.132	0.434	-49.55	-2.566	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Max My	2820	0.013	0.006	0	0	0	-1.755	70.26	-0.289	-2.044	70.55	204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
Min My	530	-0.033	-0.226	0	0	0	-19.02	-66.2	-1.353	-20.38	-67.55	227 0.9(D1+D2)+W(+Z)+D
Max Mxy	650	-0.018	-0.089	0	0	0	-11.96	47.94	16.18	-28.13	64.12	203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
Min Mxy	170	0.049	-0.149	0	0	0	-18.15	4.912	-15.44	-33.59	20.36	207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

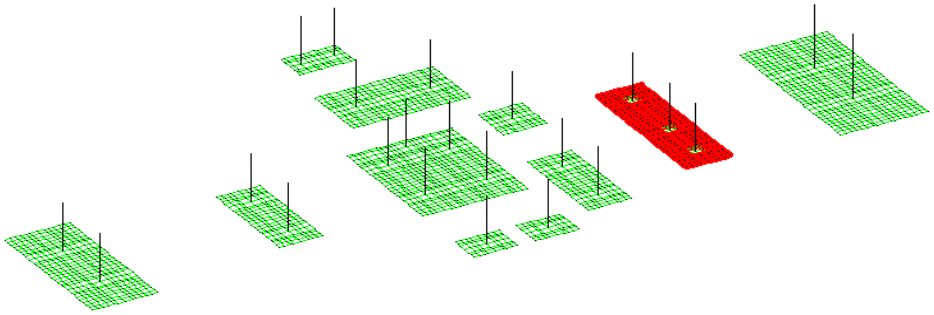
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	42.3	top face	50	444	42.3	0.014	253.0		12	251.43	200	565.8	7.978	9.972	0.131	tension control
Min Mx	-75.6	bottom face	75	417	75.64		483.8	900	16	223.49	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.159		79.5	32.57	0.001	0.444	430.7	429.9	761.8	429.9	1623	> 79.5	ok	322.4	φVc>Vu	not required
	Min Qx	0.106		53	5.422			462			429.9	1623	> 53	ok	322.4	φVc>Vu	not required

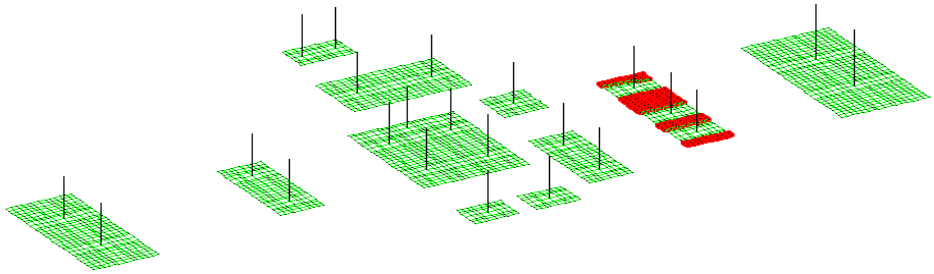
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	81.8	top face	50	444	81.79	0.014	491.2		12	230.35	200	565.8	7.978	9.972	0.131	tension control
Min My	-139.9	bottom face	75	417	139.9		901.1	900	16	223.21	200	1005.8	14.18	17.73	0.068	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.268		134	14.29	0.002	0.444	460.3	429.9	761.8	429.9	452	> 134	ok	322.4	φVc>Vu	not required
	Min Qy	0.314		157	50.63			433.5			429.9	452	> 157	ok	322.4	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Design of Foundation F10:

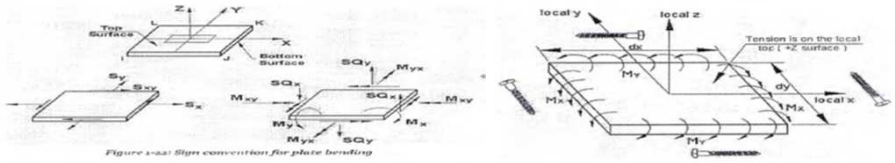


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G19 : F10_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	735	0.811	-0.407	0	0	0	99.53	166	-44.02	143.5	210	227 0.9(D1+D2)+W(+Z)+D
Min Qx	255	-0.811	-0.408	0	0	0	99.67	166.1	0	99.67	166.1	227 0.9(D1+D2)+W(+Z)+D
Max Qy	135	0.165	0.578	0	0	0	-92.75	-155.7	-4.269	-97.02	-160	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Min Qy	135	-0.274	-0.822	0	0	0	106.6	239.1	17.01	123.6	256.1	227 0.9(D1+D2)+W(+Z)+D
Max Sx	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Min Sx	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Max Sy	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Min Sy	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Max Sxy	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Min Sxy	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Max Mx	615	0.275	-0.822	0	0	0	106.6	239.1	-17.05	123.7	256.1	227 0.9(D1+D2)+W(+Z)+D
Min Mx	573	-0.066	0.334	0	0	0	-99.37	-85.97	-13.42	-104.8	-99.39	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max My	135	-0.274	-0.822	0	0	0	106.6	239.1	17.01	123.6	256.1	227 0.9(D1+D2)+W(+Z)+D
Min My	39	0.211	-0.461	0	0	0	-22.96	-169.6	30.91	-53.86	-200.5	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Mxy	159	0.518	-0.101	0	0	0	-35.46	-93.46	79.25	-114.7	-172.7	227 0.9(D1+D2)+W(+Z)+D
Min Mxy	639	-0.518	-0.102	0	0	0	-35.59	-93.56	-79.44	-115	-173	227 0.9(D1+D2)+W(+Z)+D

PLATE CENTER STRESS ; G20 : F10_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	975	0.218	0.062	0	0	0	12.91	74.9	-13.56	26.47	88.47	227 0.9(D1+D2)+W(+Z)+D
Min Qx	495	-0.218	0.062	0	0	0	12.99	74.99	0	12.99	74.99	227 0.9(D1+D2)+W(+Z)+D
Max Qy	147	0.06	0.25	0	0	0	-65.84	-59.09	5.002	-70.84	-64.1	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Min Qy	147	-0.094	-0.342	0	0	0	82.67	89.82	-4.654	87.32	94.47	227 0.9(D1+D2)+W(+Z)+D
Max Sx	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Min Sx	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Max Sy	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Min Sy	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Max Sxy	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Min Sxy	2519	-0.001	0	0	0	0	-0.132	-0.091	0.108	-0.24	-0.199	201 1.4D+1.4(D1+D2)
Max Mx	147	-0.094	-0.342	0	0	0	82.67	89.82	-4.654	87.32	94.47	227 0.9(D1+D2)+W(+Z)+D
Min Mx	147	0.06	0.25	0	0	0	-65.84	-59.09	5.002	-70.84	-64.1	218 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
Max My	147	-0.089	-0.332	0	0	0	72.14	92.03	-4.055	76.2	96.09	215 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
Min My	459	0.168	0.019	0	0	0	-20.52	-83.51	6.87	-27.39	-90.38	223 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
Max Mxy	393	-0.031	0.19	0	0	0	6.961	13.6	37.87	44.84	51.47	227 0.9(D1+D2)+W(+Z)+D
Min Mxy	873	0.031	0.19	0	0	0	6.886	13.49	-37.96	44.85	51.45	227 0.9(D1+D2)+W(+Z)+D

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		600	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

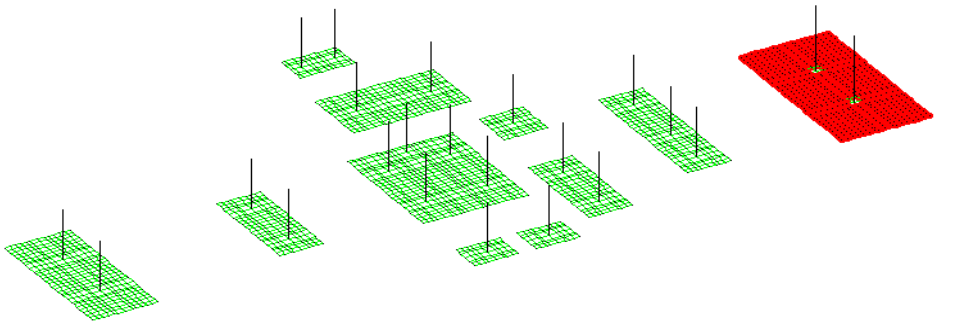
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required		Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2		mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	123.7	top face	50	542	123.7	0.014	608.4		16	330.59	200	1005.8	14.18	17.73	0.089	tension control
Min Mx	-104.8	bottom face	75	517	104.8		540.2		16	372.38	200	1005.8	14.18	17.73	0.084	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.218		130.8	26.47	0.002	0.542	558.8	530.1	929.9	530.14	1985	> 130.8	ok	397.6	φVc>Vu	not required
	Min Qx	0.218		130.8	12.99			606.4			530.14	1985	> 130.8	ok	397.6	φVc>Vu	not required

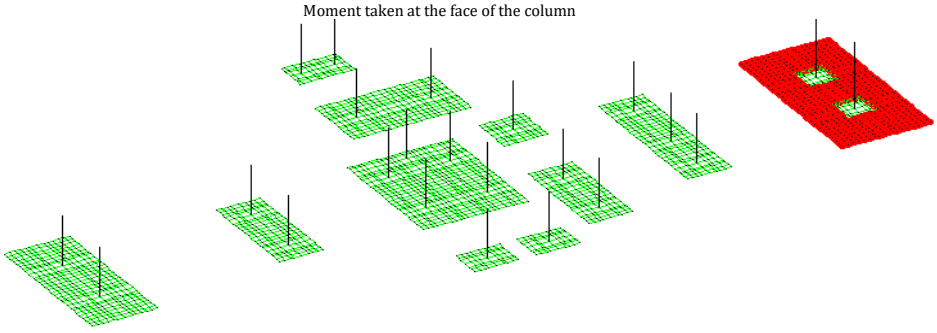
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	256.1	top face	50	540	256.1	0.014	1276.0	1080	20	246.31	200	1571.5	22.16	27.7	0.055	tension control
Min My	-200.5	bottom face	75	515	200.5		1045.0		20	291.01	200	1571.5	22.16	27.7	0.053	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.25		150	64.1	0.003	0.54	534.7	530.1	929.9	530.14	556	> 150	ok	397.6	φVc>Vu	not required
	Min Qy	0.342		205.2	94.47			533.2			530.14	556	> 205.2	ok	397.6	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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Stair Case Foundation Base Pressure :

refer staad file : MD-216-3510-EG-CV-CAL-1281_stair fdn in CCD area

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	105	101	0	51.701	0
Min Px	105	101	0	51.701	0
Max Py	120	102	0	167.666	0
Min Py	416	102	0	0	0
Max Pz	105	101	0	51.701	0
Min Pz	105	101	0	51.701	0

Gross Maximum Base pressure, p_{max} = 167.7 kN/m^2

Net Allowable Safe Bearing Capacity = 980 kN/m^2 (considered)

Depth of footing from FGL = 2 m

Unit weight of soil = 19 kN/m^3

Gross Safe Bearing Capacity = 1018 kN/m^2

> 167.7 kN/m^2 safe

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Design of Foundation F11:

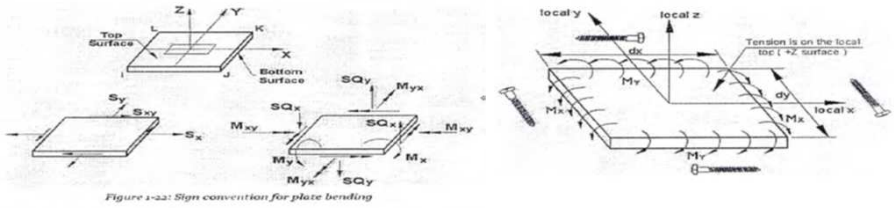


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G1 : F11_MOMENT At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	14	0.547	-0.234	0	0	0	201.6	-46.26	-30.96	232.6	-77.22	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min Qx	101	-0.546	0.225	0	0	0	202.9	-45.4	0	202.9	-45.4	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Max Qy	33	-0.058	0.588	0	0	0	3.763	-81.04	71.99	75.76	-153	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min Qy	15	-0.056	-0.587	0	0	0	4.095	-80.42	-72.03	76.12	-152.5	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Max Sx	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Min Sx	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Max Sy	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Min Sy	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Max Sxy	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Min Sxy	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Max Mx	2	-0.428	0.227	0	0	0	534.4	22.83	-35.82	570.2	58.65	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min Mx	81	-0.279	-0.115	0	0	0	-239.2	-23.89	-14.23	-245.4	-38.12	203 1.4(D1+D2)+1.4T(-VE)+1.7L1
Max My	105	0.062	-0.466	0	0	0	427.1	50.26	-56.42	483.5	106.7	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min My	28	-0.128	0.272	0	0	0	-8.488	-88.78	12.5	-20.99	-101.3	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Max Mxy	84	0.052	-0.578	0	0	0	4.529	-79.05	72.17	76.7	-151.2	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min Mxy	102	0.051	0.578	0	0	0	4.602	-78.93	-72.32	76.92	-151.3	202 1.4(D1+D2)+1.4T(+VE)+1.7L1

PLATE CENTER STRESS ; G2 : F11_SHEAR At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	268	0.389	0.042	0	0	0	141.7	-2.812	12.31	154	-15.12	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min Qx	251	-0.391	-0.04	0	0	0	142.5	-2.651	0	142.5	-2.651	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Max Qy	34	-0.024	0.509	0	0	0	-0.97	-52.76	22.34	-23.31	-75.09	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min Qy	16	-0.024	-0.511	0	0	0	-0.937	-52.55	-22.46	-23.39	-75.01	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Max Sx	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Min Sx	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Max Sy	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Min Sy	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Max Sxy	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Min Sxy	444	-0.013	0.001	0	0	0	10.79	-0.198	-0.284	11.07	-0.482	201 1.4(D1+D2)
Max Mx	165	-0.164	0.065	0	0	0	435.1	30.9	-1.248	436.3	32.15	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min Mx	180	-0.122	0.034	0	0	0	-179.2	-20.64	-0.71	-179.9	-21.35	203 1.4(D1+D2)+1.4T(-VE)+1.7L1
Max My	180	0.166	-0.065	0	0	0	433.6	31.85	-0.721	434.4	32.58	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min My	28	-0.128	0.272	0	0	0	-8.488	-88.78	12.5	-20.99	-101.3	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Max Mxy	85	0.023	-0.504	0	0	0	-0.893	-50.87	22.44	-23.34	-73.32	202 1.4(D1+D2)+1.4T(+VE)+1.7L1
Min Mxy	103	0.023	0.505	0	0	0	-0.887	-50.82	-22.51	-23.4	-73.33	202 1.4(D1+D2)+1.4T(+VE)+1.7L1

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	75	mm
	top face	50	mm
Thicknes of slab =		700	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

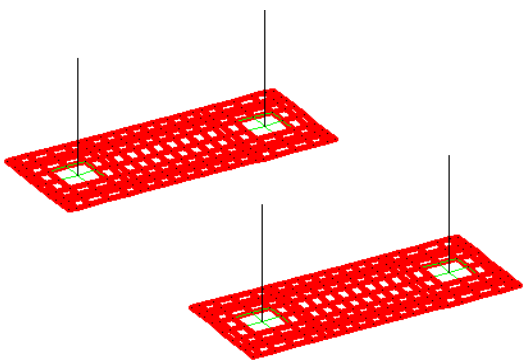
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	570.2	top face	50	640	570.2	0.014	2421.6	1260	20	129.78	100	3142.9	44.31	55.39	0.032	tension control
Min Mx	-245.4	bottom face	75	615	245.4		1068.8		20	249.43	200	1571.5	22.16	27.7	0.064	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.389		272.3	154	0.005	0.64	666.3	659.2	1098	659.24	2369	> 272.3	ok	494.4	φVc>Vu	not required
	Min Qx	0.391		273.7	142.5			671.5			659.24	2369	> 273.7	ok	494.4	φVc>Vu	not required

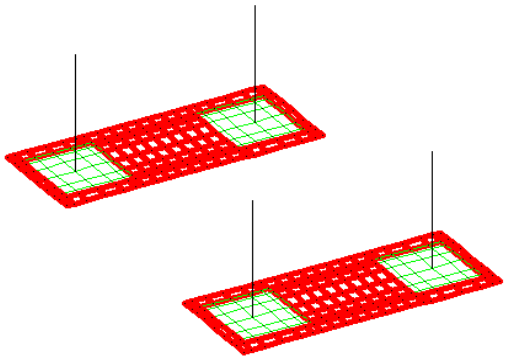
Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt. Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required		Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,}$ provided	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2		mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	106.7	top face	50	642	106.7	0.014	441.7		16	319.27	200	1005.8	14.18	17.73	0.106	tension control
Min My	-101.3	bottom face	75	617	101.3		436.4		16	319.27	200	1005.8	14.18	17.73	0.101	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.509		356.3	75.09	0.002	0.642	768.1	659.2	1098	659.24	682	> 356.3	ok	494.4	φVc>Vu	not required
	Min Qy	0.511		357.7	75.01			768.9			659.24	682	> 357.7	ok	494.4	φVc>Vu	not required



Moment taken at the face of the column



Shear taken at a distance d from face of the column

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7.3. Design of Pit

Foundation Base Pressure :

refer staad file : MD-216-3510-EG-CV-CAL-1281_pit in CCD area

	Node	L/C	Horizontal Fx KN/m2	Vertical Fy KN/m2	Horizontal Fz KN/m2
Max Px	1	101	0	37.439	0
Min Px	1	101	0	37.439	0
Max Py	1	103	0	62.408	0
Min Py	27	101	0	34.436	0
Max Pz	1	101	0	37.439	0
Min Pz	1	101	0	37.439	0

Gross Maximum Base pressure, p_{max} = 62.4 KN/m^2

Net Allowable Safe Bearing Capacity = 980 KN/m^2 (considered)

Depth of footing from FGL = 2 m

Unit weight of soil = 19 KN/m^3

Gross Safe Bearing Capacity = 1018 KN/m^2

> 62.4 KN/m^2 safe

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Design of Pit Wall :

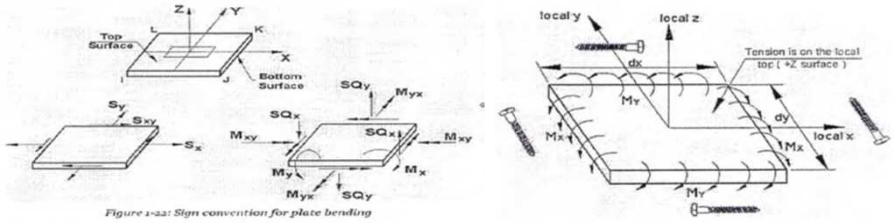


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards inside of pit

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G2 : WALL At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	164	0.066	-0.006	-0.11	-0.05	0	-0.652	0.318	-0.043	-0.695	0.361	202 1.4DL + 1.7(SDC+SP)
Min Qx	95	-0.066	-0.006	-0.11	-0.05	0	-0.652	0.318	0	-0.652	0.318	202 1.4DL + 1.7(SDC+SP)
Max Qy	148	0.006	0.092	-0.03	-0.12	-0.016	-0.157	-2.302	0.235	-0.392	-2.537	202 1.4DL + 1.7(SDC+SP)
Min Qy	226	-0.01	-0.011	-0.04	-0.01	0.002	0.884	0.109	-0.183	1.067	0.292	202 1.4DL + 1.7(SDC+SP)
Max Sx	155	0.016	0.01	0.028	-0.06	-0.018	0.052	-0.376	-0.024	0.076	-0.4	201 1.4DL
Min Sx	167	-0.066	-0.006	-0.11	-0.05	0	-0.652	0.318	0.043	-0.695	0.361	202 1.4DL + 1.7(SDC+SP)
Max Sy	251	-0.011	0.001	-0.01	-0.01	0.003	-0.358	-0.026	-0.211	-0.569	-0.237	202 1.4DL + 1.7(SDC+SP)
Min Sy	148	0.007	0.026	0.005	-0.13	-0.006	-0.539	-2.206	-0.244	-0.783	-2.45	201 1.4DL
Max Sxy	74	-0.003	0.04	-0.03	-0.07	0.027	-0.311	-1.252	-0.365	-0.676	-1.617	202 1.4DL + 1.7(SDC+SP)
Min Sxy	149	0.003	0.04	-0.03	-0.07	-0.027	-0.311	-1.252	0.365	-0.676	-1.617	202 1.4DL + 1.7(SDC+SP)
Max Mx	166	-0.02	-0.005	-0.1	-0.04	-0.001	1.967	1.117	0.042	2.009	1.159	202 1.4DL + 1.7(SDC+SP)
Min Mx	170	0.061	-0.002	-0.1	-0.05	-0.003	-0.691	0.143	0.151	7.158	0.294	202 1.4DL + 1.7(SDC+SP)
Max My	160	-0.019	0.004	-0.09	-0.06	-0.005	1.805	1.343	0.219	2.024	1.562	202 1.4DL + 1.7(SDC+SP)
Min My	148	0.006	0.092	-0.03	-0.12	-0.016	-0.157	-2.302	0.235	-0.392	-2.537	202 1.4DL + 1.7(SDC+SP)
Max Mxy	155	-0.036	0.005	-0.07	-0.07	-0.022	-0.351	0.072	0.589	-0.94	0.661	202 1.4DL + 1.7(SDC+SP)
Min Mxy	80	0.036	0.005	-0.07	-0.07	0.022	-0.351	0.072	-0.589	-0.94	0.661	202 1.4DL + 1.7(SDC+SP)

PLATE CENTER STRESS ; G2 : WALL At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	164	0.066	-0.006	-0.11	-0.05	0	-0.652	0.318	-0.043	-0.695	0.361	202 1.4DL + 1.7(SDC+SP)
Min Qx	95	-0.066	-0.006	-0.11	-0.05	0	-0.652	0.318	0	-0.652	0.318	202 1.4DL + 1.7(SDC+SP)
Max Qy	148	0.006	0.092	-0.03	-0.12	-0.016	-0.157	-2.302	0.235	-0.392	-2.537	202 1.4DL + 1.7(SDC+SP)
Min Qy	226	-0.01	-0.011	-0.04	-0.01	0.002	0.884	0.109	-0.183	1.067	0.292	202 1.4DL + 1.7(SDC+SP)
Max Sx	155	0.016	0.01	0.028	-0.06	-0.018	0.052	-0.376	-0.024	0.076	-0.4	201 1.4DL
Min Sx	167	-0.066	-0.006	-0.11	-0.05	0	-0.652	0.318	0.043	-0.695	0.361	202 1.4DL + 1.7(SDC+SP)
Max Sy	251	-0.011	0.001	-0.01	-0.01	0.003	-0.358	-0.026	-0.211	-0.569	-0.237	202 1.4DL + 1.7(SDC+SP)
Min Sy	148	0.007	0.026	0.005	-0.13	-0.006	-0.539	-2.206	-0.244	-0.783	-2.45	201 1.4DL
Max Sxy	74	-0.003	0.04	-0.03	-0.07	0.027	-0.311	-1.252	-0.365	-0.676	-1.617	202 1.4DL + 1.7(SDC+SP)
Min Sxy	149	0.003	0.04	-0.03	-0.07	-0.027	-0.311	-1.252	0.365	-0.676	-1.617	202 1.4DL + 1.7(SDC+SP)
Max Mx	166	-0.02	-0.005	-0.1	-0.04	-0.001	1.967	1.117	0.042	2.009	1.159	202 1.4DL + 1.7(SDC+SP)
Min Mx	170	0.061	-0.002	-0.1	-0.05	-0.003	-0.691	0.143	0.151	-0.842	0.294	202 1.4DL + 1.7(SDC+SP)
Max My	160	-0.019	0.004	-0.09	-0.06	-0.005	1.805	1.343	0.219	2.024	1.562	202 1.4DL + 1.7(SDC+SP)
Min My	148	0.006	0.092	-0.03	-0.12	-0.016	-0.157	-2.302	0.235	-0.392	-2.537	202 1.4DL + 1.7(SDC+SP)
Max Mxy	155	-0.036	0.005	-0.07	-0.07	-0.022	-0.351	0.072	0.589	-0.94	0.661	202 1.4DL + 1.7(SDC+SP)
Min Mxy	80	0.036	0.005	-0.07	-0.07	0.022	-0.351	0.072	-0.589	-0.94	0.661	202 1.4DL + 1.7(SDC+SP)

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	inner face	50	mm
	outer face	50	mm
Thicknes of slab =		200	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

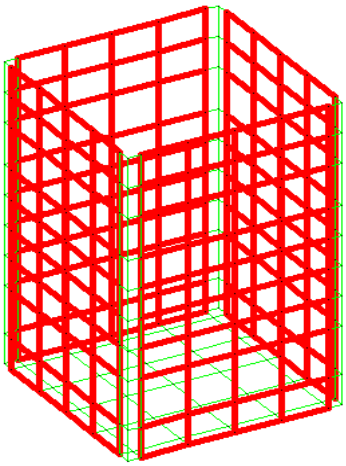
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	2.0	inner face	50	144	2.009	0.014	37.0	360	12	314.29	200	565.8	7.978	9.972	0.04	tension control
Min Mx	7.2	inner face	50	144	7.158		132.4		12	314.29	200	565.8	7.978	9.972	0.04	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.066		13.2	0.695	0.004	0.144	162.6	145.9	247.1	145.93	531	> 13.2	ok	109.4	φVc>Vu	not required
	Min Qx	0.066		13.2	0.652			164.3			145.93	531	> 13.2	ok	109.4	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	1.6	inner face	50	144	1.562	0.014	28.7	360	12	314.29	200	565.8	7.978	9.972	0.04	tension control
Min My	-2.5	outer face	50	144	2.537		46.7		12	314.29	200	565.8	7.978	9.972	0.04	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.092		18.4	2.537	0.007	0.144	146.4	145.9	247.1	145.93	152	> 18.4	ok	109.4	φVc>Vu	not required
	Min Qy	0.011		2.2	0.292			146.7			145.93	152	> 2.2	ok	109.4	φVc>Vu	not required



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Design of Pit Base :

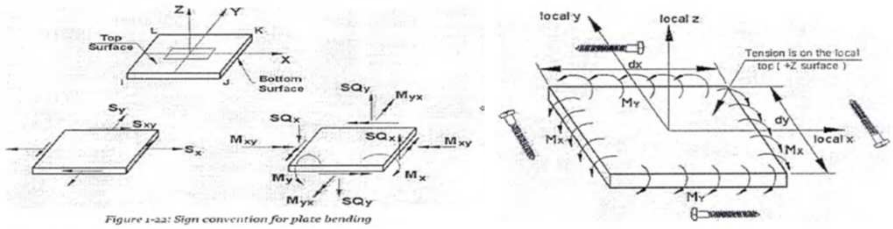


Figure 1-22: Sign convention for plate bending

Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS ; G1 : FDN At the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY (local) MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	20	0.069	-0.003	0	0	0	-0.283	0.768	0.386	-0.669	1.154	202 1.4DL + 1.7(SDC+SP)
Min Qx	23	-0.069	-0.003	0	0	0	-0.283	0.768	0	-0.283	0.768	202 1.4DL + 1.7(SDC+SP)
Max Qy	10	-0.003	0.069	0	0	0	0.768	-0.283	0.386	1.154	-0.669	202 1.4DL + 1.7(SDC+SP)
Min Qy	28	-0.003	-0.069	0	0	0	0.768	-0.283	-0.386	1.154	-0.669	202 1.4DL + 1.7(SDC+SP)
Max Sx	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Min Sx	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Max Sy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Min Sy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Max Sxy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Min Sxy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Max Mx	21	0.018	-0.018	0	0	0	2.561	2.561	0.25	2.811	2.811	201 1.4DL
Min Mx	11	-0.025	0.025	0	0	0	-0.329	-0.329	0.695	6.976	-1.024	202 1.4DL + 1.7(SDC+SP)
Max My	21	0.018	-0.018	0	0	0	2.561	2.561	0.25	2.811	2.811	201 1.4DL
Min My	11	-0.025	0.025	0	0	0	-0.329	-0.329	0.695	-1.024	-1.024	202 1.4DL + 1.7(SDC+SP)
Max Mxy	26	0.028	-0.028	0	0	0	0.298	0.298	1.133	1.431	1.431	201 1.4DL
Min Mxy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL

PLATE CENTER STRESS ; G1 : FDN At a distance "d" from the face of the column

Load condition	Plate	Shear		Membrane			Bending Moment			M + I M_{xy} I		L/C
		SQX (local) Mpa	SQY(local)MPa	SX (local) MPa	SY (local) MPa	SXY(local)MPa	Mx (kN-m/m)	My (kN-m/m)	Mxy (kN-m/m)	Mx (kN-m/m)	My (kN-m/m)	
Max Qx	20	0.069	-0.003	0	0	0	-0.283	0.768	0.386	-0.669	1.154	202 1.4DL + 1.7(SDC+SP)
Min Qx	23	-0.069	-0.003	0	0	0	-0.283	0.768	0	-0.283	0.768	202 1.4DL + 1.7(SDC+SP)
Max Qy	10	-0.003	0.069	0	0	0	0.768	-0.283	0.386	1.154	-0.669	202 1.4DL + 1.7(SDC+SP)
Min Qy	28	-0.003	-0.069	0	0	0	0.768	-0.283	-0.386	1.154	-0.669	202 1.4DL + 1.7(SDC+SP)
Max Sx	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Min Sx	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Max Sy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Min Sy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Max Sxy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Min Sxy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL
Max Mx	21	0.018	-0.018	0	0	0	2.561	2.561	0.25	2.811	2.811	201 1.4DL
Min Mx	11	-0.025	0.025	0	0	0	-0.329	-0.329	0.695	-1.024	-1.024	202 1.4DL + 1.7(SDC+SP)
Max My	21	0.018	-0.018	0	0	0	2.561	2.561	0.25	2.811	2.811	201 1.4DL
Min My	11	-0.025	0.025	0	0	0	-0.329	-0.329	0.695	-1.024	-1.024	202 1.4DL + 1.7(SDC+SP)
Max Mxy	26	0.028	-0.028	0	0	0	0.298	0.298	1.133	1.431	1.431	201 1.4DL
Min Mxy	29	-0.028	-0.028	0	0	0	0.298	0.298	-1.133	1.431	1.431	201 1.4DL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		200	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

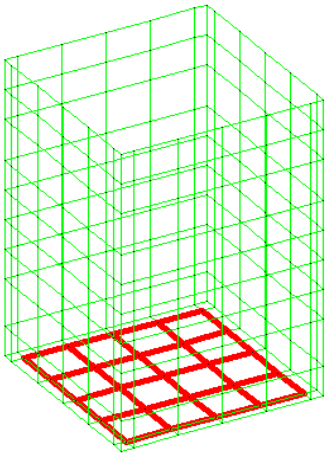
Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max Mx	2.8	top face	50	144	2.811	0.014	51.8	360	12	314.29	200	565.8	7.978	9.972	0.04	tension control
Min Mx	7.0	top face	50	144	6.976		129.0		12	314.29	200	565.8	7.978	9.972	0.04	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{ca}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qx	0.069		13.8	0.669	0.004	0.144	164.9	145.9	247.1	145.93	531	> 13.8	ok	109.4	φVc>Vu	not required
	Min Qx	0.069		13.8	0.283			203.8			145.93	531	> 13.8	ok	109.4	φVc>Vu	not required

Reinforcement Along Y:

Force Values			Depth		Reinforcement Calculation											
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	$a=f_y/0.85f_c' bA_s$	A_s required	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	$c = a/\beta_1$	$\varepsilon_t = \varepsilon_{cu}/c *(d-c)$	Remarks
	kN-m/m		mm	mm	kN-m/m	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
Max My	2.8	top face	50	144	2.811	0.014	51.8	360	12	314.29	200	565.8	7.978	9.972	0.04	tension control
Min My	-1.0	bottom face	50	144	1.024		18.8		12	314.29	200	565.8	7.978	9.972	0.04	tension control

Shear Check		Qx		Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		kN	kN-m/m		m^2	kN	kN	kN	kN	kN			kN		
Load condition	Max Qy	0.069		13.8	0.669	0.007	0.144	164.9	145.9	247.1	145.93	152	> 13.8	ok	109.4	φVc>Vu	not required
	Min Qy	0.069		13.8	0.669			164.9			145.93	152	> 13.8	ok	109.4	φVc>Vu	not required

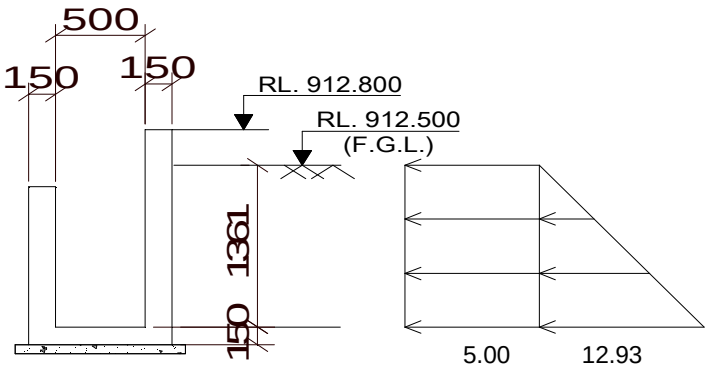


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7.4. Design of Drain



Drain cross section

Pressure diagram

Design Input :

Clear height of pit from FGL =	1361	mm
Clear width of Pit =	500	mm
Vertical wall thickness =	150	mm
Bottom slab thickness =	150	mm
Pressure coefficient, C_p =	0.5	
Surcharge load at side of pit, P =	10	kN/m^2
Dry unit weight of soil, $\gamma_{dry\ soil}$ =	19	kN/m^3

Load Calculation :

1 SOIL PRESSURE(DRY CONDITION) (H)

Lateral earth pressure at side wall due to dry soil = C_p x $\gamma_{dry\ soil}$ x h 12.93 kN/m^2

2 SURCHARGE PRESSURE (L)

Lateral earth pressure at side wall due to surcharged load = C_p x P 5.00 kN/m^2

Moment due lateral pressure -

5*1.361*1.361/2+0.5*12.9295*1.361*1.361/3 = 8.62 kN-m

Depth of section=	150	mm	Factored moment =	14.66	kN-m
Effective depth =	75	mm	As =	544.99	mm^2
Dia of bar =	12	mm	As min = 0.18/100*150*1000	270	mm^2
Grade of concrete =	fc'35		Spacing required =	207.52	mm c/c
Grade of reinf. =	fy420				
Strength Reduction Factor,φ :	0.9		Spacing provided	150	mm c/c

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Check for shear :

Shear force at a distance d = 5*1.286 + 0.5*19*1.286*0.5*1.286 = 14.29 kN

Factored shear force = 24.29 kN

$V_{c1} = (0.16\lambda\sqrt{f_c'} + 17\rho_w V_u d_s / M_{us}) b_w d_s$	72.6	kN					
$V_{c2} = (0.16\lambda\sqrt{f_c'} + 17\rho_w) b_w d_s$	83.8	kN	A_s=	753.98	mm^2	M_{us}=	14.66 kN-m
$V_{c3} = 0.29\lambda\sqrt{f_c'} b_w d_s$	128.7	kN	ρ_w=	0.01005		b_w =	1000 mm
V_c = Minimum of V_{c1}, V_{c2} and V_{c3}	72.6	kN	λ=	1		d_s =	75 mm
Strength Reduction Factor,φ :	0.75						
$V_u \leq \varphi (V_c + 0.66\sqrt{f_c'} b_w d_s)$	274.1	kN	ok				

φV_c = 54.4 kN

Remarks : shear reinforcement not reqd

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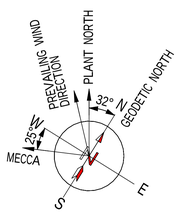
ATTACHMENT 1 OTE INPUT (CIVIL GUIDE DRAWING)

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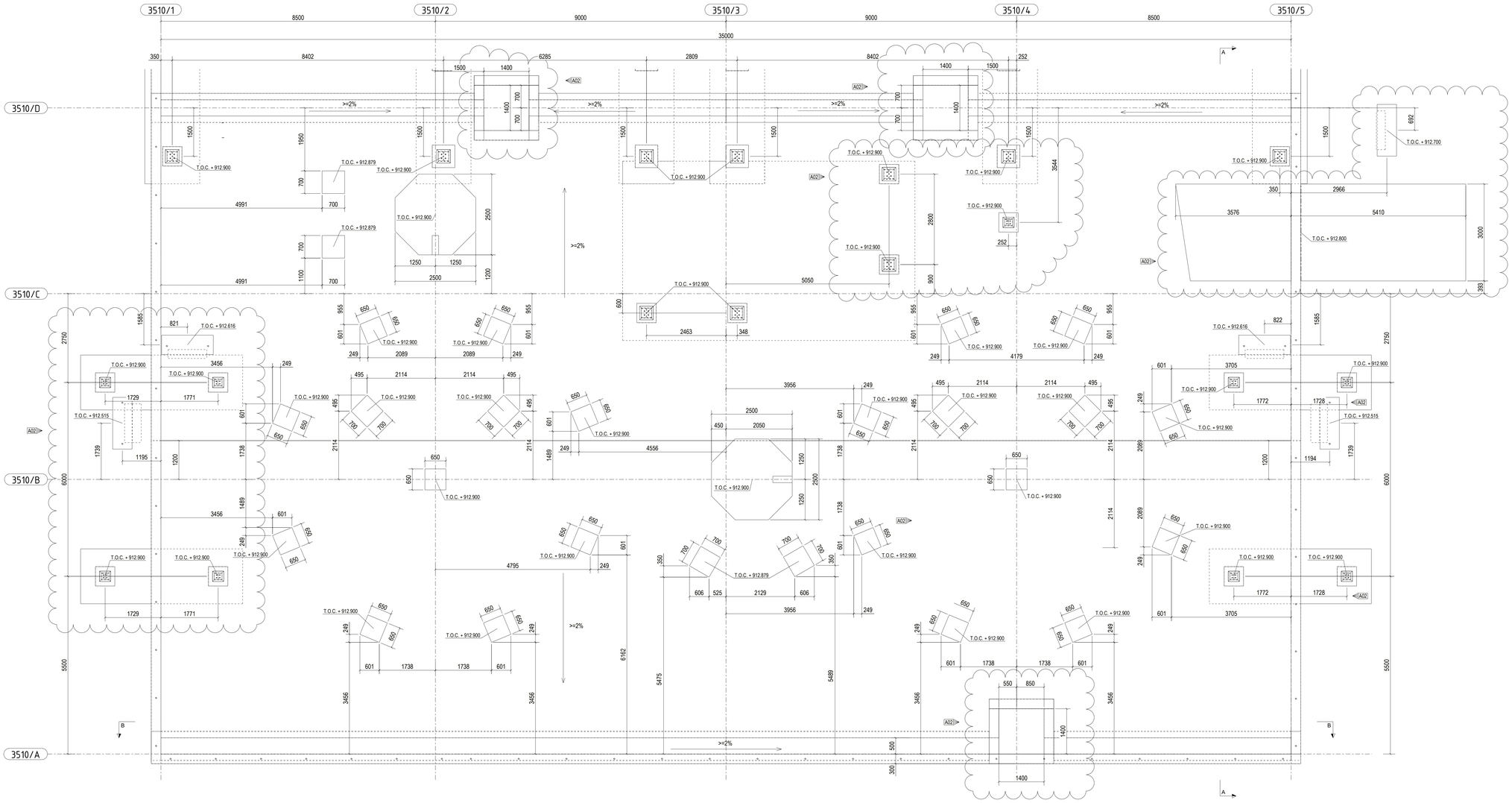
DRAWING NUMBER MD-216-3510-EG-ME-GAD-1025 REV A01



Node No.	LC/CO	Support Forces [kN] P_x	P_y	P_z	Support Moments [kNm] M_x	M_y	M_z
P101	LC2 - Live Load -4 kN/m^2	0,00	-2,37	-14,55	0,00	0,00	-0,02
	LC3 - Wind load +x direction	0,00	-0,01	-16,29	0,00	0,00	-0,01
	LC4 - Wind load -x direction	-2,54	0,00	14,03	0,00	0,00	-0,14
	LC5 - Wind load +y direction	0,00	5,66	10,09	0,00	0,00	-0,01
	LC6 - Wind load -y direction	0,00	-1,08	-8,24	0,00	0,00	0,00
	LC7 - Temperature T max +20	0,04	-1,65	-0,09	0,00	0,00	0,29
	LC8 - Temperature T min -20	-0,02	0,83	0,05	0,00	0,00	-0,15
	LC9 - Self Weight + Secondary Structures	0,00	-1,70	-15,73	0,00	0,00	-0,02
P102	LC2 - Live Load -4 kN/m^2	2,94	-0,05	-20,79	0,00	0,00	0,00
	LC3 - Wind load +x direction	11,49	0,00	-14,55	0,00	0,00	0,00
	LC4 - Wind load -x direction	-12,87	0,00	8,07	0,00	0,00	-0,02
	LC5 - Wind load +y direction	-0,20	0,00	-1,75	0,00	0,00	0,00
	LC6 - Wind load -y direction	0,30	-2,46	-0,24	0,00	0,00	0,00
	LC7 - Temperature T max +20	287,58	0,00	0,93	0,00	0,00	0,00
	LC8 - Temperature T min -20	-143,79	0,00	-0,46	0,00	0,00	0,00
	LC9 - Self Weight + Secondary Structures	1,89	-0,04	-18,29	0,00	0,00	0,00
P103	LC2 - Live Load -4 kN/m^2	0,71	-4,14	-52,73	0,00	0,00	-0,03
	LC3 - Wind load +x direction	16,77	0,00	16,18	0,00	0,00	-0,06
	LC4 - Wind load -x direction	-12,39	-0,02	-14,03	0,00	0,00	-0,01
	LC5 - Wind load +y direction	-3,78	10,99	5,14	0,00	0,00	0,15
	LC6 - Wind load -y direction	3,71	-9,71	-1,24	0,00	0,00	-0,14
	LC7 - Temperature T max +20	-0,05	-5,02	-0,35	0,00	0,00	-0,07
	LC8 - Temperature T min -20	0,03	2,51	0,17	0,00	0,00	0,04
	LC9 - Self Weight + Secondary Structures	0,55	-3,25	-47,09	0,00	0,00	-0,02
P104	LC2 - Live Load -4 kN/m^2	-3,68	0,12	-33,60	0,00	0,00	-0,11
	LC3 - Wind load +x direction	4,97	0,00	15,16	0,00	0,00	0,00
	LC4 - Wind load -x direction	-0,04	0,00	-8,08	0,00	0,00	0,00
	LC5 - Wind load +y direction	0,03	-0,01	-9,52	0,00	0,00	0,01
	LC6 - Wind load -y direction	-0,05	-2,95	9,77	0,00	0,00	-1,07
	LC7 - Temperature T max +20	-287,57	0,67	2,66	0,00	0,00	0,00
	LC8 - Temperature T min -20	143,79	-0,34	-1,33	0,00	0,00	0,00
	LC9 - Self Weight + Secondary Structures	-2,46	0,08	-28,29	0,00	0,00	-0,07
P105	LC2 - Self Weight + Secondary Structures	0,58	0,00	-28,57	0,00	0,00	0,00
	LC3 - Tank	-0,07	0,00	-38,46	0,00	0,00	0,00
	LC4 - Live Load -5 kN/m^2	0,85	0,00	-37,00	0,00	0,00	0,00
	LC5 - Live Load -10 kN/m^2	1,73	0,00	-69,43	0,00	0,00	0,00
	LC6 - Live Load -15 kN/m^2	2,63	0,00	-101,51	0,00	0,00	-0,01
	LC7 - Temperature +20	-3,33	-0,02	0,59	0,00	0,00	0,02
	LC8 - Temperature -20	1,70	-0,02	-0,19	0,00	0,00	0,03
	LC9 - Wind load +x direction	-18,89	0,00	-22,92	0,00	0,00	0,00
	LC10 - Wind load -x direction	-16,44	0,00	20,65	0,00	0,00	0,00
	LC11 - Wind load +y direction	1,61	0,37	9,90	0,00	0,00	-0,10
	LC12 - Wind load -y direction	-1,88	-0,17	-12,35	0,00	0,00	0,16
P106	LC2 - Self Weight + Secondary Structures	-4,48	0,00	-43,19	0,00	0,00	0,00
	LC3 - Tank	-8,95	-0,01	-67,80	0,00	0,00	0,00
	LC4 - Live Load -5 kN/m^2	-6,48	0,00	-56,98	0,00	0,00	0,00
	LC5 - Live Load -10 kN/m^2	-10,28	0,00	-95,19	0,00	0,00	0,00
	LC6 - Live Load -15 kN/m^2	-14,03	0,00	-133,08	0,00	0,00	0,00
	LC7 - Temperature +20	21,32	0,00	1,47	0,00	0,00	-0,02
	LC8 - Temperature -20	1,20	-0,03	0,24	0,00	0,00	-0,03
	LC9 - Wind load +x direction	-1,54	0,00	-49,18	0,00	0,00	0,00
	LC10 - Wind load -x direction	1,03	0,00	39,30	0,00	0,00	0,00
	LC11 - Wind load +y direction	-4,84	0,07	7,50	0,00	0,00	0,29
	LC12 - Wind load -y direction	5,94	-0,31	-8,32	0,00	0,00	-0,34
P107	LC2 - Self Weight + Secondary Structures	0,34	0,01	-5,64	0,00	0,00	0,00
	LC3 - Tank	0,00	0,00	0,04	0,00	0,00	0,00
	LC4 - Live Load -5 kN/m^2	0,83	0,02	-11,53	0,00	0,00	0,01
	LC5 - Live Load -10 kN/m^2	0,84	0,02	-11,62	0,00	0,00	0,00
	LC6 - Live Load -15 kN/m^2	0,84	0,02	-11,70	0,00	0,00	0,00
	LC7 - Temperature +20	16,70	-0,03	-8,57	0,00	0,00	0,00
	LC8 - Temperature -20	1,55	-0,05	-0,33	0,00	0,00	0,20
	LC9 - Wind load +x direction	0,13	0,00	-4,74	0,00	0,00	0,00
	LC10 - Wind load -x direction	-0,15	0,00	4,67	0,00	0,00	0,00
	LC11 - Wind load +y direction	-0,14	0,27	3,09	0,00	0,00	0,01
	LC12 - Wind load -y direction	0,13	-0,33	-3,17	0,00	0,00	-0,01
P108	LC2 - Self Weight + Secondary Structures	-0,59	0,00	-21,72	0,00	0,00	0,00
	LC3 - Tank	-0,05	0,00	-27,98	0,00	0,00	0,00
	LC4 - Live Load -5 kN/m^2	-0,87	0,00	-24,02	0,00	0,00	0,00
	LC5 - Live Load -10 kN/m^2	-1,79	0,00	-49,10	0,00	0,00	0,00
	LC6 - Live Load -15 kN/m^2	-2,72	0,00	-72,08	0,00	0,00	0,00
	LC7 - Temperature +20	3,00	0,02	-15,14	0,00	0,00	-0,02
	LC8 - Temperature -20	-1,59	0,02	-23,16	0,00	0,00	-0,03
	LC9 - Wind load +x direction	0,31	0,00	22,99	0,00	0,00	0,00
	LC10 - Wind load -x direction	-0,20	0,00	-20,46	0,00	0,00	0,00
	LC11 - Wind load +y direction	-0,18	-0,12	-29,52	0,00	0,00	0,22
	LC12 - Wind load -y direction	0,22	0,17	43,07	0,00	0,00	-0,26
P109	LC2 - Self Weight + Secondary Structures	4,40	0,96	-14,75	0,00	0,00	0,02
	LC3 - Tank	8,74	0,99	-16,42	0,00	0,00	0,03
	LC4 - Live Load -5 kN/m^2	6,36	2,04	-18,70	0,00	0,00	0,03
	LC5 - Live Load -10 kN/m^2	10,05	2,53	-35,16	0,00	0,00	0,02
	LC6 - Live Load -15 kN/m^2	13,72	2,92	-50,32	0,00	0,00	0,02
	LC7 - Temperature +20	-21,24	-13,53	28,24	0,00	0,00	-0,63
	LC8 - Temperature -20	-1,10	-21,70	36,80	0,00	0,00	-0,29
	LC9 - Wind load +x direction	29,79	0,06	49,07	0,00	0,00	0,04
	LC10 - Wind load -x direction	-22,71	0,18	-39,57	0,00	0,00	-0,04
	LC11 - Wind load +y direction	3,95	-34,81	64,99	0,00	0,00	2,42
	LC12 - Wind load -y direction	-5,03	47,35	-86,10	0,00	0,00	-2,79
P110	LC2 - Self Weight + Secondary Structures	-0,33	0,01	-5,62	0,00	0,00	0,00
	LC3 - Tank	0,04	0,00	0,06	0,00	0,00	0,00
	LC4 - Live Load -5 kN/m^2	-0,84	0,02	-11,50	0,00	0,00	0,00
	LC5 - Live Load -10 kN/m^2	-0,83	0,02	-11,56	0,00	0,00	0,00
	LC6 - Live Load -15 kN/m^2	-0,82	0,02	-11,63	0,00	0,00	0,00
	LC7 - Temperature +20	-16,53	0,04	-8,16	0,00	0,00	0,01
	LC8 - Temperature -20	-1,25	0,06	-0,31	0,00	0,00	0,01
	LC9 - Wind load +x direction	4,57	0,00	4,75	0,00	0,00	0,00
	LC10 - Wind load -x direction	-4,54	0,00	-4,74	0,00	0,00	0,00
	LC11 - Wind load +y direction	-2,28	0,04	-0,31	0,00	0,00	0,04
	LC12 - Wind load -y direction	2,80	-0,04	0,92	0,00	0,00	-0,05
P111	LC2 - Live Load - 5 kN/m^2	0,76	-0,19	-72,60	0,00	0,00	0,00
	LC3 - Wind Load +X Direction	1,26	-0,01	-2,05	0,00	0,00	-0,01
	LC4 - Wind Load -X Direction	-1,12	0,01	1,88	0,00	0,00	0,01
	LC5 - Wind Load +Y Direction	-0,29	3,69	170,77	0,00	0,00	-0,02
	LC6 - Wind Load -Y Direction	0,39	0,71	-319,90	0,00	0,00	0,03
	LC7 - Tank	-0,06	-0,01	2,46	0,00	0,00	0,00
	LC8 - Temperature +20	11,56	2,28	-15,50	0,00	0,00	-0,07
	LC10 - Live Load -10 kN/m^2	1,47	-0,43	-143,23	0,00	0,00	-0,01
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	1,50	-0,42	-143,67	0,00	0,00	-0,01
	LC13 - Self weight + Secondary steel struct	0,33	-0,10	-40,31	0,00	0,00	0,00
	LC14 - Temperature -20	-11,56	-2,28	15,50	0,00	0,00	0,07
P112	LC2 - Live Load - 5 kN/m^2	0,76	1,01	-64,92	0,00	0,00	0,02
	LC3 - Wind Load +X Direction	1,28	-0,28	-1,14	0,00	0,00	0,02
	LC4 - Wind Load -X Direction	-1,13	0,02	1,76	0,00	0,00	-0,02
	LC5 - Wind Load +Y Direction	0,17	94,78	-171,26	0,00	0,00	-0,05
	LC6 - Wind Load -Y Direction	-0,30	-177,56	320,90	0,00	0,00	0,03
	LC7 - Tank	-0,04	0,85	-2,14	0,00	0,00	0,00
	LC8 - Temperature +20	11,22	-11,64	2,89	0,00	0,00	0,18
	LC10 - Live Load -10 kN/m^2	1,76	0,75	-146,81	0,00	0,00	0,04
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	1,78	0,60	-146,42	0,00	0,00	0,05
	LC13 - Self weight + Secondary steel struct	0,39	0,05	-39,21	0,00	0,00	0,01
	LC14 - Temperature -20	-11,22	11,64	-2,89	0,00	0,00	-0,18
P113	LC2 - Live Load - 5 kN/m^2	-0,45	-0,62	-162,50	0,54	-0,20	0,00
	LC3 - Wind Load +X Direction	6,24	-0,06	0,13	0,16	8,98	0,00
	LC4 - Wind Load -X Direction	-4,04	-0,06	-1,14	0,06	-5,82	0,00
	LC5 - Wind Load +Y Direction	-2,03	33,21	-10,13	-60,30	-3,35	0,00
	LC6 - Wind Load -Y Direction	3,30	-35,27	11,48	91,86	5,48	0,02
	LC7 - Tank	-0,31	0,53	-1,04	-1,32	-0,46	0,00
	LC8 - Temperature +20	25,83	-9,18	4,88	11,99	37,03	0,00
	LC10 - Live Load -10 kN/m^2	-0,69	0,00	-312,61	-0,61	-0,22	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	-0,56	-0,38	-312,87	0,05	-0,02	0,00
	LC13 - Self weight + Secondary steel struct	-0,09	-0,15	-73,58	0,26	0,04	0,00
	LC14 - Temperature -20	-25,83	9,18	-4,88	-11,99	-37,03	0,00
P114	LC2 - Live Load - 5 kN/m^2	0,49	-0,46	-143,76	0,44	0,64	0,00
	LC3 - Wind Load +X Direction	6,27	-0,06	-0,47	0,15	9,01	0,00
	LC4 - Wind Load -X Direction	-4,05	-0,05	-0,48	0,04	-5,83	0,00
	LC5 - Wind Load +Y Direction	1,51	19,32	10,00	-50,63	2,47	-0,01
	LC6 - Wind Load -Y Direction	-2,55	-42,10	-12,00	94,57	-4,20	0,02
	LC7 - Tank	-0,19	0,46	0,15	-1,25	-0,26	0,00
	LC8 - Temperature +20	24,46	14,07	5,62	-22,02	35,10	0,00
	LC10 - Live Load -10 kN/m^2	1,15	0,07	-329,24	-0,58	1,53	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	1,25	-0,22	-328,42	0,02	1,67	0,00
	LC13 - Self weight + Secondary steel struct	0,28	-0,11	-76,76	0,24	0,37	0,00
	LC14 - Temperature -20	-24,46	-14,07	-5,62	22,02	-35,10	0,00
P115	LC2 - Live Load - 5 kN/m^2	-1,28	-0,01	-11,62	0,00	0,00	0,00
	LC3 - Wind Load +X Direction	21,56	-0,03	-22,69	0,00	0,00	0,00
	LC4 - Wind Load -X Direction	-14,26	-0,01	13,98	0,00	0,00	0,00
	LC5 - Wind Load +Y Direction	0,59	4,84	-8,41	0,00	0,00	0,00
	LC6 - Wind Load -Y Direction	-1,58	-4,84	9,61	0,00	0,00	0,00
	LC7 - Tank	0,08	0,04	-0,22	0,00	0,00	0,00
	LC8 - Temperature +20	26,70	-2,09	-32,60	0,00	0,00	0,00
	LC10 - Live Load -10 kN/m^2	-0,51	0,00	-12,59	0,00	0,00	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	0,89	0,01	-19,84	0,00	0,00	0,00
	LC13 - Self weight + Secondary steel struct	0,21	-0,02	-8,70	0,00	0,00	0,00
	LC14 - Temperature -20	-26,70	0,09	32,60	0,00	0,00	0,00
P116	LC2 - Live Load - 5 kN/m^2	-0,01	1,96	-63,66	0,00	0,00	0,00
	LC3 - Wind Load +X Direction	0,00	0,36	-2,48	0,00	0,00	0,00
	LC4 - Wind Load -X Direction	0,00	-0,36	2,34	0,00	0,00	0,00
	LC5 - Wind Load +Y Direction	0,00	-1,34	38,92	0,00	0,00	0,00
	LC6 - Wind Load -Y Direction	0,00	1,44	-55,43	0,00	0,00	0,00
	LC7 - Tank	0,00	-0,16	-65,73	0,00	0,00	0,00
	LC8 - Temperature +20	-0,03	-0,05	6,19	0,00	0,00	0,01
	LC10 - Live Load -10 kN/m^2	-0,01	2,75	-114,54	0,00	0,00	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	-0,01	2,25	-90,90	0,00	0,00	0,00
	LC13 - Self weight + Secondary steel struct	0,00	1,57	-33,73	0,00	0,00	0,00
	LC14 - Temperature -20	0,03	0,05	-6,19	0,00	0,00	-0,01
P117	LC2 - Live Load - 5 kN/m^2	0,00	-1,51	-13,98	0,00	0,00	0,01
	LC3 - Wind Load +X Direction	0,00	-0,13	28,05	0,00	0,00	0,01
	LC4 - Wind Load -X Direction	0,00	0,25	-19,84	0,00	0,00	0,01
	LC5 - Wind Load +Y Direction	0,00	15,58	-36,27	0,00	0,00	0,00
	LC6 - Wind Load -Y Direction	0,00	-25,76	51,14	0,00	0,00	0,00
	LC7 - Tank	0,00	0,42	-17,30	0,00	0,00	0,02
	LC8 - Temperature +20	0,03	0,66	55,81	0,00	0,00	0,20
	LC10 - Live Load -10 kN/m^2	0,00	-2,01	-28,09	0,00	0,00	0,02
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	0,00	-1,68	-26,45	0,00	0,00	0,01
	LC13 - Self weight + Secondary steel struct	0,00	-1,30	-4,53	0,00	0,00	0,00
	LC14 - Temperature -20	-0,03	-0,66	-55,81	0,00	0,00	-0,20
P118	LC2 - Live Load - 5 kN/m^2	0,03	-6,10	-70,60	0,00	0,00	0,00
	LC3 - Wind Load +X Direction	0,12	-0,10	-22,21	0,00	0,00	0,00
	LC4 - Wind Load -X Direction	-9,37	0,12	22,42	0,00	0,00	0,00
	LC5 - Wind Load +Y Direction	0,02	10,46	19,16	0,00	0,00	0,00
	LC6 - Wind Load -Y Direction	-0,02	-9,21	-23,77	0,00	0,00	0,00
	LC7 - Tank	-0,03	-9,89	-91,09	0,00	0,00	0,00
	LC8 - Temperature +20	0,09	-1,56	19,91	0,00	0,00	0,03
	LC10 - Live Load -10 kN/m^2	0,05	-13,25	-139,38	0,00	0,00	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	0,09	-9,76	-97,69	0,00	0,00	0,00
	LC13 - Self weight + Secondary steel struct	0,08	-1,41	-26,59	0,00	0,00	0,00
	LC14 - Temperature -20	-0,09	1,56	-19,91	0,00	0,00	-0,03
P119	LC2 - Live Load - 5 kN/m^2	-0,64	-0,32	-180,67	-0,54	-0,99	0,00
	LC3 - Wind Load +X Direction	7,21	-0,09	-1,16	0,16	9,78	0,00
	LC4 - Wind Load -X Direction	-10,16	-0,12	-0,26	0,13	-8,03	0,00
	LC5 - Wind Load +Y Direction	-0,25	53,56	63,15	-79,78	-0,30	0,00
	LC6 - Wind Load -Y Direction	0,28	-80,87	-115,81	127,43	0,30	0,00
	LC7 - Tank	-0,28	1,51	-56,99	-2,81	-0,39	0,00
	LC8 - Temperature +20	-1,12	-14,99	10,40	18,82	-4,26	0,00
	LC10 - Live Load -10 kN/m^2	-1,12	-0,68	-349,15	-1,74	-1,69	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	-1,01	-0,17	-326,13	-1,31	-1,53	0,00
	LC13 - Self weight + Secondary steel struct	-0,19	-0,43	-79,28	0,28	-0,30	0,00
	LC14 - Temperature -20	1,12	14,99	-10,40	-18,82	4,26	0,00
P120	LC2 - Live Load - 5 kN/m^2	-0,71	2,24	-92,38	-1,84	-0,89	0,00
	LC3 - Wind Load +X Direction	6,48	-0,10	-1,51	0,16	9,30	0,00
	LC4 - Wind Load -X Direction	-3,73	-0,11	-0,09	0,12	-5,37	0,00
	LC5 - Wind Load +Y Direction	0,39	49,99	-80,52	-77,06	-0,75	0,00
	LC6 - Wind Load -Y Direction	0,88	-85,84	137,15	130,44	1,23	0,00
	LC7 - Tank	-0,29	2,25	-3,00	-3,17	-0,42	0,00
	LC8 - Temperature +20	-4,02	14,96	-0,13	-20,23	-5,89	0,00
	LC10 - Live Load -10 kN/m^2	-1,56	5,67	-206,13	-4,95	-1,97	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	-1,42	3,77	-206,12	-3,32	-1,77	0,00
	LC13 - Self weight + Secondary steel struct	-0,29	0,66	-53,70	-0,28	-0,36	0,00
	LC14 - Temperature -20	4,02	-14,96	0,13	20,23	5,89	0,00
P121	LC2 - Live Load - 5 kN/m^2	-1,46	-0,30	-95,27	0,00	0,00	-0,46
	LC3 - Wind Load +X Direction	24,41	-0,05	24,22	0,00	0,00	-0,02
	LC4 - Wind Load -X Direction	-14,97	0,02	-24,36	0,00	0,00	0,01
	LC5 - Wind Load +Y Direction	-1,14	6,02	12,27	0,00	0,00	0,30
	LC6 - Wind Load -Y Direction	1,70	-0,45	-12,04	0,00	0,00	-0,75
	LC7 - Tank	-1,68	-0,45	-63,78	0,00	0,00	-0,72
	LC8 - Temperature +20	-13,41	-0,32	-21,58	0,00	0,00	-0,16
	LC10 - Live Load -10 kN/m^2	-2,55	-0,62	-171,32	0,00	0,00	-0,97
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	-1,83	-0,47	-150,76	0,00	0,00	-0,71
	LC13 - Self weight + Secondary steel struct	-0,43	-0,07	-46,24	0,00	0,00	-0,11
	LC14 - Temperature -20	13,41	0,32	21,58	0,00	0,00	0,16
P122	LC2 - Live Load - 5 kN/m^2	-0,12	0,38	-138,40	-1,30	-0,32	0,00
	LC3 - Wind Load +X Direction	13,91	0,23	0,03	-0,27	12,70	0,00
	LC4 - Wind Load -X Direction	-4,09	0,08	1,32	-0,11	-5,56	0,00
	LC5 - Wind Load +Y Direction	0,61	51,08	60,23	-75,24	0,67	0,00
	LC6 - Wind Load -Y Direction	-0,95	-78,80	-103,78	119,46	-1,13	0,00
	LC7 - Tank	-0,08	1,65	-49,65	-2,25	-0,27	0,00
	LC8 - Temperature +20	-9,78	-12,49	7,55	16,05	-17,77	0,00
	LC10 - Live Load -10 kN/m^2	-0,17	0,84	-250,28	-2,75	-0,50	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	-0,15	1,46	-232,23	-2,59	-0,38	0,00
	LC13 - Self weight + Secondary steel struct	-0,05	0,07	-62,67	-0,20	-0,09	0,00
	LC14 - Temperature -20	9,78	12,49	-7,55	-16,05	17,77	0,00
P123	LC2 - Live Load - 5 kN/m^2	0,46	3,34	-78,82	-2,86	0,13	0,00
	LC3 - Wind Load +X Direction	6,67	0,24	1,13	-0,28	9,52	0,00
	LC4 - Wind Load -X Direction	-3,70	0,08	0,05	-0,11	-5,28	0,00
	LC5 - Wind Load +Y Direction	-1,20	45,03	-72,53	-71,94	-1,67	0,00
	LC6 - Wind Load -Y Direction	1,97	-74,14	116,96	118,56	2,73	0,00
	LC7 - Tank	-0,35	1,38	-2,93	-2,14	-0,50	0,00
	LC8 - Temperature +20	-14,07	16,83	3,82	-22,70	-20,00	0,00
	LC10 - Live Load -10 kN/m^2	0,86	6,19	-183,87	-5,56	0,21	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	1,08	4,84	-181,26	-4,40	0,47	0,00
	LC13 - Self weight + Secondary steel struct	0,25	1,15	-46,45	-0,77	0,13	0,00
	LC14 - Temperature -20	14,07	-16,83	-3,82	22,70	20,00	0,00
P124	LC2 - Live Load - 5 kN/m^2	0,19	0,91	-162,64	-0,88	-0,03	0,00
	LC3 - Wind Load +X Direction	6,71	-0,06	-1,65	0,09	9,95	0,00
	LC4 - Wind Load -X Direction	-3,40	0,06	-0,38	-0,10	-5,04	0,00
	LC5 - Wind Load +Y Direction	1,81	29,49	-8,51	-54,94	2,54	0,01
	LC6 - Wind Load -Y Direction	-3,09	-31,36	6,71	82,79	-4,32	-0,01
	LC7 - Tank	-0,33	0,41	-2,56	-1,05	-0,45	0,00
	LC8 - Temperature +20	-35,09	-7,66	1,76	10,02	-52,10	0,00
	LC10 - Live Load -10 kN/m^2	0,63	1,55	-305,67	-1,90	0,21	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	0,77	1,42	-307,58	-1,50	0,40	0,00
	LC13 - Self weight + Secondary steel struct	0,02	0,51	-74,90	-0,36	-0,04	0,00
	LC14 - Temperature -20	35,09	7,66	-1,76	-10,02	52,10	0,00
P125	LC2 - Live Load - 5 kN/m^2	-0,33	0,89	-124,47	-0,67	-0,42	0,00
	LC3 - Wind Load +X Direction	7,05	-0,05	-0,85	0,07	10,15	0,00
	LC4 - Wind Load -X Direction	-3,58	0,06	-0,46	-0,09	-5,15	0,00
	LC5 - Wind Load +Y Direction	-2,29	16,83	8,76	-46,47	-3,61	0,01
	LC6 - Wind Load -Y Direction	3,72	-36,12	-8,42	85,23	5,90	-0,01
	LC7 - Tank	-0,38	0,35	-0,02	-1,00	-0,56	0,00
	LC8 - Temperature +20	-35,47	14,73	2,37	-23,46	-50,93	0,00
	LC10 - Live Load -10 kN/m^2	-0,50	1,47	-284,17	-1,56	-0,66	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	-0,38	1,40	-284,36	-1,16	-0,48	0,00
	LC13 - Self weight + Secondary steel struct	-0,08	0,50	-66,91	-0,24	-0,09	0,00
	LC14 - Temperature -20	35,47	-14,73	-2,37	23,46	50,93	0,00
P126	LC2 - Live Load - 5 kN/m^2	-1,98	-0,14	-77,10	0,00	0,00	0,00
	LC3 - Wind Load +X Direction	2,48	0,00	4,23	0,00	0,00	0,00
	LC4 - Wind Load -X Direction	-1,02	0,00	-1,07	0,00	0,00	0,00
	LC5 - Wind Load +Y Direction	0,38	3,69	164,56	0,00	0,00	0,02
	LC6 - Wind Load -Y Direction	-0,62	0,70	-308,14	0,00	0,00	-0,02
	LC7 - Tank	-0,08	-0,01	1,92	0,00	0,00	0,00
	LC8 - Temperature +20	-19,54	2,29	-16,27	0,00	0,00	0,01
	LC10 - Live Load -10 kN/m^2	-4,07	-0,34	-148,87	0,00	0,00	0,00
	LC12 - Live Load -15 kN/m^2 and -10 kN/m	-4,11	-0,34	-150,05	0,00	0,00	0,00
	LC13 - Self weight + Secondary steel struct	-0,87	-0,08	-41,96	0,00	0,00	0,00
	LC14 - Temperature -20	19,54	-2,29	16,27	0,00	0,00	-0,01
P127	LC2 - Live Load - 5 kN/m^2	-0,89	-1,09	-58,83	0,00	0,00	-0,16
	LC3 - Wind Load +X Direction	2,17	0,17	3,06	0,00	0,00	0,17
	LC4 - Wind Load -X Direction	-0,79	0,03	-0,90	0,00	0,00	-0,08
	LC5 - Wind Load +Y Direction	-0,34	90,96	-165,92	0,00	0,00	0,01
	LC6 - Wind Load -Y Direction	0,60	-168,75	307,60	0,00	0,00	0,09
	LC7 - Tank	-0,08	0,65	-1,76	0,00	0,00	-0,01
	LC8 - Temperature +20	-14,56	-7,82	-5,72	0,00	0,00	-1,57
	LC10 - Live Load -10 kN/m^2	-1,96	-1,98	-137,85	0,00	0,00	-0,35
	LC12 - Live Load -15 k

DRAWING NUMBER: MD-216-3510-EG-CV-GAD-1001
REV: A02

LEVEL +912.500
1:50



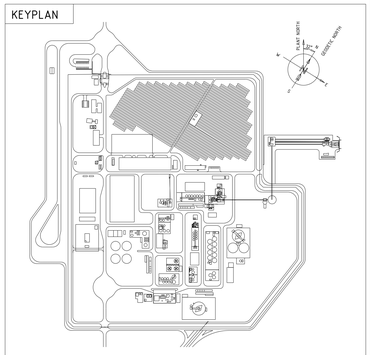
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L)
FOR PROCESS AREA 3510 = RL(+)912.500 m

CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3510-EG-ST-GAD-1001	PLAN AT EL. +912.500
MD-216-3510-EG-ME-GAD-1019	SECTION B-B
MD-216-3510-EG-ME-GAD-1017	PLAN VIEW
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3510-EG-ST-GAD-1002	3D VIEW AT EL. +912.500
MD-216-3510-EG-CV-GAD-1002	PLAN AT EL. +912.500 - GRID D-D/1-5
MD-216-3510-EG-ME-GAD-1018	SECTION A-A
MD-216-3510-EG-ME-GAD-1016	PLAN AT EL. +912.500

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	30.03.2020	ISSUED FOR DETAIL DESIGN, RAMP ADDED, STAIRS, EQUIP. FOUND. , CHANNELS UPDATED	BabHam	MakLai	LuiPiz	SouCha	
A01	11.12.2019	ISSUED FOR APPROVAL	BabHam	MakLai	LuiPiz	SouCha	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER	019045_450_C_G_24938_001	REVISION	1
OUTOTEC DRAWING NUMBER	900707_GAU01_ST61_CTA04_35101	DOCUMENT ID	OU100667795
		REVISION	1

DRAWN	BabHam	11.12.2019
CHECKED	MakLai	11.12.2019
ENGINEER	LuiPiz	11.12.2019
APPROVED	SouCha	11.12.2019
EMPLOYER		
SCALE	1:50	A0

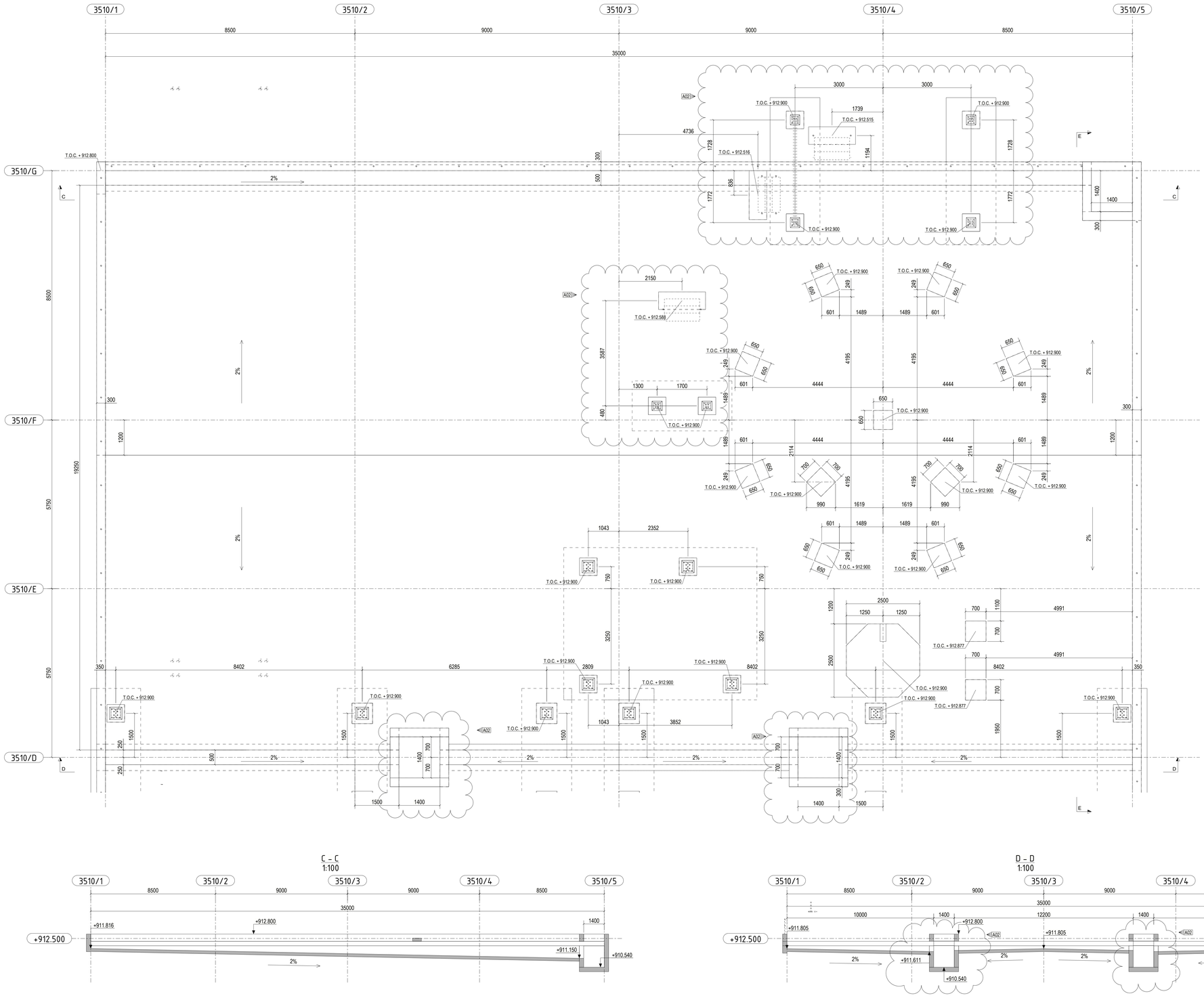
MANSOURAH - MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
CCD CIRCUIT
PLAN AT EL. +912.500 - GRID A-D/1-5

DRAWING NUMBER	REV
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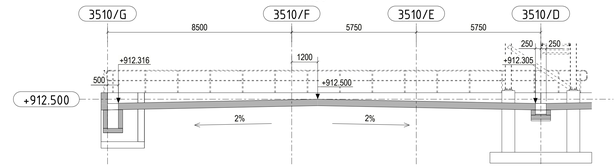
Tekla Structures

DRAWING NUMBER: MD-216-3510-EG-CV-GAD-1002
REV: A02

LEVEL +912.500
1:50

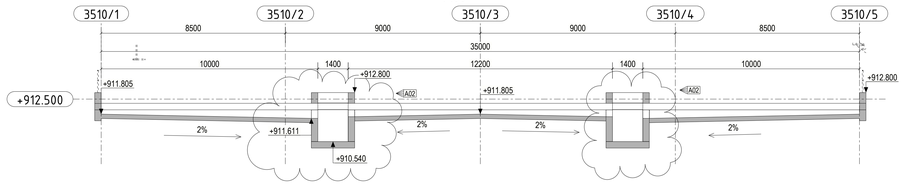


E – E
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C – C
1:100

D – D
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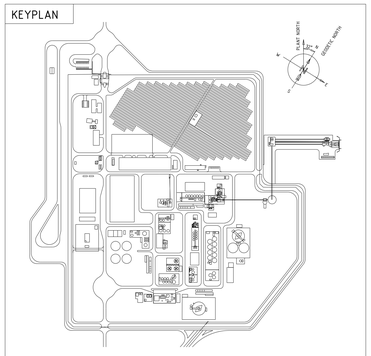
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L)
FOR PROCESS AREA 3510 = RL(+)912.500 m

- CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
- CONCRETE FLOOR
- CONCRETE FLOOR WITH ACID RESISTANT COATING
- CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
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4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

KEYPLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3510-EG-ST-GAD-1001	PLAN AT EL. +912.500	MD-216-3510-EG-ST-GAD-1002	3D VIEW AT EL. +912.500
MD-216-3510-EG-ME-GAD-1019	SECTION B-B	MD-216-3510-EG-CV-GAD-1001	PLAN AT EL. +912.500 – GRID A-D/1-5
MD-216-3510-EG-ME-GAD-1017	PLAN VIEW	MD-216-3510-EG-ME-GAD-1018	SECTION A-A
MD-216-1000-EG-ST-CRT-0001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN BASIS	MD-216-3510-EG-ME-GAD-1016	PLAN AT EL. +912.500

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A02	30.03.2020	ISSUED FOR DETAIL DESIGN, STAIRCASES MOVED, CHANNELS UPDATED	NikJan	MikLai	LuiPiz	SouCha	
A01	11.12.2019	ISSUED FOR APPROVAL	NikJan	MikLai	LuiPiz	SouCha	

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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24939_001 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707_GAU01_ST61_CTA04_35102 — DOCUMENT ID: OU100667796 — REVISION: 1

DRAWN	NikJan	11.12.1970
CHECKED	MikLai	11.12.2019
ENGINEER	LuiPiz	11.12.2019
APPROVED	SouCha	11.12.2019
EMPLOYER		
SCALE	1:50	A0

MANSOURAH – MASSARAH GOLD PROJECT
CIVIL WORKS GUIDE DRAWING
CCD CIRCUIT
PLAN AT EL. +912.500 – GRID D-G/1-5

DRAWING NUMBER: MD-216-3510-EG-CV-GAD-1002 — REV: A02

Tekla Structures

ATTACHMENT 2 STAAD OUTPUT: MD-216-3510-EG-CV-CAL-1281_PIPE RACK FDN IN CCD AREA

PROPRIETARY & CONFIDENTIAL

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*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    MAY 20, 2020                 *
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     4. END JOB INFORMATION
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1829. 2748 3180 1745 1743 3178; 2749 3181 1746 1744 3179; 2750 3182 1765 1745 3180
1830. 2751 3183 1766 1746 3181; 2752 3184 1783 1765 3182; 2753 3185 1784 1766 3183

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1831. 2754 3178 1743 1819 3186; 2755 3179 1744 1820 3187; 2756 3186 1819 1837 3188
1832. 2757 3187 1820 1838 3189; 2758 3188 1837 1855 3190; 2759 3189 1838 1856 3191
1833. 2760 1729 3194 3192 1727; 2761 1730 3195 3193 1728; 2762 1757 3196 3194 1729
1834. 2763 1758 3197 3195 1730; 2764 1775 3198 3196 1757; 2765 1776 3199 3197 1758
1835. 2766 1727 3192 3200 1811; 2767 1728 3193 3201 1812; 2768 1811 3200 3202 1829
1836. 2769 1812 3201 3203 1830; 2770 1829 3202 3204 1847; 2771 1830 3203 3205 1848
1837. 2773 3207 3208 3209 3210; 2774 3206 3211 3208 3207; 2775 3212 3213 3211 3206
1838. 2776 3214 3215 3213 3212; 2777 3216 3217 3215 3214; 2778 3208 3218 3219 3209
1839. 2779 3211 3220 3218 3208; 2780 3213 3221 3220 3211; 2781 3215 3222 3221 3213
1840. 2782 3217 3223 3222 3215; 2783 3218 3224 3225 3219; 2784 3220 3226 3224 3218
1841. 2785 3221 3227 3226 3220; 2786 3222 3228 3227 3221; 2787 3223 3229 3228 3222
1842. 2788 3230 3207 3210 3231; 2789 3232 3206 3207 3230; 2790 3233 3212 3206 3232
1843. 2791 3234 3214 3212 3233; 2792 3235 3216 3214 3234; 2793 3236 3230 3231 3237
1844. 2794 3238 3232 3230 3236; 2795 3239 3233 3232 3238; 2796 3240 3234 3233 3239
1845. 2797 3241 3235 3234 3240; 2798 3242 3236 3237 3243; 2799 3244 3238 3236 3242
1846. 2800 3245 3239 3238 3244; 2801 3246 3240 3239 3245; 2802 3247 3241 3240 3246
1847. 2803 3248 3249 314 308; 2804 3249 3250 464 314; 2805 3250 3251 590 464
1848. 2806 3252 3248 308 842; 2807 3253 3252 842 968; 2808 3254 3253 968 1094
1849. 2809 3210 3209 3249 3248; 2810 3209 3219 3250 3249; 2811 3219 3225 3251 3250
1850. 2812 3231 3210 3248 3252; 2813 3237 3231 3252 3253; 2814 3243 3237 3253 3254
1851. 2815 500 3256 3255 494; 2816 506 3257 3256 500; 2817 512 3258 3257 506
1852. 2818 518 3259 3258 512; 2819 524 3260 3259 518; 2820 530 3261 3260 524
1853. 2821 536 3262 3261 530; 2822 542 3263 3262 536; 2823 548 3264 3263 542
1854. 2824 554 3265 3264 548; 2825 560 3266 3265 554; 2826 566 3267 3266 560
1855. 2827 572 3268 3267 566; 2828 578 3269 3268 572; 2829 584 3270 3269 578
1856. 2830 590 3271 3270 584; 2831 494 3255 3272 596; 2832 596 3272 3273 602
1857. 2833 602 3273 3274 608; 2834 608 3274 3275 614; 2835 3224 3276 3277 3225
1858. 2836 3226 3278 3276 3224; 2837 3227 3279 3278 3226; 2838 3228 3280 3279 3227
1859. 2839 3229 3281 3280 3228; 2840 3251 3282 3271 590; 2841 3225 3277 3282 3251
1860. 2842 3285 1010 1002 3283; 2843 3286 1016 1010 3285; 2844 3287 1022 1016 3286
1861. 2845 3288 1028 1022 3287; 2846 3289 1034 1028 3288; 2847 3290 1040 1034 3289
1862. 2848 3291 1046 1040 3290; 2849 3292 1052 1046 3291; 2850 3293 1058 1052 3292
1863. 2851 3294 1064 1058 3293; 2852 3295 1070 1064 3294; 2853 3284 1003 1070 3295
1864. 2854 3296 1076 1003 3284; 2855 3297 1082 1076 3296; 2856 3298 1088 1082 3297
1865. 2857 3299 1094 1088 3298; 2858 3283 1002 1100 3300; 2859 3300 1100 1106 3301
1866. 2860 3301 1106 1112 3302; 2861 3302 1112 1118 3303; 2862 3304 3242 3243 3305
1867. 2863 3306 3244 3242 3304; 2864 3307 3245 3244 3306; 2865 3308 3246 3245 3307
1868. 2866 3309 3247 3246 3308; 2867 3310 3254 1094 3299; 2868 3305 3243 3254 3310
1869. 2869 3311 3312 3217 3216; 2870 3312 3313 3223 3217; 2871 3313 3314 3229 3223
1870. 2872 3315 3311 3216 3235; 2873 3316 3315 3235 3241; 2874 3317 3316 3241 3247
1871. 2875 3314 3318 3281 3229; 2876 3319 3317 3247 3309
1872. START GROUP DEFINITION
1873. ELEMENT
1874. _F1_MOMENT 34 40 46 52 58 64 70 76 82 88 106 112 118 130 136 142 148 154 160 -
1875. 166 172 178 184 190 196 202 208 214 220 226 232 238 244 250 256 262 268 274 -
1876. 280 286 292 298 304 310 316 322 328 334 340 346 352 358 364 370 376 382 388 -
1877. 394 400 406 412 418 424 430 436 442 448 454 460 466 472 478 484 490 496 502 -
1878. 514 520 526 532 538 544 550 556 562 568 586 592 598 610 616 622 628 634 640 -
1879. 646 652 658 664 670 676 682 688 694 700 706 712 718 724 730 736 742 748 754 -
1880. 760 766 772 778 784 790 796 802 808 814 820 826 832 838 844 850 856 862 868 -
1881. 874 880 886 892 898 904 910 916 922 928 934 940 946 952 958 964 970 976 982 -
1882. 1684 1686 1688 1690 1692 1694 1696 1698 1700 1702 1704 1706 1708 1710 1712 -
1883. 1714 1716 1718 1720 1722 1724 TO 1743 2256 TO 2295 2520 TO 2591 2616 TO 2639
1884. _F1_SHEAR 46 52 58 64 70 76 118 142 166 172 178 184 190 196 238 262 286 292 -
1885. 298 304 310 316 358 382 388 394 400 406 412 418 424 430 436 442 448 454 460 -
1886. 466 472 478 484 490 496 502 526 532 538 544 550 556 598 622 646 652 658 664 -
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1887. 670 676 718 742 766 772 778 784 790 796 838 862 868 874 880 886 892 898 904 -
1888. 910 916 922 928 934 940 946 952 958 964 970 976 982 1684 1686 1688 1690 1692 -
1889. 1694 1696 1698 1700 1702 1704 1706 1708 1710 1712 1714 1716 1718 1720 1722 -
1890. 1724 TO 1743 2256 TO 2295 2520 TO 2591 2616 TO 2639
1891. _F2_MOMENT 35 41 47 53 59 65 71 77 83 89 107 113 119 131 137 143 149 155 161 -
1892. 167 173 179 185 191 197 203 209 215 221 227 233 239 245 251 257 263 269 275 -
1893. 281 287 293 299 305 311 317 323 329 335 341 347 353 359 365 371 377 383 389 -
1894. 395 401 407 413 419 425 431 437 443 449 455 461 467 473 479 485 491 497 503 -
1895. 515 521 527 533 539 545 551 557 563 569 587 593 599 611 617 623 629 635 641 -
1896. 647 653 659 665 671 677 683 689 695 701 707 713 719 725 731 737 743 749 755 -
1897. 761 767 773 779 785 791 797 803 809 815 821 827 833 839 845 851 857 863 869 -
1898. 875 881 887 893 899 905 911 917 923 929 935 941 947 953 959 965 971 977 983
1899. _F2_SHEAR 47 53 59 65 71 77 119 143 167 173 179 185 191 197 239 263 287 293 -
1900. 299 305 311 317 359 383 389 395 401 407 413 419 425 431 437 443 449 455 461 -
1901. 467 473 479 485 491 497 503 527 533 539 545 551 557 599 623 647 653 659 665 -
1902. 671 677 719 743 767 773 779 785 791 797 839 863 869 875 881 887 893 899 905 -
1903. 911 917 923 929 935 941 947 953 959 965 971 977 983
1904. _F3_MOMENT 36 37 42 43 48 49 54 55 60 61 66 67 72 73 78 79 84 85 90 91 108 -
1905. 109 114 115 120 121 132 133 138 139 144 145 150 151 156 157 162 163 168 169 -
1906. 174 175 180 181 186 187 192 193 198 199 204 205 210 211 216 217 222 223 228 -
1907. 229 234 235 240 241 246 247 252 253 258 259 264 265 270 271 276 277 282 283 -
1908. 288 289 294 295 300 301 306 307 312 313 318 319 324 325 330 331 336 337 342 -
1909. 343 348 349 354 355 360 361 366 367 372 373 378 379 384 385 390 391 396 397 -
1910. 402 403 408 409 414 415 420 421 426 427 432 433 438 439 444 445 450 451 456 -
1911. 457 462 463 468 469 474 475 480 481 486 487 492 493 498 499 504 505 516 517 -
1912. 522 523 528 529 534 535 540 541 546 547 552 553 558 559 564 565 570 571 588 -
1913. 589 594 595 600 601 612 613 618 619 624 625 630 631 636 637 642 643 648 649 -
1914. 654 655 660 661 666 667 672 673 678 679 684 685 690 691 696 697 702 703 708 -
1915. 709 714 715 720 721 726 727 732 733 738 739 744 745 750 751 756 757 762 763 -
1916. 768 769 774 775 780 781 786 787 792 793 798 799 804 805 810 811 816 817 822 -
1917. 823 828 829 834 835 840 841 846 847 852 853 858 859 864 865 870 871 876 877 -
1918. 882 883 888 889 894 895 900 901 906 907 912 913
1919. _F3_MOMENT 918 919 924 925 930 931 936 937 942 943 948 949 954 955 960 961 -
1920. 966 967 972 973 978 979 984 985 1606 1607 1932 TO 2045 2168 TO 2207
1921. _F3_SHEAR 48 49 54 55 60 61 66 67 72 73 78 79 120 121 144 145 168 169 174 -
1922. 175 180 181 186 187 192 193 198 199 240 241 264 265 288 289 294 295 300 301 -
1923. 306 307 312 313 318 319 360 361 384 385 390 391 396 397 402 403 408 409 414 -
1924. 415 420 421 426 427 432 433 438 439 444 445 450 451 456 457 462 463 468 469 -
1925. 474 475 480 481 486 487 493 499 505 528 529 534 535 540 541 546 547 552 553 -
1926. 558 559 600 601 624 625 648 649 654 655 660 661 666 667 672 673 678 679 720 -
1927. 721 744 745 768 769 774 775 780 781 786 787 792 793 798 799 840 841 864 865 -
1928. 870 871 876 877 882 883 888 889 894 895 900 901 906 907 912 913 918 919 924 -
1929. 925 930 931 936 937 942 943 948 949 954 955 960 961 966 967 972 978 984 1932 -
1930. 1933 TO 1937 1939 TO 1946 1948 TO 1963 1966 TO 2001 2010 TO 2045 2170 TO 2175 -
1931. 2177 TO 2184 2188 TO 2207
1932. _F4_MOMENT 989 TO 992 995 999 1002 TO 1009 1012 1013 1016 TO 1023 1026 1027 -
1933. 1045 TO 1048 1051 1055 1058 TO 1065 1068 1069 1072 TO 1079 1082 1083
1934. _F4_SHEAR 990 991 1004 1005 1016 TO 1023 1026 1027 1046 1047 1060 1061 1072 -
1935. 1073 TO 1079 1082 1083
1936. _F5_MOMENT 1102 TO 1107 1110 TO 1163 1166 TO 1171 1174 TO 1227 1812 TO 1827 -
1937. 1834 TO 1849 1856 TO 1871 1878 TO 1893 2670 TO 2681 2686 TO 2697 -
1938. 2702 TO 2713 2718 TO 2729 2736 TO 2747
1939. _F5_SHEAR 1106 1107 1114 1115 1122 1123 1130 1131 1138 1139 1146 TO 1163 1170 -
1940. 1171 1178 1179 1186 1187 1194 1195 1202 1203 1210 TO 1227 1812 TO 1827 1834 -
1941. 1835 TO 1849 1856 TO 1871 1878 TO 1893 2670 TO 2681 2686 TO 2697 2702 TO 2713 -
1942. 2718 TO 2729 2736 TO 2747
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1943. _F6_MOMENT 1230 1232 1234 1240 1242 1244 1246 1248 1250 1252 1256 1258 1260 -
1944. 1262 1264 1266 1268 1270 1274 1276 1278 1280 1282 1284 1286 1288 1292 1294 -
1945. 1296 1298 1302 1304 1306 1312 1314 1316 1318 1320 1322 1324 1328 1330 1332 -
1946. 1334 1336 1338 1340 1342 1346 1348 1350 1352 1354 1356 1358 1360 1364 1366 -
1947. 1368 1370
1948. _F6_SHEAR 1234 1244 1252 1262 1270 1280 1282 1284 1286 1288 1292 1294 1296 -
1949. 1298 1306 1316 1324 1334 1342 1352 1354 1356 1358 1360 1364 1366 1368 1370
1950. _F7_MOMENT 1231 1233 1235 1241 1243 1245 1247 1249 1251 1253 1257 1259 1261 -
1951. 1263 1265 1267 1269 1271 1275 1277 1279 1281 1283 1285 1287 1289 1293 1295 -
1952. 1297 1299 1303 1305 1307 1313 1315 1317 1319 1321 1323 1325 1329 1331 1333 -
1953. 1335 1337 1339 1341 1343 1347 1349 1351 1353 1355 1357 1359 1361 1365 1367 -
1954. 1369 1371 1501 TO 1503 1505 TO 1507 1509 TO 1511 1513 TO 1563 1620 TO 1643
1955. _F7_SHEAR 1235 1245 1253 1263 1271 1281 1283 1285 1287 1289 1293 1295 1297 -
1956. 1299 1307 1317 1325 1335 1343 1353 1355 1357 1359 1361 1365 1367 1369 1371 -
1957. 1503 1507 1511 1515 1519 1523 1527 1531 TO 1539 1543 1547 1551 1555 TO 1563 -
1958. 1620 TO 1643
1959. _F8_MOMENT 1374 TO 1377 1382 TO 1385 1388 TO 1393 1396 TO 1401 1404 TO 1409 -
1960. 1412 TO 1417 1438 TO 1441 1446 TO 1449 1452 TO 1457 1460 TO 1465 -
1961. 1468 TO 1473 1476 TO 1481 2748 TO 2771
1962. _F8_SHEAR 1376 1377 1384 1385 1392 1393 1400 1401 1404 TO 1409 1412 TO 1417 -
1963. 1440 1441 1448 1449 1456 1457 1464 1465 1468 TO 1473 1476 TO 1481 -
1964. 2748 TO 2771
1965. _F9_MOMENT 38 44 50 56 62 68 74 80 86 92 110 116 122 134 140 146 152 158 164 -
1966. 170 176 182 188 194 200 206 212 218 224 230 236 242 248 254 260 266 272 278 -
1967. 284 290 296 302 308 314 320 326 332 338 344 350 356 362 368 374 380 386 518 -
1968. 524 530 536 542 548 554 560 566 572 590 596 602 614 620 626 632 638 644 650 -
1969. 656 662 668 674 680 686 692 698 704 710 716 722 728 734 740 746 752 758 764 -
1970. 770 776 782 788 794 800 806 812 818 824 830 836 842 848 854 860 866 2773 -
1971. 2776 TO 2788 2791 TO 2876
1972. _F9_SHEAR 50 56 62 68 74 80 122 146 170 176 182 188 194 200 242 266 290 296 -
1973. 302 308 314 320 362 380 386 530 536 542 548 554 560 602 626 650 656 662 668 -
1974. 674 680 722 746 770 776 782 788 794 800 842 866 2803 TO 2808 2818 TO 2823 -
1975. 2830 2833 2834 2840 2845 TO 2850 2857 2861 2867 2869 TO 2876
1976. _F10_MOMENT 39 45 51 57 63 69 75 81 87 93 111 117 123 135 141 147 153 159 -
1977. 165 171 177 183 189 195 201 207 213 219 225 231 237 243 249 255 261 267 273 -
1978. 279 285 291 297 303 309 315 321 327 333 339 345 351 357 363 369 375 381 387 -
1979. 393 399 405 411 417 423 429 435 441 447 453 459 465 471 477 483 489 495 501 -
1980. 507 519 525 531 537 543 549 555 561 567 573 591 597 603 615 621 627 633 639 -
1981. 645 651 657 663 669 675 681 687 693 699 705 711 717 723 729 735 741 747 753 -
1982. 759 765 771 777 783 789 795 801 807 813 819 825 831 837 843 849 855 861 867 -
1983. 873 879 885 891 897 903 909 915 921 927 933 939 945 951 957 963 969 975 981 -
1984. 987 1645 1647 1649 1651 1653 1655 1657 1659 1661 1663 1665 1667 1669 1671 -
1985. 1673 1675 1677 1679 1681 1683 1685 1687 1689 1691 1693 1695 1697 1699 1701 -
1986. 1703 1705 1707 1709 1711 1713 1715 1717 1719 1721 1723 2128 TO 2167 2208 -
1987. 2209 TO 2255 2296 TO 2519
1988. _F10_SHEAR 51 57 63 69 75 81 123 147 171 177 183 189 195 201 243 267 291 297 -
1989. 303 309 315 321 363 387 393 399 405 411 417 423 429 435 441 447 453 459 465 -
1990. 471 477 483 489 495 501 507 531 537 543 549 555 561 603 627 651 657 663 669 -
1991. 675 681 723 747 771 777 783 789 795 801 843 867 873 879 885 891 897 903 909 -
1992. 915 921 927 933 939 945 951 957 963 969 975 981 987 1645 1647 1649 1651 1653 -
1993. 1655 1657 1659 1661 1663 1665 1667 1669 1671 1673 1675 1677 1679 1681 1683 -
1994. 1685 1687 1689 1691 1693 1695 1697 1699 1701 1703 1705 1707 1709 1711 1713 -
1995. 1715 1717 1719 1721 1723 2128 TO 2167 2208 TO 2255 2296 TO 2519
1996. MEMBER
1997. _C1 1 TO 7 9 TO 19 22 TO 25 2772
1998. END GROUP DEFINITION
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1999. ELEMENT PROPERTY
2000. 28 TO 32 34 TO 38 40 TO 44 46 TO 50 52 TO 56 58 TO 62 64 TO 68 70 TO 74 76 -
2001. 77 TO 80 82 TO 86 88 TO 92 94 TO 98 100 TO 104 106 TO 110 112 TO 116 -
2002. 118 TO 122 124 TO 128 130 TO 134 136 TO 140 142 TO 146 148 TO 152 -
2003. 154 TO 158 160 TO 164 166 TO 170 172 TO 176 178 TO 182 184 TO 188 -
2004. 190 TO 194 196 TO 200 202 TO 206 208 TO 212 214 TO 218 220 TO 224 -
2005. 226 TO 230 232 TO 236 238 TO 242 244 TO 248 250 TO 254 256 TO 260 -
2006. 262 TO 266 268 TO 272 274 TO 278 280 TO 284 286 TO 290 292 TO 296 -
2007. 298 TO 302 304 TO 308 310 TO 314 316 TO 320 322 TO 326 328 TO 332 -
2008. 334 TO 338 340 TO 344 346 TO 350 352 TO 356 358 TO 362 364 TO 368 -
2009. 370 TO 374 376 TO 380 382 TO 386 388 TO 391 394 TO 397 400 TO 403 -
2010. 406 TO 409 412 TO 415 418 TO 421 424 TO 427 430 TO 433 436 TO 439 -
2011. 442 TO 445 448 TO 451 454 TO 457 460 TO 463 466 TO 469 472 TO 475 -
2012. 478 TO 481 484 TO 487 490 TO 493 496 TO 499 502 TO 505 508 TO 512 -
2013. 514 TO 518 520 TO 524 526 TO 530 532 TO 536 538 TO 542 544 TO 548 -
2014. 550 TO 554 556 TO 560 562 TO 566 568 TO 572 574 TO 578 580 TO 584 -
2015. 586  THICKNESS 0.5
2016. 2815 TO 2834 THICKNESS 0.5
2017. 2773 TO 2790 2803 TO 2805 2809 TO 2811 2835 TO 2841 2869 TO 2871 -
2018. 2875 THICKNESS 0.5
2019. 587 TO 590 592 TO 596 598 TO 602 604 TO 608 610 TO 614 616 TO 620 622 TO 626 -
2020. 628 TO 632 634 TO 638 640 TO 644 646 TO 650 652 TO 656 658 TO 662 -
2021. 664 TO 668 670 TO 674 676 TO 680 682 TO 686 688 TO 692 694 TO 698 -
2022. 700 TO 704 706 TO 710 712 TO 716 718 TO 722 724 TO 728 730 TO 734 -
2023. 736 TO 740 742 TO 746 748 TO 752 754 TO 758 760 TO 764 766 TO 770 -
2024. 772 TO 776 778 TO 782 784 TO 788 790 TO 794 796 TO 800 802 TO 806 -
2025. 808 TO 812 814 TO 818 820 TO 824 826 TO 830 832 TO 836 838 TO 842 -
2026. 844 TO 848 850 TO 854 856 TO 860 862 TO 866 868 TO 871 874 TO 877 -
2027. 880 TO 883 886 TO 889 892 TO 895 898 TO 901 904 TO 907 910 TO 913 -
2028. 916 TO 919 922 TO 925 928 TO 931 934 TO 937 940 TO 943 946 TO 949 -
2029. 952 TO 955 958 TO 961 964 TO 967 970 TO 973 976 TO 979 982 TO 985 -
2030. 1100 TO 1227 1229 1231 1233 1235 1239 1241 1243 1245 1247 1249 1251 1253 -
2031. 1257 1259 1261 1263 1265 1267 1269 1271 1275 1277 1279 1281 1283 1285 1287 -
2032. 1289 1293 1295 1297 1299 1301 1303 1305 1307 1311 1313 1315 1317 1319 1321 -
2033. 1323 1325 1329 1331 1333 1335 1337 1339 1341 1343 1347 1349 1351 1353 1355 -
2034. 1357 1359 1361 1365 1367 1369 1371 1500 TO 1563 1604 TO 1609 1620 TO 1643 -
2035. 1684 1686 1688 THICKNESS 0.5
2036. 2842 TO 2861 THICKNESS 0.5
2037. 2791 TO 2802 2806 TO 2808 2812 TO 2814 2862 TO 2868 2872 TO 2874 -
2038. 2876 THICKNESS 0.5
2039. 1690 1692 1694 1696 1698 1700 1702 1704 1706 1708 1710 1712 1714 1716 1718 -
2040. 1720 1722 1724 TO 1743 1812 TO 1827 1834 TO 1849 1856 TO 1871 1878 TO 1893 -
2041. 1932 TO 2045 2168 TO 2207 2256 TO 2295 2520 TO 2591 2616 TO 2639 -
2042. 2670 TO 2681 2686 TO 2697 2702 TO 2713 2718 TO 2729 2736 TO 2746 -
2043. 2747 THICKNESS 0.5
2044. 33 39 45 51 57 63 69 75 81 87 93 99 105 111 117 123 129 135 141 147 153 159 -
2045. 165 171 177 183 189 195 201 207 213 219 225 231 237 243 249 255 261 267 273 -
2046. 279 285 291 297 303 309 315 321 327 333 339 345 351 357 363 369 375 381 387 -
2047. 393 399 405 411 417 423 429 435 441 447 453 459 465 471 477 483 489 495 501 -
2048. 507 513 519 525 531 537 543 549 555 561 567 573 579 585 591 597 603 609 615 -
2049. 621 627 633 639 645 651 657 663 669 675 681 687 693 699 705 711 717 723 729 -
2050. 735 741 747 753 759 765 771 777 783 789 795 801 807 813 819 825 831 837 843 -
2051. 849 855 861 867 873 879 885 891 897 903 909 915 921 927 933 939 945 951 957 -
2052. 963 969 975 981 987 1645 1647 1649 1651 1653 1655 1657 1659 1661 1663 1665 -
2053. 1667 1669 1671 1673 1675 1677 1679 1681 1683 1685 1687 1689 1691 1693 1695 -
2054. 1697 1699 1701 1703 1705 1707 1709 1711 1713 1715 1717 1719 1721 1723 2128 -
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2055. 2129 TO 2167 2208 TO 2255 2296 TO 2519 THICKNESS 0.6
2056. 988 TO 995 998 999 1002 TO 1009 1012 1013 1016 TO 1023 1026 1027 1044 TO 1051 -
2057. 1054 1055 1058 TO 1065 1068 1069 1072 TO 1079 1082 1083 1228 1230 1232 1234 -
2058. 1238 1240 1242 1244 1246 1248 1250 1252 1256 1258 1260 1262 1264 1266 1268 -
2059. 1270 1274 1276 1278 1280 1282 1284 1286 1288 1292 1294 1296 1298 1300 1302 -
2060. 1304 1306 1310 1312 1314 1316 1318 1320 1322 1324 1328 1330 1332 1334 1336 -
2061. 1338 1340 1342 1346 1348 1350 1352 1354 1356 1358 1360 1364 1366 1368 1370 -
2062. 1372 TO 1377 1380 TO 1385 1388 TO 1393 1396 TO 1401 1404 TO 1409 -
2063. 1412 TO 1417 1436 TO 1441 1444 TO 1449 1452 TO 1457 1460 TO 1465 -
2064. 1468 TO 1473 1476 TO 1481 2748 TO 2771 THICKNESS 0.3
2065. DEFINE MATERIAL START
2066. ISOTROPIC CONCRETE
2067. E 2.17184E+007
2068. POISSON 0.17
2069. DENSITY 24
2070. ALPHA 5.5E-006
2071. DAMP 0.05
2072. TYPE CONCRETE
2073. STRENGTH FCU 27578.9
2074. END DEFINE MATERIAL
2075. MEMBER PROPERTY AMERICAN
2076. 1 TO 7 9 TO 19 22 TO 25 2772 PRIS YD 0.7 ZD 0.7
2077. CONSTANTS
2078. MATERIAL CONCRETE ALL
2079. SUPPORTS
2080. 28 TO 391 393 TO 397 399 TO 403 405 TO 409 411 TO 415 417 TO 421 423 TO 427 -
2081. 429 TO 433 435 TO 439 441 TO 445 447 TO 451 453 TO 457 459 TO 463 -
2082. 465 TO 469 471 TO 475 477 TO 481 483 TO 487 489 TO 493 495 TO 499 -
2083. 501 TO 505 507 TO 871 873 TO 877 879 TO 883 885 TO 889 891 TO 895 -
2084. 897 TO 901 903 TO 907 909 TO 913 915 TO 919 921 TO 925 927 TO 931 -
2085. 933 TO 937 939 TO 943 945 TO 949 951 TO 955 957 TO 961 963 TO 967 -
2086. 969 TO 973 975 TO 979 981 TO 985 987 TO 995 998 999 1002 TO 1009 1012 1013 -
2087. 1016 TO 1023 1026 1027 1044 TO 1051 1054 1055 1058 TO 1065 1068 1069 1072 -
2088. 1073 TO 1079 1082 1083 1100 TO 1235 1238 TO 1253 1256 TO 1271 1274 TO 1289 -
2089. 1292 TO 1307 1310 TO 1325 1328 TO 1343 1346 TO 1361 1364 TO 1377 -
2090. 1380 TO 1385 1388 TO 1393 1396 TO 1401 1404 TO 1409 1412 TO 1417 -
2091. 1436 TO 1441 1444 TO 1449 1452 TO 1457 1460 TO 1465 1468 TO 1473 -
2092. 1476 TO 1481 1500 TO 1563 1604 TO 1609 1620 TO 1643 1645 1647 1649 1651 1653 -
2093. 1655 1657 1659 1661 1663 1665 1667 1669 1671 1673 1675 1677 1679 1681 1683 -
2094. 1684 TO 1743 1812 TO 1827 1834 TO 1849 1856 TO 1871 1878 TO 1893 1932 TO 2045 -
2095. 2128 TO 2591 2616 TO 2639 2670 TO 2681 2686 TO 2697 2702 TO 2713 -
2096. 2718 TO 2729 2736 TO 2771 2773 TO -
2097. 2872  PLATE MAT DIRECT Y SUBGRADE 117600 PRINT COMPRESSION
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JOINT	INFLUENCE AREA
100	0.122501
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2309	0.025000
2310	0.025000
2311	0.031250
2312	0.031250
2313	0.015625

2314	0.031250
2357	0.087500
2358	0.087500
2359	0.062500
2360	0.062500
2361	0.050000
2362	0.050000
2363	0.050000
2364	0.062500
2365	0.040000
2366	0.031250
2367	0.062500
2368	0.062500
2369	0.050000
2370	0.050000
2371	0.062500
2372	0.050000
2373	0.031250
2374	0.036125
2375	0.087500
2376	0.087500
2377	0.062500
2378	0.062500
2379	0.050000
2380	0.050000
2381	0.050000
2382	0.062500
2383	0.040000
2384	0.031250
2385	0.062500
2386	0.062500
2387	0.050000
2388	0.050000
2389	0.062500
2390	0.050000
2391	0.031250
2392	0.036125
2393	0.087500
2394	0.061125
2395	0.065000
2396	0.122500
2397	0.085575
2398	0.091000
2399	0.087500
2400	0.061125
2401	0.065000
2402	0.070000
2403	0.048900
2404	0.052000
2405	0.087500
2406	0.061125
2407	0.065000
2408	0.087499
2409	0.061124
2410	0.064999
2411	0.087502

2412	0.061126
2413	0.065001
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2415	0.061126
2416	0.065001
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2418	0.061125
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2423	0.087500
2424	0.061125
2425	0.065000
2426	0.070000
2427	0.048900
2428	0.052000
2429	0.087500
2430	0.061125
2431	0.065000
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2433	0.085575
2434	0.091000
2435	0.087500
2436	0.061125
2437	0.065000
2438	0.070000
2439	0.048900
2440	0.052000
2441	0.087500
2442	0.061125
2443	0.065000
2444	0.087500
2445	0.061125
2446	0.065000
2447	0.087500
2448	0.061125
2449	0.065000
2537	0.087500
2538	0.062500
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2541	0.062500
2542	0.062500
2543	0.050000
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2545	0.062500
2546	0.062500
2547	0.062500
2548	0.087500
2549	0.062500
2550	0.050000
2551	0.062500
2552	0.062500
2553	0.062500
2554	0.050000

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2556	0.062500
2557	0.062500
2558	0.062500
2559	0.087500
2560	0.062500
2561	0.050000
2562	0.062500
2563	0.062500
2564	0.062500
2565	0.050000
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2569	0.062500
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2571	0.062500
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2573	0.062500
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2575	0.062500
2576	0.050000
2577	0.062500
2578	0.062500
2579	0.062500
2580	0.062500
2581	0.026250
2582	0.018337
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2584	0.026250
2585	0.026250
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2587	0.018750
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2590	0.015000
2591	0.018750
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2593	0.009375
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2605	0.031250
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2608	0.025000
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2613	0.031250
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2615	0.025000
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2617	0.025000
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2619	0.018062
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2624	0.087500
2625	0.062500
2626	0.050000
2627	0.062500
2628	0.062500
2629	0.062500
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2631	0.062500
2632	0.062500
2633	0.062501
2634	0.050001
2635	0.062500
2636	0.062500
2637	0.050000
2638	0.062500
2639	0.062500
2640	0.062500
2641	0.050000
2642	0.062500
2643	0.062500
2644	0.062500
2645	0.062500
2646	0.062500
2647	0.062500
2648	0.087500
2649	0.062500
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2651	0.062500
2652	0.062500
2653	0.062500
2654	0.062500
2655	0.062500
2656	0.062500
2657	0.062501
2658	0.050001
2659	0.062500
2660	0.087500
2661	0.062500
2662	0.050000
2663	0.062500
2664	0.062500
2665	0.062500
2666	0.050000

2667	0.062500
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2669	0.062500
2670	0.062500
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2672	0.062500
2673	0.087500
2674	0.062500
2675	0.050000
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2678	0.062500
2679	0.050000
2680	0.062500
2681	0.062500
2682	0.062500
2683	0.062500
2684	0.087500
2685	0.062500
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2687	0.062500
2688	0.062500
2689	0.062500
2690	0.050000
2691	0.062500
2692	0.062500
2693	0.062500
2694	0.062500
2695	0.087500
2696	0.062500
2697	0.050000
2698	0.062500
2699	0.062500
2700	0.062500
2701	0.050000
2702	0.062500
2703	0.062500
2704	0.062500
2705	0.062500
2706	0.087500
2707	0.062500
2708	0.050000
2709	0.062500
2710	0.062500
2711	0.062500
2712	0.050000
2713	0.062500
2714	0.062500
2715	0.062500
2716	0.062500
2717	0.087500
2718	0.062500
2719	0.050000
2720	0.062500
2721	0.062500
2722	0.062500

2723	0.050000
2724	0.062500
2725	0.062500
2726	0.062500
2727	0.062500
2728	0.062500
2729	0.062500
2730	0.087500
2731	0.062500
2732	0.050000
2733	0.062500
2734	0.062500
2735	0.062500
2736	0.050000
2737	0.062500
2738	0.062500
2739	0.062500
2740	0.062500
2741	0.062500
2742	0.062500
2743	0.087500
2744	0.062500
2745	0.050000
2746	0.062500
2747	0.062500
2748	0.062500
2749	0.050000
2750	0.062500
2751	0.062500
2752	0.062500
2753	0.062500
2754	0.062500
2755	0.062500
2756	0.087500
2757	0.062500
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2759	0.062500
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2761	0.062500
2762	0.050000
2763	0.062500
2764	0.062500
2765	0.062500
2766	0.062500
2767	0.062500
2768	0.062500
2769	0.043750
2770	0.031250
2771	0.025000
2772	0.031250
2773	0.031250
2774	0.031250
2775	0.025000
2776	0.031250
2777	0.031250
2778	0.031250

2779	0.031250
2780	0.031250
2781	0.031250
2782	0.087500
2783	0.062500
2784	0.050000
2785	0.062500
2786	0.062500
2787	0.062500
2788	0.050000
2789	0.062500
2790	0.062500
2791	0.062500
2792	0.062500
2793	0.062500
2794	0.062500
2795	0.087500
2796	0.062500
2797	0.050000
2798	0.062500
2799	0.062500
2800	0.062500
2801	0.050000
2802	0.062500
2803	0.062500
2804	0.062500
2805	0.062500
2806	0.062500
2807	0.062500
2808	0.043750
2809	0.031250
2810	0.025000
2811	0.031250
2812	0.031250
2813	0.031250
2814	0.025000
2815	0.031250
2816	0.031250
2817	0.031250
2818	0.031250
2819	0.031250
2820	0.031250
2821	0.087500
2822	0.062500
2823	0.050000
2824	0.062500
2825	0.062500
2826	0.062500
2827	0.062500
2828	0.062500
2829	0.062500
2830	0.062501
2831	0.050001
2832	0.062500
2833	0.087500
2834	0.062500

2835	0.050000
2836	0.062500
2837	0.062500
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2839	0.050000
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2849	0.062500
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2851	0.062500
2852	0.062500
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2855	0.031250
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2857	0.031250
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2864	0.025001
2865	0.031250
2866	0.043750
2867	0.031250
2868	0.025000
2869	0.031250
2870	0.031250
2871	0.031250
2872	0.025000
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2877	0.031250
2878	0.031250
2879	0.031250
2880	0.031250
2881	0.031250
2882	0.031250
2883	0.015625
2884	0.031250
2885	0.031250
2886	0.015625
2887	0.087500
2888	0.087500
2889	0.062500
2890	0.050000

2891	0.062500
2892	0.062500
2893	0.062500
2894	0.062500
2895	0.062500
2896	0.062500
2897	0.062501
2898	0.050001
2899	0.062500
2900	0.062500
2901	0.050000
2902	0.062500
2903	0.062500
2904	0.062500
2905	0.050000
2906	0.062500
2907	0.062500
2908	0.062500
2909	0.062500
2910	0.062500
2911	0.062500
2912	0.062500
2913	0.062500
2914	0.062500
2915	0.062500
2916	0.031250
2917	0.062500
2918	0.062500
2919	0.031250
2920	0.043750
2921	0.043750
2922	0.031250
2923	0.025000
2924	0.031250
2925	0.031250
2926	0.031250
2927	0.031250
2928	0.031250
2929	0.031250
2930	0.031251
2931	0.025001
2932	0.031250
2933	0.031250
2934	0.025000
2935	0.031250
2936	0.031250
2937	0.031250
2938	0.025000
2939	0.031250
2940	0.031250
2941	0.031250
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2943	0.031250
2944	0.031250
2945	0.031250
2946	0.031250

2947	0.031250
2948	0.031250
2949	0.015625
2950	0.031250
2951	0.031250
2952	0.015625
2953	0.043750
2954	0.031250
2955	0.025000
2956	0.031250
2957	0.031250
2958	0.031250
2959	0.031250
2960	0.031250
2961	0.031250
2962	0.031251
2963	0.025001
2964	0.031250
2965	0.043750
2966	0.031250
2967	0.025000
2968	0.031250
2969	0.031250
2970	0.031250
2971	0.025000
2972	0.031250
2973	0.031250
2974	0.031250
2975	0.031250
2976	0.031250
2977	0.031250
2978	0.043750
2979	0.043750
2980	0.031250
2981	0.025000
2982	0.031250
2983	0.031250
2984	0.031250
2985	0.031250
2986	0.031250
2987	0.031250
2988	0.031251
2989	0.025001
2990	0.031250
2991	0.031250
2992	0.025000
2993	0.031250
2994	0.031250
2995	0.031250
2996	0.025000
2997	0.031250
2998	0.031250
2999	0.031250
3000	0.031250
3001	0.031250
3002	0.031250

3003	0.087500
3004	0.062500
3005	0.050000
3006	0.062500
3007	0.062500
3008	0.062500
3009	0.050000
3010	0.062500
3011	0.062500
3012	0.062500
3013	0.062500
3014	0.031250
3015	0.031250
3016	0.043750
3017	0.031250
3018	0.025000
3019	0.031250
3020	0.031250
3021	0.031250
3022	0.025000
3023	0.031250
3024	0.031250
3025	0.031250
3026	0.031250
3027	0.015625
3028	0.015625
3055	0.087500
3056	0.062500
3057	0.050000
3058	0.062500
3059	0.062500
3060	0.062500
3061	0.050000
3062	0.062500
3063	0.062500
3064	0.062500
3065	0.062500
3066	0.031250
3067	0.031250
3068	0.043750
3069	0.031250
3070	0.025000
3071	0.031250
3072	0.031250
3073	0.031250
3074	0.025000
3075	0.031250
3076	0.031250
3077	0.031250
3078	0.031250
3079	0.015625
3080	0.015625
3114	0.087500
3115	0.062500
3116	0.050000
3117	0.062500

3118	0.062500
3119	0.062500
3120	0.050000
3121	0.062500
3122	0.062500
3123	0.062500
3124	0.031250
3125	0.062500
3126	0.031250
3131	0.112875
3132	0.080625
3133	0.064500
3134	0.080625
3135	0.080625
3136	0.080625
3137	0.064500
3138	0.080625
3139	0.080625
3140	0.080625
3141	0.040313
3142	0.080625
3143	0.040313
3148	0.087500
3149	0.062500
3150	0.050000
3151	0.062500
3152	0.062500
3153	0.062500
3154	0.050000
3155	0.062500
3156	0.062500
3157	0.062500
3158	0.031250
3159	0.062500
3160	0.031250
3165	0.112875
3166	0.080625
3167	0.064500
3168	0.080625
3169	0.080625
3170	0.080625
3171	0.064500
3172	0.080625
3173	0.080625
3174	0.080625
3175	0.040313
3176	0.080625
3177	0.040313
3178	0.043750
3179	0.043750
3180	0.031250
3181	0.031250
3182	0.025000
3183	0.025000
3184	0.015625
3185	0.015625

3186	0.031250
3187	0.031250
3188	0.025000
3189	0.025000
3190	0.015625
3191	0.015625
3192	0.043750
3193	0.043750
3194	0.031250
3195	0.031250
3196	0.025000
3197	0.025000
3198	0.015625
3199	0.015625
3200	0.031250
3201	0.031250
3202	0.025000
3203	0.025000
3204	0.015625
3205	0.015625
3206	0.122500
3207	0.087500
3208	0.062502
3209	0.056250
3210	0.078750
3211	0.087513
3212	0.087500
3213	0.062502
3214	0.070000
3215	0.050000
3216	0.087500
3217	0.062500
3218	0.050000
3219	0.045000
3220	0.070000
3221	0.050000
3222	0.040000
3223	0.050000
3224	0.062500
3225	0.056250
3226	0.087500
3227	0.062500
3228	0.050000
3229	0.046875
3230	0.062498
3231	0.056250
3232	0.087487
3233	0.062498
3234	0.050000
3235	0.053125
3236	0.050000
3237	0.045000
3238	0.070000
3239	0.050000
3240	0.040000
3241	0.025000

3242	0.062500
3243	0.056250
3244	0.087500
3245	0.062500
3246	0.050000
3247	0.031250
3248	0.096250
3249	0.068750
3250	0.055000
3251	0.068750
3252	0.068750
3253	0.055000
3254	0.068750
3255	0.043750
3256	0.031250
3257	0.025000
3258	0.031250
3259	0.031250
3260	0.031250
3261	0.031250
3262	0.031250
3263	0.031250
3264	0.031251
3265	0.025001
3266	0.031250
3267	0.043750
3268	0.031250
3269	0.025000
3270	0.031250
3271	0.031250
3272	0.031250
3273	0.025000
3274	0.031250
3275	0.015625
3276	0.031250
3277	0.028125
3278	0.043750
3279	0.031250
3280	0.025000
3281	0.015625
3282	0.034375
3283	0.043750
3284	0.043750
3285	0.031250
3286	0.025000
3287	0.031250
3288	0.031250
3289	0.031250
3290	0.031250
3291	0.031250
3292	0.031250
3293	0.031251
3294	0.025001
3295	0.031250
3296	0.031250
3297	0.025000

3298	0.031250
3299	0.031250
3300	0.031250
3301	0.025000
3302	0.031250
3303	0.015625
3304	0.031250
3305	0.028125
3306	0.043750
3307	0.031250
3308	0.025000
3309	0.015625
3310	0.034375
3311	0.043750
3312	0.031250
3313	0.025000
3314	0.015625
3315	0.021875
TOTAL	145.264633

2098. 2873 PLATE MAT DIRECT Y SUBGRADE 117600 PRINT COMPRESSION

JOINT	INFLUENCE AREA
3235	0.009375
3241	0.009375
3315	0.009375
3316	0.009375
TOTAL	0.037500

2099. 2874 PLATE MAT DIRECT Y SUBGRADE 117600 PRINT COMPRESSION

JOINT	INFLUENCE AREA
3241	0.015625
3247	0.015625
3316	0.015625
3317	0.015625
TOTAL	0.062500

2100. 2875 PLATE MAT DIRECT Y SUBGRADE 117600 PRINT COMPRESSION

JOINT	INFLUENCE AREA
3229	0.015625
3281	0.015625
3314	0.015625
3318	0.015625
TOTAL	0.062500

2101. 2876 PLATE MAT DIRECT Y SUBGRADE 117600 PRINT COMPRESSION

JOINT	INFLUENCE AREA
3247	0.015625
3309	0.015625

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        3317          0.015625
        3319          0.015625
      TOTAL           0.062500
 2102. DEFINE REFERENCE LOADS
  **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
             MASTER/SLAVE OR IF UNCONNECTED JOINTS.

**WARNING, 1 or more joints are in multiple SUPPORT commands, First Joint  3235
The subsequent entries will add to support direction specs. See Tech Ref Manual.
 2103. LOAD R1 LOADTYPE DEAD  TITLE SELF WEIGHT (D)
 2104. SELFWEIGHT Y -1
 2105. ELEMENT LOAD
 2106. 28 29 34 35 40 41 46 47 52 53 58 59 64 65 70 71 76 77 82 83 88 89 94 95 100 -
 2107. 101 106 107 112 113 118 119 124 125 130 131 136 137 142 143 148 149 154 155 -
 2108. 160 161 166 167 172 173 178 179 184 185 190 191 196 197 202 203 208 209 214 -
 2109. 215 220 221 226 227 232 233 238 239 244 245 250 251 256 257 262 263 268 269 -
 2110. 274 275 280 281 286 287 292 293 298 299 304 305 310 311 316 317 322 323 328 -
 2111. 329 334 335 340 341 346 347 352 353 358 359 364 365 370 371 376 377 382 383 -
 2112. 388 389 394 395 400 401 406 407 412 413 418 419 424 425 430 431 436 437 442 -
 2113. 443 448 449 454 455 460 461 466 467 472 473 478 479 484 485 490 491 496 497 -
 2114. 502 503 508 509 514 515 520 521 526 527 532 533 538 539 544 545 550 551 556 -
 2115. 557 562 563 568 569 574 575 580 581 586 587 592 593 598 599 604 605 610 611 -
 2116. 616 617 622 623 628 629 634 635 640 641 646 647 652 653 658 659 664 665 670 -
 2117. 671 676 677 682 683 688 689 694 695 700 701 706 707 712 713 718 719 724 725 -
 2118. 730 731 736 737 742 743 748 749 754 755 760 761 766 767 772 773 778 779 784 -
 2119. 785 790 791 796 797 802 803 808 809 814 815 820 821 826 827 832 833 838 839 -
 2120. 844 845 850 851 856 857 862 863 PR GY -32.3
 2121. 868 874 880 886 892 898 904 910 916 922 928 934 940 946 952 958 964 970 976 -
 2122. 982 1684 1686 1688 1690 1692 1694 1696 1698 1700 1702 1704 1706 1708 1710 -
 2123. 1712 1714 1716 1718 1720 1722 1724 TO 1743 2256 TO 2295 2520 TO 2591 2616 -
 2124. 2617 TO 2639 PR GY -32.3
 2125. 868 869 874 875 880 881 886 887 892 893 898 899 904 905 910 911 916 917 922 -
 2126. 923 928 929 934 935 940 941 946 947 952 953 958 959 964 965 970 971 976 977 -
 2127. 982 983 1684 1686 1688 1690 1692 1694 1696 1698 1700 1702 1704 1706 1708 -
 2128. 1710 1712 1714 1716 1718 1720 1722 1724 TO 1743 2263 TO 2265 2273 TO 2275 -
 2129. 2283 TO 2285 2293 TO 2295 2520 TO 2567 2575 TO 2579 2587 TO 2591 -
 2130. 2623 TO 2627 2635 TO 2639 PR GY -32.3
 2131. 30 TO 32 36 TO 38 42 TO 44 48 TO 50 54 TO 56 60 TO 62 66 TO 68 72 TO 74 78 -
 2132. 79 TO 80 84 TO 86 90 TO 92 96 TO 98 102 TO 104 108 TO 110 114 TO 116 -
 2133. 120 TO 122 126 TO 128 132 TO 134 138 TO 140 144 TO 146 150 TO 152 -
 2134. 156 TO 158 162 TO 164 168 TO 170 174 TO 176 180 TO 182 186 TO 188 -
 2135. 192 TO 194 198 TO 200 204 TO 206 210 TO 212 216 TO 218 222 TO 224 -
 2136. 228 TO 230 234 TO 236 240 TO 242 246 TO 248 252 TO 254 258 TO 260 -
 2137. 264 TO 266 270 TO 272 276 TO 278 282 TO 284 288 TO 290 294 TO 296 -
 2138. 300 TO 302 306 TO 308 312 TO 314 318 TO 320 324 TO 326 330 TO 332 -
 2139. 336 TO 338 342 TO 344 348 TO 350 354 TO 356 360 TO 362 366 TO 368 -
 2140. 372 TO 374 378 TO 380 384 TO 386 390 391 396 397 402 403 408 409 414 415 -
 2141. 420 421 426 427 432 433 438 439 444 445 450 451 456 457 462 463 468 469 474 -
 2142. 475 480 481 486 487 492 493 498 499 504 505 510 TO 512 516 TO 518 -
 2143. 522 TO 524 528 TO 530 534 TO 536 540 TO 542 546 TO 548 552 TO 554 -
 2144. 558 TO 560 564 TO 566 570 TO 572 576 TO 578 582 TO 584 588 TO 590 -
 2145. 594 TO 596 600 TO 602 606 TO 608 612 TO 614 618 TO 620 624 625 -
 2146. 2773  PR GY -32.3
 2147. 2805 2815 TO 2834 2840 PR GY -32.3
 2148. 2774 TO 2792 2803 TO 2806 2809 TO 2812 2835 TO 2841 2869 TO 2872 -
 2149. 2875 PR GY -32.3
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2150. 626 630 TO 632 636 TO 638 642 TO 644 648 TO 650 654 TO 656 660 TO 662 666 -
2151. 667 TO 668 672 TO 674 678 TO 680 684 TO 686 690 TO 692 696 TO 698 702 TO 704 -
2152. 708 TO 710 714 TO 716 720 TO 722 726 TO 728 732 TO 734 738 TO 740 -
2153. 744 TO 746 750 TO 752 756 TO 758 762 TO 764 768 TO 770 774 TO 776 -
2154. 780 TO 782 786 TO 788 792 TO 794 798 TO 800 804 TO 806 810 TO 812 -
2155. 816 TO 818 822 TO 824 828 TO 830 834 TO 836 840 TO 842 846 TO 848 -
2156. 852 TO 854 858 TO 860 864 TO 866 870 871 876 877 882 883 888 889 894 895 -
2157. 900 901 906 907 912 913 918 919 924 925 930 931 936 937 942 943 948 949 954 -
2158. 955 960 961 966 967 972 973 978 979 984 985 1604 TO 1609 1932 TO 2045 2168 -
2159. 2169 TO 2207 2793 TO 2802 2807 2808 2813 2814 2842 TO 2868 2873 2874 -
2160. 2876 PR GY -32.3
2161. 33 39 45 51 57 63 69 75 81 87 93 99 105 111 117 123 129 135 141 147 153 159 -
2162. 165 171 177 183 189 195 201 207 213 219 225 231 237 243 249 255 261 267 273 -
2163. 279 285 291 297 303 309 315 321 327 333 339 345 351 357 363 369 375 381 387 -
2164. 393 399 405 411 417 423 429 435 441 447 453 459 465 471 477 483 489 495 501 -
2165. 507 513 519 525 531 537 543 549 555 561 567 573 579 585 591 597 603 609 615 -
2166. 621 627 633 639 645 651 657 663 669 675 681 687 693 699 705 711 717 723 729 -
2167. 735 741 747 753 759 765 771 777 783 789 795 801 807 813 819 825 831 837 843 -
2168. 849 855 861 867 873 879 885 891 897 903 909 915 921 927 933 939 945 951 957 -
2169. 963 969 975 981 987 1645 1647 1649 1651 1653 1655 1657 1659 1661 1663 1665 -
2170. 1667 1669 1671 1673 1675 1677 1679 1681 1683 1685 1687 1689 1691 1693 1695 -
2171. 1697 1699 1701 1703 1705 1707 1709 1711 1713 1715 1717 1719 1721 1723 2128 -
2172. 2129 TO 2167 2208 TO 2255 2296 TO 2519 PR GY -30.4
2173. 988 TO 995 998 999 1002 TO 1009 1012 1013 1016 TO 1023 1026 1027 1044 TO 1051 -
2174. 1054 1055 1058 TO 1065 1068 1069 1072 TO 1079 1082 1083 1100 TO 1235 1238 -
2175. 1239 TO 1253 1256 TO 1271 1274 TO 1289 1292 TO 1307 1310 TO 1325 1328 TO 1343 -
2176. 1346 TO 1361 1364 TO 1377 1380 TO 1385 1388 TO 1393 1396 TO 1401 -
2177. 1404 TO 1409 1412 TO 1417 1436 TO 1441 1444 TO 1449 1452 TO 1457 -
2178. 1460 TO 1465 1468 TO 1473 1476 TO 1481 1500 TO 1563 1620 TO 1643 -
2179. 1812 TO 1827 1834 TO 1849 1856 TO 1871 1878 TO 1893 2670 TO 2681 -
2180. 2686 TO 2697 2702 TO 2713 2718 TO 2729 2736 TO 2771 PR GY -36.1
2181. LOAD R2 LOADTYPE LIVE  TITLE LIVE  LOAD -5KN/SQM (L1)
2182. JOINT LOAD
2183. 19 FX -1.46 FY -95.27 FZ 0.3 MX 0 MY -0.46 MZ 0
2184. 5 FX 0.76 FY -72.6 FZ 0.19 MX 0 MY 0 MZ 0
2185. 27 FX -1.98 FY -77.1 FZ 0.14 MX 0 MY 0 MZ 0
2186. 6 FX 0.76 FY -64.92 FZ -1.01 MX 0 MY 0.02 MZ 0
2187. 28 FX -0.89 FY -58.83 FZ 1.09 MX 0 MY -0.16 MZ 0
2188. 14 FX 0.03 FY -70.6 FZ 6.1 MX 0 MY 0 MZ 0
2189. 15 FX -0.64 FY -180.67 FZ 0.32 MX -0.54 MY 0 MZ 0.99
2190. 11 FX -0.01 FY -63.66 FZ -1.96 MX 0 MY 0 MZ 0
2191. 16 FX -0.71 FY -92.38 FZ -2.24 MX -1.84 MY 0 MZ 0.89
2192. 7 FX -0.45 FY -162.5 FZ 0.62 MX 0.54 MY 0 MZ 0.2
2193. 8 FX 0.49 FY -143.76 FZ 0.46 MX 0.44 MY 0 MZ -0.64
2194. 20 FX -0.12 FY -138.4 FZ -0.38 MX -1.3 MY 0 MZ 0.32
2195. 21 FX 0.46 FY -78.82 FZ -3.34 MX -2.86 MY 0 MZ -0.13
2196. 25 FX 0.19 FY -162.64 FZ -0.91 MX -0.88 MY 0 MZ 0.03
2197. 26 FX -0.33 FY -124.47 FZ -0.89 MX -0.67 MY 0 MZ 0.42
2198. 12 FX 0 FY -13.98 FZ 1.51 MX 0 MY 0.01 MZ 0
2199. 10 FX -1.28 FY -11.62 FZ 0.01 MX 0 MY 0 MZ 0
2200. 22 FX -0.87 FY -24.02 FZ 0 MX 0 MY 0 MZ 0
2201. 13 FX 0.85 FY -37 FZ 0 MX 0 MY 0 MZ 0
2202. 23 FX 6.36 FY -18.7 FZ -2.04 MX 0 MY 0.03 MZ 0
2203. 17 FX -6.48 FY -56.98 FZ 0 MX 0 MY 0 MZ 0
2204. 24 FX -0.84 FY -11.5 FZ -0.02 MX 0 MY 0 MZ 0
2205. 18 FX 0.83 FY -11.53 FZ -0.02 MX 0 MY 0 MZ 0
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2206. LOAD R3 LOADTYPE WIND  TITLE WIND LOAD +X DIRN(W +X)
2207. JOINT LOAD
2208. 19 FX 24.41 FY 24.22 FZ 0.05 MX 0 MY -0.02 MZ 0
2209. 5 FX 1.26 FY -2.05 FZ 0.01 MX 0 MY -0.01 MZ 0
2210. 27 FX 2.48 FY 4.23 FZ 0 MX 0 MY 0 MZ 0
2211. 6 FX 1.28 FY -1.14 FZ 0.28 MX 0 MY 0.02 MZ 0
2212. 28 FX 2.17 FY 3.06 FZ -0.17 MX 0 MY 0.17 MZ 0
2213. 14 FX 0.12 FY -22.21 FZ 0.1 MX 0 MY 0 MZ 0
2214. 15 FX 7.21 FY -1.16 FZ 0.09 MX 0.16 MY 0 MZ -9.78
2215. 11 FX 0 FY -2.48 FZ -0.36 MX 0 MY 0 MZ 0
2216. 16 FX 6.48 FY -1.51 FZ 0.1 MX 0.16 MY 0 MZ -9.3
2217. 7 FX 6.24 FY 0.13 FZ 0.06 MX 0.16 MY 0 MZ -8.98
2218. 8 FX 6.27 FY -0.47 FZ 0.06 MX 0.15 MY 0 MZ -9.01
2219. 20 FX 13.91 FY 0.03 FZ -0.23 MX -0.27 MY 0 MZ -12.7
2220. 21 FX 6.67 FY 1.13 FZ -0.24 MX -0.28 MY 0 MZ -9.52
2221. 25 FX 6.71 FY -1.65 FZ 0.06 MX 0.09 MY 0 MZ -9.95
2222. 26 FX 7.05 FY -0.85 FZ 0.05 MX 0.07 MY 0 MZ -10.15
2223. 12 FX 0 FY 29.05 FZ 0.13 MX 0 MY 0.01 MZ 0
2224. 10 FX 21.56 FY -22.69 FZ 0.03 MX 0 MY 0 MZ 0
2225. 22 FX 0.31 FY 22.99 FZ 0 MX 0 MY 0 MZ 0
2226. 13 FX 18.89 FY -22.92 FZ 0 MX 0 MY 0 MZ 0
2227. 23 FX 29.79 FY 49.07 FZ -0.06 MX 0 MY 0.04 MZ 0
2228. 17 FX -1.54 FY -49.18 FZ 0 MX 0 MY 0 MZ 0
2229. 24 FX 4.57 FY 4.75 FZ 0 MX 0 MY 0 MZ 0
2230. 18 FX 0.13 FY -4.74 FZ 0 MX 0 MY 0 MZ 0
2231. LOAD R4 LOADTYPE WIND  TITLE WIND LOAD -X DIRN(W -X)
2232. JOINT LOAD
2233. 19 FX -14.97 FY -24.36 FZ -0.02 MX 0 MY 0.01 MZ 0
2234. 5 FX -1.12 FY 1.88 FZ -0.01 MX 0 MY 0.01 MZ 0
2235. 27 FX -1.02 FY -1.07 FZ 0 MX 0 MY 0 MZ 0
2236. 6 FX -1.13 FY 1.76 FZ -0.02 MX 0 MY -0.02 MZ 0
2237. 28 FX -0.79 FY -0.9 FZ -0.03 MX 0 MY -0.08 MZ 0
2238. 14 FX -9.37 FY 22.42 FZ -0.12 MX 0 MY 0 MZ 0
2239. 15 FX -10.16 FY -0.26 FZ 0.12 MX 0.13 MY 0 MZ 8.03
2240. 11 FX 0 FY 2.34 FZ 0.36 MX 0 MY 0 MZ 0
2241. 16 FX -3.73 FY -0.09 FZ 0.11 MX 0.12 MY 0 MZ 5.37
2242. 7 FX -4.04 FY -1.14 FZ 0.06 MX 0.06 MY 0 MZ 5.82
2243. 8 FX -4.05 FY -0.48 FZ 0.05 MX 0.04 MY 0 MZ 5.83
2244. 20 FX -4.09 FY 1.32 FZ -0.08 MX -0.11 MY 0 MZ 5.56
2245. 21 FX -3.7 FY 0.05 FZ -0.08 MX -0.11 MY 0 MZ 5.28
2246. 25 FX -3.4 FY -0.38 FZ -0.06 MX -0.1 MY 0 MZ 5.04
2247. 26 FX -3.58 FY -0.46 FZ -0.06 MX -0.09 MY 0 MZ 5.15
2248. 12 FX 0 FY -19.84 FZ -0.25 MX 0 MY 0.01 MZ 0
2249. 10 FX -14.26 FY 13.98 FZ 0.01 MX 0 MY 0 MZ 0
2250. 22 FX -0.2 FY -20.46 FZ 0 MX 0 MY 0 MZ 0
2251. 13 FX -16.44 FY 20.65 FZ 0 MX 0 MY 0 MZ 0
2252. 23 FX -22.71 FY -39.57 FZ -0.18 MX 0 MY -0.04 MZ 0
2253. 17 FX 1.03 FY 39.3 FZ 0 MX 0 MY 0 MZ 0
2254. 24 FX -4.54 FY -4.74 FZ 0 MX 0 MY 0 MZ 0
2255. 18 FX -0.15 FY 4.67 FZ 0 MX 0 MY 0 MZ 0
2256. LOAD R5 LOADTYPE WIND  TITLE WIND LOAD +Z DIRN(W -Z)
2257. JOINT LOAD
2258. 19 FX -1.14 FY 12.27 FZ -6.02 MX 0 MY 0.3 MZ 0
2259. 5 FX -0.29 FY 170.77 FZ -3.69 MX 0 MY -0.02 MZ 0
2260. 27 FX 0.38 FY 164.56 FZ -3.69 MX 0 MY 0.02 MZ 0
2261. 6 FX 0.17 FY -171.26 FZ -94.78 MX 0 MY -0.05 MZ 0
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2262. 28 FX -0.34 FY -165.92 FZ -90.96 MX 0 MY 0.01 MZ 0
2263. 14 FX 0.02 FY 19.16 FZ -10.46 MX 0 MY 0 MZ 0
2264. 15 FX -0.25 FY 63.15 FZ -53.56 MX -79.78 MY 0 MZ 0.3
2265. 11 FX 0 FY 38.92 FZ 1.34 MX 0 MY 0 MZ 0
2266. 16 FX -0.55 FY -80.52 FZ -49.99 MX -77.06 MY 0 MZ 0.75
2267. 7 FX -2.03 FY -10.13 FZ -33.21 MX -60.3 MY 0 MZ 3.35
2268. 8 FX 1.51 FY 10 FZ -19.32 MX -50.63 MY -0.01 MZ -2.47
2269. 20 FX 0.61 FY 60.23 FZ -51.08 MX -75.24 MY 0 MZ -0.67
2270. 21 FX -1.2 FY -72.53 FZ -45.03 MX -71.94 MY 0 MZ 1.67
2271. 25 FX 1.81 FY -8.51 FZ -29.49 MX -54.94 MY 0.01 MZ -2.54
2272. 26 FX -2.29 FY 8.76 FZ -16.83 MX -46.47 MY 0.01 MZ 3.61
2273. 12 FX 0 FY -36.27 FZ -15.58 MX 0 MY 0 MZ 0
2274. 10 FX 0.59 FY -8.41 FZ -4.84 MX 0 MY 0 MZ 0
2275. 22 FX -0.18 FY -29.52 FZ 0.12 MX 0 MY 0.22 MZ 0
2276. 13 FX 1.61 FY 9.9 FZ -0.37 MX 0 MY -0.1 MZ 0
2277. 23 FX 3.95 FY 64.99 FZ 34.81 MX 0 MY 2.42 MZ 0
2278. 17 FX -4.84 FY 7.5 FZ -0.07 MX 0 MY 0.29 MZ 0
2279. 24 FX -2.28 FY -0.31 FZ -0.04 MX 0 MY 0.04 MZ 0
2280. 18 FX -0.14 FY 3.09 FZ -0.27 MX 0 MY 0.01 MZ 0
2281. LOAD R6 LOADTYPE WIND TITLE WIND LOAD -Z DIRN(W +Z)
2282. JOINT LOAD
2283. 19 FX 1.7 FY -12.04 FZ 0.45 MX 0 MY -0.75 MZ 0
2284. 5 FX 0.39 FY -319.9 FZ -0.71 MX 0 MY 0.03 MZ 0
2285. 27 FX -0.62 FY -305.14 FZ -0.7 MX 0 MY -0.02 MZ 0
2286. 6 FX -0.3 FY 320.9 FZ 177.56 MX 0 MY 0.03 MZ 0
2287. 28 FX 0.6 FY 307.6 FZ 168.75 MX 0 MY 0.09 MZ 0
2288. 14 FX -0.02 FY -23.77 FZ 9.21 MX 0 MY 0 MZ 0
2289. 15 FX 0.28 FY -115.81 FZ 80.87 MX 127.43 MY 0 MZ -0.3
2290. 11 FX 0 FY -55.43 FZ -1.44 MX 0 MY 0 MZ 0
2291. 16 FX 0.88 FY 137.15 FZ 85.84 MX 130.44 MY 0 MZ -1.23
2292. 7 FX 3.3 FY 11.48 FZ 35.27 MX 91.86 MY 0.02 MZ -5.48
2293. 8 FX -2.55 FY -12 FZ 42.1 MX 94.57 MY 0.02 MZ 4.2
2294. 20 FX -0.95 FY -103.78 FZ 78.8 MX 119.46 MY 0 MZ 1.13
2295. 21 FX 1.97 FY 116.96 FZ 74.14 MX 118.56 MY 0 MZ -2.73
2296. 25 FX -3.09 FY 6.71 FZ 31.36 MX 82.79 MY -0.01 MZ 4.32
2297. 26 FX 3.72 FY -8.42 FZ 36.12 MX 85.23 MY -0.01 MZ -5.9
2298. 12 FX 0 FY 51.14 FZ 25.76 MX 0 MY 0 MZ 0
2299. 10 FX -1.58 FY 9.61 FZ 4.84 MX 0 MY 0 MZ 0
2300. 22 FX 0.22 FY 43.07 FZ -0.17 MX 0 MY -0.26 MZ 0
2301. 13 FX -1.88 FY -12.35 FZ 0.17 MX 0 MY 0.16 MZ 0
2302. 23 FX -5.03 FY -86.1 FZ -47.35 MX 0 MY -2.79 MZ 0
2303. 17 FX 5.94 FY -8.32 FZ 0.31 MX 0 MY -0.34 MZ 0
2304. 24 FX 2.8 FY 0.92 FZ 0.04 MX 0 MY -0.05 MZ 0
2305. 18 FX 0.13 FY -3.17 FZ 0.33 MX 0 MY -0.01 MZ 0
2306. LOAD R7 LOADTYPE DEAD TITLE TANK (D1)
2307. JOINT LOAD
2308. 19 FX -1.68 FY -63.78 FZ 0.45 MX 0 MY -0.72 MZ 0
2309. 5 FX -0.06 FY 2.46 FZ 0.01 MX 0 MY 0 MZ 0
2310. 27 FX -0.08 FY 1.92 FZ 0.01 MX 0 MY 0 MZ 0
2311. 6 FX -0.04 FY -2.14 FZ -0.85 MX 0 MY 0 MZ 0
2312. 28 FX -0.08 FY -1.76 FZ -0.65 MX 0 MY -0.01 MZ 0
2313. 14 FX -0.03 FY -91.09 FZ 9.89 MX 0 MY 0 MZ 0
2314. 15 FX -0.28 FY -56.99 FZ -1.51 MX -2.81 MY 0 MZ 0.39
2315. 11 FX 0 FY -65.73 FZ 0.16 MX 0 MY 0 MZ 0
2316. 16 FX -0.29 FY -3 FZ -2.25 MX -3.17 MY 0 MZ 0.42
2317. 7 FX -0.31 FY -1.04 FZ -0.53 MX -1.32 MY 0 MZ 0.46

2318. 8 FX -0.19 FY 0.15 FZ -0.46 MX -1.25 MY 0 MZ 0.26
2319. 20 FX -0.08 FY -49.65 FZ -1.65 MX -2.25 MY 0 MZ 0.27
2320. 21 FX -0.35 FY -2.93 FZ -1.38 MX -2.14 MY 0 MZ 0.5
2321. 25 FX -0.33 FY -2.56 FZ -0.41 MX -1.05 MY 0 MZ 0.45
2322. 26 FX -0.38 FY -0.02 FZ -0.35 MX -1 MY 0 MZ 0.56
2323. 12 FX 0 FY -17.3 FZ -0.42 MX 0 MY 0.02 MZ 0
2324. 10 FX 0.08 FY -0.22 FZ -0.04 MX 0 MY 0 MZ 0
2325. 22 FX -0.05 FY -27.98 FZ 0 MX 0 MY 0 MZ 0
2326. 13 FX -0.07 FY -38.46 FZ 0 MX 0 MY 0 MZ 0
2327. 23 FX 8.74 FY -16.42 FZ -0.99 MX 0 MY 0.03 MZ 0
2328. 17 FX -8.95 FY -67.8 FZ 0.01 MX 0 MY 0 MZ 0
2329. 24 FX 0.04 FY 0.06 FZ 0 MX 0 MY 0 MZ 0
2330. 18 FX 0 FY 0.04 FZ 0 MX 0 MY 0 MZ 0
2331. LOAD R8 LOADTYPE TEMPERATURE TITLE TEMPERATURE +20(T +VE)
2332. JOINT LOAD
2333. 19 FX -13.41 FY -21.58 FZ 0.32 MX 0 MY -0.16 MZ 0
2334. 5 FX 11.56 FY -15.5 FZ -2.28 MX 0 MY -0.07 MZ 0
2335. 27 FX -19.54 FY -16.27 FZ -2.29 MX 0 MY 0.01 MZ 0
2336. 6 FX 11.22 FY 2.89 FZ 11.64 MX 0 MY 0.18 MZ 0
2337. 28 FX -14.56 FY -5.72 FZ 7.82 MX 0 MY -1.57 MZ 0
2338. 14 FX 0.09 FY 19.91 FZ 1.56 MX 0 MY 0.03 MZ 0
2339. 15 FX -1.12 FY 10.4 FZ 14.99 MX 18.82 MY 0 MZ 4.26
2340. 11 FX -0.03 FY 6.19 FZ 0.05 MX 0 MY 0.01 MZ 0
2341. 16 FX -4.02 FY -0.13 FZ -14.96 MX -20.23 MY 0 MZ 5.89
2342. 7 FX 25.83 FY 4.88 FZ 9.18 MX 11.99 MY 0 MZ -37.03
2343. 8 FX 24.46 FY 5.62 FZ -14.07 MX -22.02 MY 0 MZ -35.1
2344. 20 FX -9.78 FY 7.55 FZ 12.49 MX 16.05 MY 0 MZ 17.77
2345. 21 FX -14.07 FY 3.82 FZ -16.83 MX -22.7 MY 0 MZ 20
2346. 25 FX -35.09 FY 1.76 FZ 7.66 MX 10.02 MY 0 MZ 52.1
2347. 26 FX -35.47 FY 2.37 FZ -14.73 MX -23.46 MY 0 MZ 50.93
2348. 12 FX 0.03 FY 55.81 FZ -0.66 MX 0 MY 0.2 MZ 0
2349. 10 FX 26.7 FY -32.6 FZ 0.09 MX 0 MY 0 MZ 0
2350. 22 FX 3 FY -15.14 FZ -0.02 MX 0 MY -0.02 MZ 0
2351. 13 FX -3.33 FY 0.59 FZ 0.02 MX 0 MY 0.02 MZ 0
2352. 23 FX -21.24 FY 28.24 FZ 13.53 MX 0 MY -0.63 MZ 0
2353. 17 FX 21.32 FY 1.47 FZ 0 MX 0 MY -0.02 MZ 0
2354. 24 FX -16.53 FY -8.16 FZ -0.04 MX 0 MY 0.01 MZ 0
2355. 18 FX 16.7 FY -8.57 FZ 0.03 MX 0 MY 0 MZ 0
2356. LOAD R10 LOADTYPE LIVE TITLE LIVE LOAD -10KN/SQM (L2)
2357. JOINT LOAD
2358. 19 FX -2.55 FY -171.32 FZ 0.62 MX 0 MY -0.97 MZ 0
2359. 5 FX 1.47 FY -143.23 FZ 0.43 MX 0 MY -0.01 MZ 0
2360. 27 FX -4.07 FY -148.87 FZ 0.34 MX 0 MY 0 MZ 0
2361. 6 FX 1.76 FY -146.81 FZ -0.75 MX 0 MY 0.04 MZ 0
2362. 28 FX -1.96 FY -137.85 FZ 1.98 MX 0 MY -0.35 MZ 0
2363. 14 FX 0.05 FY -139.38 FZ 13.25 MX 0 MY 0 MZ 0
2364. 15 FX -1.12 FY -349.15 FZ 0.68 MX -1.74 MY 0 MZ 1.69
2365. 11 FX -0.01 FY -114.54 FZ -2.75 MX 0 MY 0 MZ 0
2366. 16 FX -1.56 FY -206.13 FZ -5.67 MX -4.95 MY 0 MZ 1.97
2367. 7 FX -0.69 FY -312.61 FZ 0 MX -0.61 MY 0 MZ 0.22
2368. 8 FX 1.15 FY -329.24 FZ -0.07 MX -0.58 MY 0 MZ -1.53
2369. 20 FX -0.17 FY -250.28 FZ -0.84 MX -2.75 MY 0 MZ 0.5
2370. 21 FX 0.86 FY -183.87 FZ -6.19 MX -5.56 MY 0 MZ -0.21
2371. 25 FX 0.63 FY -305.87 FZ -1.55 MX -1.9 MY 0 MZ -0.21
2372. 26 FX -0.5 FY -284.17 FZ -1.47 MX -1.56 MY 0 MZ 0.66
2373. 12 FX 0 FY -28.09 FZ 2.01 MX 0 MY 0.02 MZ 0

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2374. 10 FX -0.51 FY -12.59 FZ 0 MX 0 MY 0 MZ 0
2375. 22 FX -1.79 FY -49.1 FZ 0 MX 0 MY 0 MZ 0
2376. 13 FX 1.73 FY -69.43 FZ 0 MX 0 MY 0 MZ 0
2377. 23 FX 10.05 FY -35.16 FZ -2.53 MX 0 MY 0.02 MZ 0
2378. 17 FX -10.28 FY -95.19 FZ 0 MX 0 MY 0 MZ 0
2379. 24 FX -0.83 FY -11.56 FZ -0.02 MX 0 MY 0 MZ 0
2380. 18 FX 0.84 FY -11.62 FZ -0.02 MX 0 MY 0 MZ 0
2381. LOAD R12 LOADTYPE LIVE  TITLE LIVE  LOAD -15KN/SQM (L3)
2382. JOINT LOAD
2383. 19 FX -1.83 FY -150.76 FZ 0.47 MX 0 MY -0.71 MZ 0
2384. 5 FX 1.5 FY -143.67 FZ 0.42 MX 0 MY -0.01 MZ 0
2385. 27 FX -4.11 FY -150.05 FZ 0.34 MX 0 MY 0 MZ 0
2386. 6 FX 1.78 FY -146.42 FZ -0.6 MX 0 MY 0.05 MZ 0
2387. 28 FX -1.95 FY -137.77 FZ 2.32 MX 0 MY -0.35 MZ 0
2388. 14 FX 0.09 FY -97.69 FZ 9.76 MX 0 MY 0 MZ 0
2389. 15 FX -1.01 FY -326.13 FZ 0.17 MX -1.31 MY 0 MZ 1.53
2390. 11 FX -0.01 FY -90.9 FZ -2.25 MX 0 MY 0 MZ 0
2391. 16 FX -1.42 FY -206.12 FZ -3.77 MX -3.32 MY 0 MZ 1.77
2392. 7 FX -0.56 FY -312.87 FZ 0.38 MX 0.05 MY 0 MZ 0.02
2393. 8 FX 1.25 FY -328.42 FZ 0.22 MX 0.02 MY 0 MZ -1.67
2394. 20 FX -0.15 FY -232.23 FZ -1.46 MX -2.59 MY 0 MZ 0.38
2395. 21 FX 1.08 FY -181.26 FZ -4.84 MX -4.4 MY 0 MZ -0.47
2396. 25 FX 0.77 FY -307.58 FZ -1.42 MX -1.5 MY 0 MZ -0.4
2397. 26 FX -0.38 FY -284.36 FZ -1.4 MX -1.16 MY 0 MZ 0.48
2398. 12 FX 0 FY -26.45 FZ 1.68 MX 0 MY 0.01 MZ 0
2399. 10 FX 0.89 FY -19.84 FZ -0.01 MX 0 MY 0 MZ 0
2400. 22 FX -2.72 FY -72.08 FZ 0 MX 0 MY 0 MZ 0
2401. 13 FX 2.63 FY -101.51 FZ 0 MX 0 MY -0.01 MZ 0
2402. 23 FX 13.72 FY -50.32 FZ -2.92 MX 0 MY 0.02 MZ 0
2403. 17 FX -14.03 FY -133.08 FZ 0 MX 0 MY 0 MZ 0
2404. 24 FX -0.82 FY -11.63 FZ -0.02 MX 0 MY 0 MZ 0
2405. 18 FX 0.84 FY -11.7 FZ -0.02 MX 0 MY 0 MZ 0
2406. LOAD R13 LOADTYPE DEAD  TITLE SELF WEIGHT+SECONDARY STEEL STRUCTURE(D2)
2407. JOINT LOAD
2408. 19 FX -0.43 FY -46.24 FZ 0.07 MX 0 MY -0.11 MZ 0
2409. 5 FX 0.33 FY -40.31 FZ 0.1 MX 0 MY 0 MZ 0
2410. 27 FX -0.87 FY -41.96 FZ 0.08 MX 0 MY 0 MZ 0
2411. 6 FX 0.39 FY -39.21 FZ -0.05 MX 0 MY 0.01 MZ 0
2412. 28 FX -0.42 FY -36.21 FZ 0.84 MX 0 MY -0.07 MZ 0
2413. 14 FX 0.08 FY -26.59 FZ 1.41 MX 0 MY 0 MZ 0
2414. 15 FX -0.19 FY -79.28 FZ 0.43 MX 0.28 MY 0 MZ 0.3
2415. 11 FX 0 FY -33.73 FZ -1.57 MX 0 MY 0 MZ 0
2416. 16 FX -0.29 FY -53.7 FZ -0.66 MX -0.28 MY 0 MZ 0.36
2417. 7 FX -0.09 FY -73.58 FZ 0.15 MX 0.26 MY 0 MZ -0.04
2418. 8 FX 0.28 FY -76.76 FZ 0.11 MX 0.24 MY 0 MZ -0.37
2419. 20 FX -0.05 FY -62.67 FZ -0.07 MX -0.2 MY 0 MZ 0.09
2420. 21 FX 0.25 FY -46.45 FZ -1.15 MX -0.77 MY 0 MZ -0.13
2421. 25 FX 0.02 FY -74.9 FZ -0.51 MX -0.36 MY 0 MZ 0.04
2422. 26 FX -0.08 FY -66.91 FZ -0.5 MX -0.24 MY 0 MZ 0.09
2423. 12 FX 0 FY -4.53 FZ 1.3 MX 0 MY 0 MZ 0
2424. 10 FX 0.21 FY -8.7 FZ 0.02 MX 0 MY 0 MZ 0
2425. 22 FX -0.59 FY -21.72 FZ 0 MX 0 MY 0 MZ 0
2426. 13 FX 0.58 FY -28.57 FZ 0 MX 0 MY 0 MZ 0
2427. 23 FX 4.4 FY -14.75 FZ -0.96 MX 0 MY 0.02 MZ 0
2428. 17 FX -4.48 FY -43.19 FZ 0 MX 0 MY 0 MZ 0
2429. 24 FX -0.33 FY -5.62 FZ -0.01 MX 0 MY 0 MZ 0
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2430. 18 FX 0.34 FY -5.64 FZ -0.01 MX 0 MY 0 MZ 0
2431. LOAD R14 LOADTYPE NONE  TITLE TEMPERATURE -20(T -VE)
2432. JOINT LOAD
2433. 19 FX 13.41 FY 21.58 FZ -0.32 MX 0 MY 0.16 MZ 0
2434. 5 FX -11.56 FY 15.5 FZ 2.28 MX 0 MY 0.07 MZ 0
2435. 27 FX 19.54 FY 16.27 FZ 2.29 MX 0 MY -0.01 MZ 0
2436. 6 FX -11.22 FY -2.89 FZ -11.64 MX 0 MY -0.18 MZ 0
2437. 28 FX 14.56 FY 5.72 FZ -7.82 MX 0 MY 1.57 MZ 0
2438. 14 FX -0.09 FY -19.91 FZ -1.56 MX 0 MY -0.03 MZ 0
2439. 15 FX 1.12 FY -10.4 FZ -14.99 MX -18.82 MY 0 MZ -4.26
2440. 11 FX 0.03 FY -6.19 FZ -0.05 MX 0 MY -0.01 MZ 0
2441. 16 FX 4.02 FY 0.13 FZ 14.96 MX 20.23 MY 0 MZ -5.89
2442. 7 FX -25.83 FY -4.88 FZ -9.18 MX -11.99 MY 0 MZ 37.03
2443. 8 FX -24.46 FY -5.62 FZ 14.07 MX 22.02 MY 0 MZ 35.1
2444. 20 FX 9.78 FY -7.55 FZ -12.49 MX -16.05 MY 0 MZ -17.77
2445. 21 FX 14.07 FY -3.82 FZ 16.83 MX 22.7 MY 0 MZ -20
2446. 25 FX 35.09 FY -1.76 FZ -7.66 MX -10.02 MY 0 MZ -52.1
2447. 26 FX 35.47 FY -2.37 FZ 14.73 MX 23.46 MY 0 MZ -50.93
2448. 12 FX -0.03 FY -55.81 FZ 0.66 MX 0 MY -0.2 MZ 0
2449. 10 FX -26.7 FY 32.6 FZ -0.09 MX 0 MY 0 MZ 0
2450. 22 FX -1.59 FY -23.16 FZ -0.02 MX 0 MY -0.03 MZ 0
2451. 13 FX 1.7 FY -0.19 FZ 0.02 MX 0 MY 0.03 MZ 0
2452. 23 FX -1.1 FY 36.8 FZ 21.7 MX 0 MY -0.29 MZ 0
2453. 17 FX 1.2 FY 0.24 FZ 0.03 MX 0 MY -0.03 MZ 0
2454. 24 FX -1.25 FY -0.31 FZ -0.06 MX 0 MY 0.01 MZ 0
2455. 18 FX 1.55 FY -0.33 FZ 0.05 MX 0 MY 0 MZ 0
2456. LOAD R15 LOADTYPE SEISMIC  TITLE MAX SEISMIC FORCE
2457. JOINT LOAD
2458. 13 FX 3.11 FY -52.1 FZ -0.03 MX 0 MY 0.01 MZ 0
2459. 17 FX -9.51 FY -80.9 FZ -0.01 MX 0 MY 0.03 MZ 0
2460. 18 FX 0.83 FY -6.64 FZ -0.04 MX 0 MY 0 MZ 0
2461. 22 FX -0.71 FY -35.54 FZ -0.01 MX 0 MY 0.02 MZ 0
2462. 23 FX 17.18 FY -18.69 FZ -6.98 MX 0 MY 0.22 MZ 0
2463. 24 FX -0.18 FY -6.42 FZ -0.02 MX 0 MY 0 MZ 0
2464. 5 FX 2.18 FY 5.1 FZ -0.11 MX 0 MY 0.01 MZ 0
2465. 6 FX 2.39 FY 4.5 FZ -5.84 MX 0 MY 0.13 MZ 0
2466. 7 FX 4.81 FY 2.19 FZ -3.85 MX 4.71 MY 0.04 MZ -6.95
2467. 8 FX 5.78 FY 5.79 FZ -3.35 MX 4.09 MY 0.05 MZ -8.2
2468. 10 FX 13.73 FY 15.97 FZ -0.12 MX 0 MY 0.02 MZ 0
2469. 11 FX 0.02 FY 1.98 FZ -3.53 MX 0 MY 0.01 MZ 0
2470. 12 FX 0.01 FY 21.73 FZ -1.73 MX 0 MY 0.05 MZ 0
2471. 14 FX 0.16 FY 27.14 FZ -0.09 MX 0 MY 0.01 MZ 0
2472. 15 FX 5.5 FY 1.1 FZ -3.14 MX 3.1 MY 0 MZ -7.23
2473. 16 FX 4.83 FY 1.86 FZ -7.05 MX 2.67 MY 0 MZ -6.89
2474. 19 FX 16.29 FY 27.19 FZ -0.04 MX 0 MY 0.01 MZ 0
2475. 20 FX 5.54 FY 1.73 FZ -2.87 MX 1.87 MY 0 MZ -7.22
2476. 21 FX 5.57 FY 0.85 FZ -6.84 MX 1.96 MY 0 MZ -7.14
2477. 25 FX 4.79 FY 1.26 FZ -3.31 MX 2.32 MY 0.05 MZ -6.72
2478. 26 FX 4.81 FY 1.2 FZ -3.15 MX 2.1 MY 0.03 MZ -6.9
2479. 27 FX 1.41 FY 5.22 FZ -0.07 MX 0 MY 0.01 MZ 0
2480. 28 FX 1.07 FY 2.99 FZ -4.06 MX 0 MY 0.12 MZ 0
2481. LOAD R16 LOADTYPE SEISMIC  TITLE MIN SEISMIC FORCE
2482. JOINT LOAD
2483. 13 FX -0.92 FY -89.66 FZ 0.03 MX 0 MY -0.02 MZ 0
2484. 17 FX -15.33 FY -138.14 FZ 0.02 MX 0 MY -0.02 MZ 0
2485. 18 FX 0.4 FY -12.65 FZ 0.01 MX 0 MY 0 MZ 0
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2486. 22 FX -1.66 FY -69.13 FZ 0.02 MX 0 MY -0.01 MZ 0
2487. 23 FX 7.14 FY -50.72 FZ 2.42 MX 0 MY -0.14 MZ 0
2488. 24 FX -0.99 FY -12.78 FZ -0.01 MX 0 MY 0 MZ 0
2489. 5 FX -1.08 FY -123.86 FZ 0.46 MX 0 MY -0.02 MZ 0
2490. 6 FX -1.06 FY -126.86 FZ 4.98 MX 0 MY -0.09 MZ 0
2491. 7 FX -5.42 FY -247.31 FZ 3.76 MX -5.37 MY -0.04 MZ 7.24
2492. 8 FX -4.97 FY -262.45 FZ 3.21 MX -4.71 MY -0.05 MZ 7.12
2493. 10 FX -13.69 FY -32.81 FZ 0.12 MX 0 MY -0.02 MZ 0
2494. 11 FX -0.03 FY -132.6 FZ 0.36 MX 0 MY -0.01 MZ 0
2495. 12 FX 0 FY -49.86 FZ 4.08 MX 0 MY -0.03 MZ 0
2496. 14 FX -0.05 FY -174.29 FZ 13.35 MX 0 MY -0.01 MZ 0
2497. 15 FX -6.43 FY -299.3 FZ 3.25 MX -5.04 MY 0 MZ 8.63
2498. 16 FX -6.1 FY -170.87 FZ 2.29 MX -7.06 MY 0 MZ 8.52
2499. 19 FX -18.92 FY -200.22 FZ 0.66 MX 0 MY -0.99 MZ 0
2500. 20 FX -5.73 FY -226.9 FZ 1.54 MX -4.6 MY 0 MZ 7.71
2501. 21 FX -5.01 FY -149.98 FZ 1.67 MX -6.73 MY 0 MZ 7.12
2502. 25 FX -4.62 FY -245.36 FZ 1.72 MX -4.23 MY -0.05 MZ 6.89
2503. 26 FX -5.34 FY -223.57 FZ 1.64 MX -3.67 MY -0.03 MZ 7.62
2504. 27 FX -4.53 FY -129.04 FZ 0.34 MX 0 MY 0 MZ 0
2505. 28 FX -2.6 FY -116.24 FZ 5.74 MX 0 MY -0.39 MZ 0
2506. END DEFINE REFERENCE LOADS
2507. **D
2508. LOAD 101 (D+D1+D2)
2509. REFERENCE LOAD
2510. R7 1.0 R13 1.0 R1 1.0
2511. **D+T+L
2512. LOAD 102 (D+D1+D2)+T(+VE)+L1
2513. REFERENCE LOAD
2514. R7 1.0 R13 1.0 R8 1.0 R2 1.0 R1 1.0
2515. LOAD 103 (D+D1+D2)+T(+VE)+L2
2516. REFERENCE LOAD
2517. R7 1.0 R13 1.0 R8 1.0 R10 1.0 R1 1.0
2518. LOAD 104 (D+D1+D2)+T(+VE)+L3
2519. REFERENCE LOAD
2520. R7 1.0 R13 1.0 R8 1.0 R12 1.0 R1 1.0
2521. *
2522. LOAD 105 (D+D1+D2)+T(-VE)+L1
2523. REFERENCE LOAD
2524. R7 1.0 R13 1.0 R14 1.0 R2 1.0 R1 1.0
2525. LOAD 106 (D+D1+D2)+T(-VE)+L2
2526. REFERENCE LOAD
2527. R7 1.0 R13 1.0 R14 1.0 R10 1.0 R1 1.0
2528. LOAD 107 (D+D1+D2)+T(-VE)+L3
2529. REFERENCE LOAD
2530. R7 1.0 R13 1.0 R14 1.0 R12 1.0 R1 1.0
2531. **D+W
2532. LOAD 108 D+D1+D2+W(+X)
2533. REFERENCE LOAD
2534. R7 1.0 R13 1.0 R3 1.0 R1 1.0
2535. LOAD 109 D+D1+D2+W(-X)
2536. REFERENCE LOAD
2537. R7 1.0 R13 1.0 R4 1.0 R1 1.0
2538. LOAD 110 LOADTYPE NONE  TITLE D+D1+D2+W(-Z)
2539. REFERENCE LOAD
2540. R7 1.0 R13 1.0 R5 1.0 R1 1.0
2541. LOAD 111 LOADTYPE NONE  TITLE D+D1+D2+W(+Z)
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2542. REFERENCE LOAD
2543. R7 1.0 R13 1.0 R6 1.0 R1 1.0
2544. **D+0.75W+0.75L
2545. LOAD 112 D+D1+D2+0.75L1+0.75W(+X)
2546. REFERENCE LOAD
2547. R7 1.0 R13 1.0 R2 0.75 R3 0.75 R1 1.0
2548. LOAD 113 D+D1+D2+0.75L1+0.75W(-X)
2549. REFERENCE LOAD
2550. R7 1.0 R13 1.0 R2 0.75 R4 0.75 R1 1.0
2551. LOAD 114 LOADTYPE NONE  TITLE D+D1+D2+0.75L1+0.75W(-Z)
2552. REFERENCE LOAD
2553. R7 1.0 R13 1.0 R2 0.75 R5 0.75 R1 1.0
2554. LOAD 115 LOADTYPE NONE  TITLE D+D1+D2+0.75L1+0.75W(+Z)
2555. REFERENCE LOAD
2556. R7 1.0 R13 1.0 R2 0.75 R6 0.75 R1 1.0
2557. *
2558. LOAD 116 D+D1+D2+0.75L2+0.75W(+X)
2559. REFERENCE LOAD
2560. R7 1.0 R13 1.0 R10 0.75 R3 0.75 R1 1.0
2561. LOAD 117 D+D1+D2+0.75L2+0.75W(-X)
2562. REFERENCE LOAD
2563. R7 1.0 R13 1.0 R10 0.75 R4 0.75 R1 1.0
2564. LOAD 118 LOADTYPE NONE  TITLE D+D1+D2+0.75L2+0.75W(-Z)
2565. REFERENCE LOAD
2566. R7 1.0 R13 1.0 R10 0.75 R5 0.75 R1 1.0
2567. LOAD 119 LOADTYPE NONE  TITLE D+D1+D2+0.75L2+0.75W(+Z)
2568. REFERENCE LOAD
2569. R7 1.0 R13 1.0 R10 0.75 R6 0.75 R1 1.0
2570. *
2571. LOAD 120 D+D1+D2+0.75L3+0.75W(+X)
2572. REFERENCE LOAD
2573. R7 1.0 R13 1.0 R12 0.75 R3 0.75 R1 1.0
2574. LOAD 121 D+D1+D2+0.75L3+0.75W(-X)
2575. REFERENCE LOAD
2576. R7 1.0 R13 1.0 R12 0.75 R4 0.75 R1 1.0
2577. LOAD 122 LOADTYPE NONE  TITLE D+D1+D2+0.75L3+0.75W(-Z)
2578. REFERENCE LOAD
2579. R7 1.0 R13 1.0 R12 0.75 R5 0.75 R1 1.0
2580. LOAD 123 LOADTYPE NONE  TITLE D+D1+D2+0.75L3+0.75W(+Z)
2581. REFERENCE LOAD
2582. R7 1.0 R13 1.0 R12 0.75 R6 0.75 R1 1.0
2583. **0.6D+W
2584. LOAD 124 0.6(D1+D2)+W(+X)+D
2585. REFERENCE LOAD
2586. R7 0.6 R13 0.6 R3 1.0 R1 1.0
2587. LOAD 125 0.6(D1+D2)+W(-X)+D
2588. REFERENCE LOAD
2589. R7 0.6 R13 0.6 R4 1.0 R1 1.0
2590. LOAD 126 LOADTYPE NONE  TITLE 0.6(D1+D2)+W(-Z)+D
2591. REFERENCE LOAD
2592. R7 0.6 R13 0.6 R5 1.0 R1 1.0
2593. LOAD 127 LOADTYPE NONE  TITLE 0.6(D1+D2)+W(+Z)+D
2594. REFERENCE LOAD
2595. R7 0.6 R13 0.6 R6 1.0 R1 1.0
2596. **D+0.7E
2597. LOAD 128 D+D1+D2+0.7E_MAX
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2598. REFERENCE LOAD
2599. R7 1.0 R13 1.0 R15 0.7 R1 1.0
2600. LOAD 129 D+D1+D2+0.7E_MIN
2601. REFERENCE LOAD
2602. R7 1.0 R13 1.0 R16 0.7 R1 1.0
2603. **D+0.75*0.7E+0.75L
2604. LOAD 130 D+D1+D2+0.75L1+0.75*0.7E_MAX
2605. REFERENCE LOAD
2606. R7 1.0 R13 1.0 R2 0.75 R15 0.525 R1 1.0
2607. LOAD 131 D+D1+D2+0.75L1+0.75*0.7E_MIN
2608. REFERENCE LOAD
2609. R7 1.0 R13 1.0 R2 0.75 R16 0.525 R1 1.0
2610. LOAD 132 D+D1+D2+0.75L2+0.75*0.7E_MAX
2611. REFERENCE LOAD
2612. R7 1.0 R13 1.0 R10 0.75 R15 0.525 R1 1.0
2613. LOAD 133 D+D1+D2+0.75L2+0.75*0.7E_MIN
2614. REFERENCE LOAD
2615. R7 1.0 R13 1.0 R10 0.75 R16 0.525 R1 1.0
2616. LOAD 134 D+D1+D2+0.75L2+0.75*0.7E_MAX
2617. REFERENCE LOAD
2618. R7 1.0 R13 1.0 R12 0.75 R15 0.525 R1 1.0
2619. LOAD 135 D+D1+D2+0.75L2+0.75*0.7E_MIN
2620. REFERENCE LOAD
2621. R7 1.0 R13 1.0 R12 0.75 R16 0.525 R1 1.0
2622. **0.6D+W
2623. LOAD 136 0.6(D1+D2)+0.7E_MAX+D
2624. REFERENCE LOAD
2625. R7 0.6 R13 0.6 R15 0.7 R1 1.0
2626. LOAD 137 0.6(D1+D2)+0.7E_MIN+D
2627. REFERENCE LOAD
2628. R7 0.6 R13 0.6 R16 0.7 R1 1.0
2629. **FACTORED LOAD COMBINATIONS
2630. **1.4D
2631. LOAD 201 1.4D+1.4(D1+D2)
2632. REFERENCE LOAD
2633. R7 1.4 R13 1.4 R1 1.4
2634. **1.4(D+T)+1.7L
2635. LOAD 202 1.4(D+D1+D2)+1.4T(+VE)+1.7L1
2636. REFERENCE LOAD
2637. R7 1.4 R13 1.4 R8 1.4 R2 1.7 R1 1.4
2638. LOAD 203 1.4(D+D1+D2))+1.4T(+VE)+1.7L2
2639. REFERENCE LOAD
2640. R7 1.4 R13 1.4 R8 1.4 R10 1.7 R1 1.4
2641. LOAD 204 1.4(D+D1+D2)+1.4T(+VE)+1.7L3
2642. REFERENCE LOAD
2643. R7 1.4 R13 1.4 R8 1.4 R12 1.7 R1 1.4
2644. *
2645. LOAD 205 1.4(D+D1+D2)+1.4T(-VE)+1.7L1
2646. REFERENCE LOAD
2647. R7 1.4 R13 1.4 R14 1.4 R2 1.7 R1 1.4
2648. LOAD 206 1.4(D+D1+D2)+1.4T(-VE)+1.7L2
2649. REFERENCE LOAD
2650. R7 1.4 R13 1.4 R14 1.4 R10 1.7 R1 1.4
2651. LOAD 207 1.4(D+D1+D2)+1.4T(-VE)+1.7L3
2652. REFERENCE LOAD
2653. R7 1.4 R13 1.4 R14 1.4 R12 1.7 R1 1.4
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2654. **1.2D+0.8W
2655. LOAD 208 1.2(D+D1+D2)+0.8W(+X)
2656. REFERENCE LOAD
2657. R7 1.2 R13 1.2 R3 0.8 R1 1.2
2658. LOAD 209 1.2(D+D1+D2)+0.8W(-X)
2659. REFERENCE LOAD
2660. R7 1.2 R13 1.2 R4 0.8 R1 1.2
2661. LOAD 210 LOADTYPE NONE  TITLE 1.2(D+D1+D2)+0.8W(-Z)
2662. REFERENCE LOAD
2663. R7 1.2 R13 1.2 R5 0.8 R1 1.2
2664. LOAD 211 LOADTYPE NONE  TITLE 1.2(D+D1+D2)+0.8W(+Z)
2665. REFERENCE LOAD
2666. R7 1.2 R13 1.2 R6 0.8 R1 1.2
2667. ***1.2D+1.6W+1.0L
2668. LOAD 212 1.2(D+D1+D2)+1.0L1+1.6W(+X)
2669. REFERENCE LOAD
2670. R7 1.2 R13 1.2 R2 1.0 R3 1.6 R1 1.2
2671. LOAD 213 1.2(D+D1+D2)+1.0L1+1.6W(-X)
2672. REFERENCE LOAD
2673. R7 1.2 R13 1.2 R2 1.0 R4 1.6 R1 1.2
2674. LOAD 214 LOADTYPE NONE  TITLE 1.2(D+D1+D2)+1.0L1+1.6W(-Z)
2675. REFERENCE LOAD
2676. R7 1.2 R13 1.2 R2 1.0 R5 1.6 R1 1.2
2677. LOAD 215 LOADTYPE NONE  TITLE 1.2(D+D1+D2)+1.0L1+1.6W(+Z)
2678. REFERENCE LOAD
2679. R7 1.2 R13 1.2 R2 1.0 R6 1.6 R1 1.2
2680. *
2681. LOAD 216 1.2(D+D1+D2)+1.0L2+1.6W(+X)
2682. REFERENCE LOAD
2683. R7 1.2 R13 1.2 R10 1.0 R3 1.6 R1 1.2
2684. LOAD 217 1.2(D+D1+D2)+1.0L2+1.6W(-X)
2685. REFERENCE LOAD
2686. R7 1.2 R13 1.2 R10 1.0 R4 1.6 R1 1.2
2687. LOAD 218 LOADTYPE NONE  TITLE 1.2(D+D1+D2)+1.0L2+1.6W(-Z)
2688. REFERENCE LOAD
2689. R7 1.2 R13 1.2 R10 1.0 R5 1.6 R1 1.2
2690. LOAD 219 LOADTYPE NONE  TITLE 1.2(D+D1+D2)+1.0L2+1.6W(+Z)
2691. REFERENCE LOAD
2692. R7 1.2 R13 1.2 R10 1.0 R6 1.6 R1 1.2
2693. *
2694. LOAD 220 1.2(D+D1+D2)+1.0L3+1.6W(+X)
2695. REFERENCE LOAD
2696. R7 1.2 R13 1.2 R12 1.0 R3 1.6 R1 1.2
2697. LOAD 221 1.2(D+D1+D2)+1.0L3+1.6W(-X)
2698. REFERENCE LOAD
2699. R7 1.2 R13 1.2 R12 1.0 R4 1.6 R1 1.2
2700. LOAD 222 LOADTYPE NONE  TITLE 1.2(D+D1+D2)+1.0L3+1.6W(-Z)
2701. REFERENCE LOAD
2702. R7 1.2 R13 1.2 R12 1.0 R5 1.6 R1 1.2
2703. LOAD 223 LOADTYPE NONE  TITLE 1.2(D+D1+D2)+1.0L3+1.6W(+Z)
2704. REFERENCE LOAD
2705. R7 1.2 R13 1.2 R12 1.0 R6 1.6 R1 1.2
2706. **0.9D+1.6W
2707. LOAD 224 0.9(D1+D2)+W(+X)+D
2708. REFERENCE LOAD
2709. R7 0.9 R13 0.9 R3 1.6 R1 1.0
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2710. LOAD 225 0.9(D1+D2)+W(-X)+D
2711. REFERENCE LOAD
2712. R7 0.9 R13 0.9 R4 1.6 R1 1.0
2713. LOAD 226 LOADTYPE NONE  TITLE 0.9(D1+D2)+W(-Z)+D
2714. REFERENCE LOAD
2715. R7 0.9 R13 0.9 R5 1.6 R1 1.0
2716. LOAD 227 LOADTYPE NONE  TITLE 0.9(D1+D2)+W(+Z)+D
2717. REFERENCE LOAD
2718. R7 0.9 R13 0.9 R6 1.6 R1 1.0
2719. ***1.2D+1.0E_MAXW+1.0L
2720. LOAD 228 1.2(D+D1+D2)+1.0L1+1.0E_MAX
2721. REFERENCE LOAD
2722. R7 1.2 R13 1.2 R2 1.0 R15 1.0 R1 1.2
2723. LOAD 229 1.2(D+D1+D2)+1.0L1+1.0E_MIN
2724. REFERENCE LOAD
2725. R7 1.2 R13 1.2 R2 1.0 R16 1.0 R1 1.2
2726. LOAD 230 1.2(D+D1+D2)+1.0L2+1.0E_MAX
2727. REFERENCE LOAD
2728. R7 1.2 R13 1.2 R10 1.0 R15 1.0 R1 1.2
2729. LOAD 231 1.2(D+D1+D2)+1.0L2+1.0E_MIN
2730. REFERENCE LOAD
2731. R7 1.2 R13 1.2 R10 1.0 R16 1.0 R1 1.2
2732. LOAD 232 1.2(D+D1+D2)+1.0L3+1.0E_MAX
2733. REFERENCE LOAD
2734. R7 1.2 R13 1.2 R12 1.0 R15 1.0 R1 1.2
2735. LOAD 233 1.2(D+D1+D2)+1.0L3+1.0E_MIN
2736. REFERENCE LOAD
2737. R7 1.2 R13 1.2 R12 1.0 R16 1.0 R1 1.2
2738. **0.9D+1.0E
2739. LOAD 234 0.9(D1+D2)+1.0E_MAX+D
2740. REFERENCE LOAD
2741. R7 0.9 R13 0.9 R15 1.0 R1 1.0
2742. LOAD 235 0.9(D1+D2)+1.0E_MIN+D
2743. REFERENCE LOAD
2744. R7 0.9 R13 0.9 R16 1.0 R1 1.0
2745. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2686 NUMBER OF MEMBERS 23
NUMBER OF PLATES 2336 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2663

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 2568/ 40/ 123 DOF
TOTAL PRIMARY LOAD CASES = 72, TOTAL DEGREES OF FREEDOM = 8127
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 1000 DOUBLE KILO-WORDS

REQRD/AVAIL. DISK SPACE = 42.8/ 193886.0 MB

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

*** LOAD CASE 102 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

*** LOAD CASE 105 -- START ITERATION NO. 2
*** LOAD CASE 105 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 105

*** LOAD CASE 106 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 106

*** LOAD CASE 107 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

*** LOAD CASE 111 -- START ITERATION NO. 2
*** LOAD CASE 111 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

**NOTE-Tension/Compression converged after 1 iterations, Case= 118

**NOTE-Tension/Compression converged after 1 iterations, Case= 119

**NOTE-Tension/Compression converged after 1 iterations, Case= 120

**NOTE-Tension/Compression converged after 1 iterations, Case= 121

**NOTE-Tension/Compression converged after 1 iterations, Case= 122

**NOTE-Tension/Compression converged after 1 iterations, Case= 123

**NOTE-Tension/Compression converged after 1 iterations, Case= 124

**NOTE-Tension/Compression converged after 1 iterations, Case= 126

*** LOAD CASE 127 -- START ITERATION NO. 2
*** LOAD CASE 127 -- START ITERATION NO. 3
*** LOAD CASE 127 -- START ITERATION NO. 4
*** LOAD CASE 127 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 127

**NOTE-Tension/Compression converged after 1 iterations, Case= 128

**NOTE-Tension/Compression converged after 1 iterations, Case= 129

**NOTE-Tension/Compression converged after 1 iterations, Case= 130

**NOTE-Tension/Compression converged after 1 iterations, Case= 131

**NOTE-Tension/Compression converged after 1 iterations, Case= 132

**NOTE-Tension/Compression converged after 1 iterations, Case= 133

**NOTE-Tension/Compression converged after 1 iterations, Case= 134

**NOTE-Tension/Compression converged after 1 iterations, Case= 135

**NOTE-Tension/Compression converged after 1 iterations, Case= 136

**NOTE-Tension/Compression converged after 1 iterations, Case= 137

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

*** LOAD CASE 203 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

*** LOAD CASE 205 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 205

*** LOAD CASE 206 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 206

*** LOAD CASE 207 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

*** LOAD CASE 215 -- START ITERATION NO. 2
*** LOAD CASE 215 -- START ITERATION NO. 3
*** LOAD CASE 215 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 216

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

*** LOAD CASE 219 -- START ITERATION NO. 2
*** LOAD CASE 219 -- START ITERATION NO. 3
*** LOAD CASE 219 -- START ITERATION NO. 4
*** LOAD CASE 219 -- START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 219

**NOTE-Tension/Compression converged after 1 iterations, Case= 220

**NOTE-Tension/Compression converged after 1 iterations, Case= 221

**NOTE-Tension/Compression converged after 1 iterations, Case= 222

*** LOAD CASE 223 -- START ITERATION NO. 2
*** LOAD CASE 223 -- START ITERATION NO. 3
*** LOAD CASE 223 -- START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 223

*** LOAD CASE 224 -- START ITERATION NO. 2
*** LOAD CASE 224 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 224

**NOTE-Tension/Compression converged after 1 iterations, Case= 225

*** LOAD CASE 226 -- START ITERATION NO. 2
*** LOAD CASE 226 -- START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 226

*** LOAD CASE 227 -- START ITERATION NO. 2
*** LOAD CASE 227 -- START ITERATION NO. 3
*** LOAD CASE 227 -- START ITERATION NO. 4
*** LOAD CASE 227 -- START ITERATION NO. 5
*** LOAD CASE 227 -- START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 227

**NOTE-Tension/Compression converged after 1 iterations, Case= 228

**NOTE-Tension/Compression converged after 1 iterations, Case= 229

**NOTE-Tension/Compression converged after 1 iterations, Case= 230

**NOTE-Tension/Compression converged after 1 iterations, Case= 231

**NOTE-Tension/Compression converged after 1 iterations, Case= 232

**NOTE-Tension/Compression converged after 1 iterations, Case= 233

**NOTE-Tension/Compression converged after 1 iterations, Case= 234

**NOTE-Tension/Compression converged after 1 iterations, Case= 235

2746. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 20,2020 TIME= 14:36:23 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

ATTACHMENT 3 STAAD OUTPUT : MD-216-3510-EG-CV-CAL-1281_STAIR IN CCD AREA

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 30, 2020                *
*           Time=    16:25:39                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\1.CCD _Neutralization area\Pipe rack\CCD area\design\A02.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 20-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 101 3.5 0.4 6; 102 3.5 0.4 0; 103 0 0.4 6; 104 0 0.4 0; 105 3.5 -2 6
     9. 106 3.5 -2 0; 107 0 -2 6; 108 0 -2 0; 109 0.35 -2 0; 110 0 -2 0.35
    10. 111 0.35 -2 0.35; 112 0.6 -2 0; 113 0.6 -2 0.35; 114 0.75 -2 0
    11. 115 0.75 -2 0.35; 116 -0.35 -2 0; 117 -0.35 -2 0.35; 118 -0.6 -2 0
    12. 119 -0.6 -2 0.35; 120 -0.75 -2 0; 121 -0.75 -2 0.35; 122 0 -2 0.6
    13. 123 0.35 -2 0.6; 124 0.6 -2 0.6; 125 0.75 -2 0.6; 126 -0.35 -2 0.6
    14. 127 -0.6 -2 0.6; 128 -0.75 -2 0.6; 129 0 -2 0.75; 130 0.35 -2 0.75
    15. 131 0.6 -2 0.75; 132 0.75 -2 0.75; 133 -0.35 -2 0.75; 134 -0.6 -2 0.75
    16. 135 -0.75 -2 0.75; 136 0 -2 -0.35; 137 0.35 -2 -0.35; 138 0.6 -2 -0.35
    17. 139 0.75 -2 -0.35; 140 -0.35 -2 -0.35; 141 -0.6 -2 -0.35; 142 -0.75 -2 -0.35
    18. 143 0 -2 -0.6; 144 0.35 -2 -0.6; 145 0.6 -2 -0.6; 146 0.75 -2 -0.6
    19. 147 -0.35 -2 -0.6; 148 -0.6 -2 -0.6; 149 -0.75 -2 -0.6; 150 0 -2 -0.75
    20. 151 0.35 -2 -0.75; 152 0.6 -2 -0.75; 153 0.75 -2 -0.75; 154 -0.35 -2 -0.75
    21. 155 -0.6 -2 -0.75; 156 -0.75 -2 -0.75; 205 3.5 -2 0.35; 206 3.85 -2 0.35
    22. 207 3.85 -2 0; 208 4.1 -2 0.35; 209 4.1 -2 0; 210 4.25 -2 0.35; 211 4.25 -2 0
    23. 212 3.15 -2 0.35; 213 3.15 -2 0; 214 2.9 -2 0.35; 215 2.9 -2 0
    24. 216 2.75 -2 0.35; 217 2.75 -2 0; 218 3.5 -2 0.6; 219 3.85 -2 0.6
    25. 220 4.1 -2 0.6; 221 4.25 -2 0.6; 222 3.15 -2 0.6; 223 2.9 -2 0.6
    26. 224 2.75 -2 0.6; 225 3.5 -2 0.75; 226 3.85 -2 0.75; 227 4.1 -2 0.75
    27. 228 4.25 -2 0.75; 229 3.15 -2 0.75; 230 2.9 -2 0.75; 231 2.75 -2 0.75
    28. 232 3.85 -2 -0.35; 233 3.5 -2 -0.35; 234 4.1 -2 -0.35; 235 4.25 -2 -0.35
    29. 236 3.15 -2 -0.35; 237 2.9 -2 -0.35; 238 2.75 -2 -0.35; 239 3.85 -2 -0.6
    30. 240 3.5 -2 -0.6; 241 4.1 -2 -0.6; 242 4.25 -2 -0.6; 243 3.15 -2 -0.6
    31. 244 2.9 -2 -0.6; 245 2.75 -2 -0.6; 246 3.85 -2 -0.75; 247 3.5 -2 -0.75
    32. 248 4.1 -2 -0.75; 249 4.25 -2 -0.75; 250 3.15 -2 -0.75; 251 2.9 -2 -0.75
    33. 252 2.75 -2 -0.75; 301 1 -2 0.75; 302 1 -2 0.6; 303 1.25 -2 0.75
    34. 304 1.25 -2 0.6; 305 1.5 -2 0.75; 306 1.5 -2 0.6; 307 1.75 -2 0.75
    35. 308 1.75 -2 0.6; 309 2 -2 0.75; 310 2 -2 0.6; 311 2.25 -2 0.75
    36. 312 2.25 -2 0.6; 313 2.5 -2 0.75; 314 2.5 -2 0.6; 315 1 -2 0.35
    37. 316 1.25 -2 0.35; 317 1.5 -2 0.35; 318 1.75 -2 0.35; 319 2 -2 0.35
    38. 320 2.25 -2 0.35; 321 2.5 -2 0.35; 322 1 -2 0; 323 1.25 -2 0; 324 1.5 -2 0
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39. 325 1.75 -2 0; 326 2 -2 0; 327 2.25 -2 0; 328 2.5 -2 0; 329 1 -2 -0.35
40. 330 1.25 -2 -0.35; 331 1.5 -2 -0.35; 332 1.75 -2 -0.35; 333 2 -2 -0.35
41. 334 2.25 -2 -0.35; 335 2.5 -2 -0.35; 336 1 -2 -0.6; 337 1.25 -2 -0.6
42. 338 1.5 -2 -0.6; 339 1.75 -2 -0.6; 340 2 -2 -0.6; 341 2.25 -2 -0.6
43. 342 2.5 -2 -0.6; 343 1 -2 -0.75; 344 1.25 -2 -0.75; 345 1.5 -2 -0.75
44. 346 1.75 -2 -0.75; 347 2 -2 -0.75; 348 2.25 -2 -0.75; 349 2.5 -2 -0.75
45. 399 0 -2 1; 400 0.35 -2 1; 401 0.6 -2 1; 402 0.75 -2 1; 403 -0.35 -2 1
46. 404 -0.6 -2 1; 405 -0.75 -2 1; 406 3.5 -2 1; 407 3.85 -2 1; 408 4.1 -2 1
47. 409 4.25 -2 1; 410 3.15 -2 1; 411 2.9 -2 1; 412 2.75 -2 1; 413 1 -2 1
48. 414 1.25 -2 1; 415 1.5 -2 1; 416 1.75 -2 1; 417 2 -2 1; 418 2.25 -2 1
49. 419 2.5 -2 1; 420 0 -2 -1; 421 0.35 -2 -1; 422 0.6 -2 -1; 423 0.75 -2 -1.
50. 424 -0.35 -2 -1; 425 -0.6 -2 -1; 426 -0.75 -2 -1; 427 3.85 -2 -1.
51. 428 3.5 -2 -1; 429 4.1 -2 -1; 430 4.25 -2 -1; 431 3.15 -2 -1; 432 2.9 -2 -1.
52. 433 2.75 -2 -1; 434 1 -2 -1; 435 1.25 -2 -1; 436 1.5 -2 -1; 437 1.75 -2 -1.
53. 438 2 -2 -1; 439 2.25 -2 -1; 440 2.5 -2 -1; 441 0 -2 6.35; 442 0.35 -2 6.35
54. 443 0.35 -2 6; 444 0.6 -2 6.35; 445 0.6 -2 6; 446 0.75 -2 6.35; 447 0.75 -2 6
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56. 452 -0.75 -2 6.35; 453 -0.75 -2 6; 454 0 -2 6.6; 455 0.35 -2 6.6
57. 456 0.6 -2 6.6; 457 0.75 -2 6.6; 458 -0.35 -2 6.6; 459 -0.6 -2 6.6
58. 460 -0.75 -2 6.6; 461 0 -2 6.75; 462 0.35 -2 6.75; 463 0.6 -2 6.75
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62. 476 0 -2 5.4; 477 0.6 -2 5.4; 478 0.75 -2 5.4; 479 -0.35 -2 5.4
63. 480 -0.6 -2 5.4; 481 -0.75 -2 5.4; 482 0.35 -2 5.25; 483 0 -2 5.25
64. 484 0.6 -2 5.25; 485 0.75 -2 5.25; 486 -0.35 -2 5.25; 487 -0.6 -2 5.25
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66. 492 4.1 -2 6.35; 493 4.1 -2 6; 494 4.25 -2 6.35; 495 4.25 -2 6
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70. 508 2.75 -2 6.6; 509 3.5 -2 6.75; 510 3.85 -2 6.75; 511 4.1 -2 6.75
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74. 524 3.5 -2 5.4; 525 4.1 -2 5.4; 526 4.25 -2 5.4; 527 3.15 -2 5.4
75. 528 2.9 -2 5.4; 529 2.75 -2 5.4; 530 3.85 -2 5.25; 531 3.5 -2 5.25
76. 532 4.1 -2 5.25; 533 4.25 -2 5.25; 534 3.15 -2 5.25; 535 2.9 -2 5.25
77. 536 2.75 -2 5.25; 537 1 -2 6.75; 538 1 -2 6.6; 539 1.25 -2 6.75
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79. 544 1.75 -2 6.6; 545 2 -2 6.75; 546 2 -2 6.6; 547 2.25 -2 6.75
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86. 574 1.5 -2 5.4; 575 1.75 -2 5.4; 576 2 -2 5.4; 577 2.25 -2 5.4; 578 2.5 -2 5.4
87. 579 1 -2 5.25; 580 1.25 -2 5.25; 581 1.5 -2 5.25; 582 1.75 -2 5.25
88. 583 2 -2 5.25; 584 2.25 -2 5.25; 585 2.5 -2 5.25; 586 0 -2 7; 587 0.35 -2 7
89. 588 0.6 -2 7; 589 0.75 -2 7; 590 -0.35 -2 7; 591 -0.6 -2 7; 592 -0.75 -2 7
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91. 598 2.9 -2 7; 599 2.75 -2 7; 600 1 -2 7; 601 1.25 -2 7; 602 1.5 -2 7
92. 603 1.75 -2 7; 604 2 -2 7; 605 2.25 -2 7; 606 2.5 -2 7; 607 0.35 -2 5
93. 608 0 -2 5; 609 0.6 -2 5; 610 0.75 -2 5; 611 -0.35 -2 5; 612 -0.6 -2 5
94. 613 -0.75 -2 5; 614 3.85 -2 5; 615 3.5 -2 5; 616 4.1 -2 5; 617 4.25 -2 5

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 95. 618 3.15 -2 5; 619 2.9 -2 5; 620 2.75 -2 5; 621 1 -2 5; 622 1.25 -2 5
 96. 623 1.5 -2 5; 624 1.75 -2 5; 625 2 -2 5; 626 2.25 -2 5; 627 2.5 -2 5
 97. MEMBER INCIDENCES
 98. 7 108 104; 8 106 102; 9 105 101; 10 107 103
 99. ELEMENT INCIDENCES SHELL
100. 1 110 111 109 108; 2 111 113 112 109; 3 113 115 114 112; 4 117 110 108 116
101. 5 119 117 116 118; 6 121 119 118 120; 11 122 123 111 110; 12 123 124 113 111
102. 13 124 125 115 113; 14 126 122 110 117; 15 127 126 117 119; 16 128 127 119 121
103. 17 129 130 123 122; 18 130 131 124 123; 19 131 132 125 124; 20 133 129 122 126
104. 21 134 133 126 127; 22 135 134 127 128; 23 108 109 137 136; 24 109 112 138 137
105. 25 112 114 139 138; 26 116 108 136 140; 27 118 116 140 141; 28 120 118 141 142
106. 29 136 137 144 143; 30 137 138 145 144; 31 138 139 146 145; 32 140 136 143 147
107. 33 141 140 147 148; 34 142 141 148 149; 35 143 144 151 150; 36 144 145 152 151
108. 37 145 146 153 152; 38 147 143 150 154; 39 148 147 154 155; 40 149 148 155 156
109. 77 205 206 207 106; 78 206 208 209 207; 79 208 210 211 209; 80 212 205 106 213
110. 81 214 212 213 215; 82 216 214 215 217; 83 218 219 206 205; 84 219 220 208 206
111. 85 220 221 210 208; 86 222 218 205 212; 87 223 222 212 214; 88 224 223 214 216
112. 89 225 226 219 218; 90 226 227 220 219; 91 227 228 221 220; 92 229 225 218 222
113. 93 230 229 222 223; 94 231 230 223 224; 95 106 207 232 233; 96 207 209 234 232
114. 97 209 211 235 234; 98 213 106 233 236; 99 215 213 236 237
115. 100 217 215 237 238; 101 233 232 239 240; 102 232 234 241 239
116. 103 234 235 242 241; 104 236 233 240 243; 105 237 236 243 244
117. 106 238 237 244 245; 107 240 239 246 247; 108 239 241 248 246
118. 109 241 242 249 248; 110 243 240 247 250; 111 244 243 250 251
119. 112 245 244 251 252; 149 132 301 302 125; 150 301 303 304 302
120. 151 303 305 306 304; 152 305 307 308 306; 153 307 309 310 308
121. 154 309 311 312 310; 155 311 313 314 312; 156 313 231 224 314
122. 157 125 302 315 115; 158 302 304 316 315; 159 304 306 317 316
123. 160 306 308 318 317; 161 308 310 319 318; 162 310 312 320 319
124. 163 312 314 321 320; 164 314 224 216 321; 165 115 315 322 114
125. 166 315 316 323 322; 167 316 317 324 323; 168 317 318 325 324
126. 169 318 319 326 325; 170 319 320 327 326; 171 320 321 328 327
127. 172 321 216 217 328; 173 114 322 329 139; 174 322 323 330 329
128. 175 323 324 331 330; 176 324 325 332 331; 177 325 326 333 332
129. 178 326 327 334 333; 179 327 328 335 334; 180 328 217 238 335
130. 181 139 329 336 146; 182 329 330 337 336; 183 330 331 338 337
131. 184 331 332 339 338; 185 332 333 340 339; 186 333 334 341 340
132. 187 334 335 342 341; 188 335 238 245 342; 189 146 336 343 153
133. 190 336 337 344 343; 191 337 338 345 344; 192 338 339 346 345
134. 193 339 340 347 346; 194 340 341 348 347; 195 341 342 349 348
135. 196 342 245 252 349; 245 399 400 130 129; 246 400 401 131 130
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199. 436 536 535 619 620; 437 485 579 621 610; 438 579 580 622 621
200. 439 580 581 623 622; 440 581 582 624 623; 441 582 583 625 624
201. 442 583 584 626 625; 443 584 585 627 626; 444 585 536 620 627
202. START GROUP DEFINITION
203. ELEMENT
204. _F_11_MOMENT 2 3 5 6 11 TO 22 24 25 27 TO 40 78 79 81 TO 94 96 97 99 TO 112 -
205. 149 TO 196 245 TO 284 286 287 289 TO 302 304 305 307 TO 320 322 323 -
206. 325 TO 338 340 341 343 TO 444
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207. _F_11_SHEAR 6 16 22 28 34 40 79 85 91 97 103 109 149 TO 196 245 TO 284 290 -
208. 296 302 308 314 320 323 329 335 341 347 353 357 TO 444
209. MEMBER
210. _C1 7 TO 10
211. END GROUP DEFINITION
212. ELEMENT PROPERTY
213. 1 TO 6 11 TO 40 77 TO 112 149 TO 196 245 TO 444 THICKNESS 0.7
214. DEFINE MATERIAL START
215. ISOTROPIC CONCRETE
216. E 2.17184E+007
217. POISSON 0.17
218. DENSITY 25
219. ALPHA 5.5E-006
220. DAMP 0.05
221. TYPE CONCRETE
222. STRENGTH FCU 27578.9
223. END DEFINE MATERIAL
224. MEMBER PROPERTY AMERICAN
225. 7 TO 10 PRIS YD 0.7 ZD 0.7
226. CONSTANTS
227. MATERIAL CONCRETE ALL
228. SUPPORTS
229. 1 TO 6 11 TO 40 77 TO 112 149 TO 196 245 TO 443 -
230. 444 PLATE MAT DIRECT Y SUBGRADE 117600 PRINT COMPRESSION
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JOINT	INFLUENCE AREA
105	0.122500
106	0.122500
107	0.122500
108	0.122500
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112	0.070000
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114	0.070000
115	0.060000
116	0.105000
117	0.090000
118	0.070000
119	0.060000
120	0.026250
121	0.022500
122	0.070000
123	0.060000
124	0.040000
125	0.040000
126	0.060000
127	0.040000
128	0.015000
129	0.070000
130	0.060000
131	0.040000
132	0.040000

133	0.060000
134	0.040000
135	0.015000
136	0.105000
137	0.090000
138	0.060000
139	0.060000
140	0.090000
141	0.060000
142	0.022500
143	0.070000
144	0.060000
145	0.040000
146	0.040000
147	0.060000
148	0.040000
149	0.015000
150	0.070000
151	0.060000
152	0.040000
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154	0.060000
155	0.040000
156	0.015000
205	0.105000
206	0.090000
207	0.105000
208	0.060000
209	0.070000
210	0.022500
211	0.026250
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213	0.105000
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215	0.070000
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217	0.070000
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225	0.070000
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227	0.040000
228	0.015000
229	0.060000
230	0.040000
231	0.040000
232	0.090000
233	0.105000
234	0.060000
235	0.022500
236	0.090000

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406	0.043750
407	0.037500
408	0.025000
409	0.009375
410	0.037500
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471	0.060000
472	0.090000
473	0.060000
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608	0.043750
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611	0.037500
612	0.025000
613	0.009375

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       615          0.043750
       616          0.025000
       617          0.009375
       618          0.037500
       619          0.025000
       620          0.025000
       621          0.031250
       622          0.031250
       623          0.031250
       624          0.031250
       625          0.031250
       626          0.031250
       627          0.031250
    TOTAL          19.999994
231. DEFINE REFERENCE LOADS
**WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
          MASTER/SLAVE OR IF UNCONNECTED JOINTS.
232. LOAD R1 LOADTYPE DEAD  TITLE FOUNDATION SELF WEIGHT (D1)
233. SELFWEIGHT Y -1
234. ELEMENT LOAD
235. 1 TO 6 11 TO 40 77 TO 112 149 TO 196 245 TO 444 PR GY -25
236. LOAD R2 LOADTYPE LIVE  TITLE LIVE  LOAD -4KN/SQM (L)
237. JOINT LOAD
238. 101 FX 0 FY -14.55 FZ 2.37
239. 102 FX 2.94 FY -20.79 FZ 0.05
240. 103 FX 0.71 FY -52.73 FZ 4.14
241. 104 FX -3.68 FY -33.6 FZ -0.12
242. LOAD R3 LOADTYPE WIND  TITLE WIND LOAD +X DIRN(W +X)
243. JOINT LOAD
244. 101 FX 0 FY -16.29 FZ 0.01
245. 102 FX 11.49 FY -14.55 FZ 0
246. 103 FX 16.77 FY 16.18 FZ 0
247. 104 FX 4.97 FY 15.16 FZ 0
248. LOAD R4 LOADTYPE WIND  TITLE WIND LOAD -X DIRN(W -X)
249. JOINT LOAD
250. 101 FX -2.54 FY 14.03 FZ 0
251. 102 FX -12.87 FY 8.07 FZ 0
252. 103 FX -12.39 FY -14.03 FZ 0.02
253. 104 FX -0.04 FY -8.08 FZ 0
254. LOAD R5 LOADTYPE WIND  TITLE WIND LOAD -Z DIRN(W -Z)
255. JOINT LOAD
256. 101 FX 0 FY 10.09 FZ -5.66
257. 102 FX -0.2 FY -1.75 FZ 0
258. 103 FX -3.78 FY 5.14 FZ -10.99
259. 104 FX 0.03 FY -9.52 FZ 0.01
260. LOAD R6 LOADTYPE WIND  TITLE WIND LOAD +Z DIRN(W +Z)
261. JOINT LOAD
262. 101 FX 0 FY -8.24 FZ 1.08
263. 102 FX 0.3 FY -0.24 FZ 2.46
264. 103 FX 3.71 FY -1.24 FZ 9.71
265. 104 FX -0.05 FY 9.77 FZ 2.95
266. LOAD R7 LOADTYPE TEMPERATURE  TITLE TEMPERATURE +20(T +VE)
267. JOINT LOAD
268. 101 FX 0.04 FY -0.09 FZ 1.65
269. 102 FX 287.58 FY 0.93 FZ 0
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270. 103 FX -0.05 FY -0.35 FZ 5.02
271. 104 FX -287.57 FY 2.66 FZ -0.67
272. LOAD R8 LOADTYPE NONE  TITLE TEMPERATURE -20(T -VE)
273. JOINT LOAD
274. 101 FX -0.02 FY 0.05 FZ -0.83
275. 102 FX -143.79 FY -0.46 FZ 0
276. 103 FX 0.03 FY 0.17 FZ -2.51
277. 104 FX 143.79 FY -1.33 FZ 0.34
278. LOAD R9 LOADTYPE DEAD  TITLE SELF WEIGHT+SECONDARY STEEL STRUCTURE(D2)
279. JOINT LOAD
280. 101 FX 0 FY -15.73 FZ 1.7
281. 102 FX 1.89 FY -18.29 FZ 0.04
282. 103 FX 0.55 FY -47.09 FZ 3.25
283. 104 FX -2.46 FY -28.29 FZ -0.08
284. LOAD R10 LOADTYPE SEISMIC  TITLE MAXIMUM SEISMIC LOAD (E_MAX)
285. JOINT LOAD
286. 101 FX 0 FY -16.29 FZ 0.01
287. 102 FX 11.49 FY -14.55 FZ 0
288. 103 FX 16.77 FY 16.18 FZ 0
289. 104 FX 4.97 FY 15.16 FZ 0
290. LOAD R11 LOADTYPE SEISMIC  TITLE MINIMUM SEISMIC LOAD (E_MIN)
291. JOINT LOAD
292. 101 FX 0 FY -8.24 FZ 1.08
293. 102 FX 0.3 FY -0.24 FZ 2.46
294. 103 FX 3.71 FY -1.24 FZ 9.71
295. 104 FX -0.05 FY 9.77 FZ 2.95
296. END DEFINE REFERENCE LOADS
297. **D
298. LOAD 101 (D1+D2)
299. REFERENCE LOAD
300. R1 1.0 R9 1.0
301. *D+T+L
302. LOAD 102 (D1+D2)+T(+VE)+L
303. REFERENCE LOAD
304. R1 1.0 R2 1.0 R7 1.0 R9 1.0
305. LOAD 103 (D1+D2)+T(-VE)+L1
306. REFERENCE LOAD
307. R1 1.0 R2 1.0 R8 1.0 R9 1.0
308. *D+W
309. LOAD 104 D1+D2+W(+X)
310. REFERENCE LOAD
311. R1 1.0 R9 1.0 R3 1.0
312. LOAD 105 D1+D2+W(-X)
313. REFERENCE LOAD
314. R1 1.0 R9 1.0 R4 1.0
315. LOAD 106 LOADTYPE NONE  TITLE D1+D2+W(-Z)
316. REFERENCE LOAD
317. R1 1.0 R9 1.0 R5 1.0
318. LOAD 107 LOADTYPE NONE  TITLE D1+D2+W(+Z)
319. REFERENCE LOAD
320. R1 1.0 R9 1.0 R6 1.0
321. *D+0.75W+0.75L
322. LOAD 108 D1+D2+0.75L1+0.75W(+X)
323. REFERENCE LOAD
324. R1 1.0 R9 1.0 R2 0.75 R3 0.75
325. LOAD 109 D1+D2+0.75L1+0.75W(-X)
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REFERENCE LOAD
R1 1.0 R9 1.0 R2 0.75 R4 0.75
LOAD 110 LOADTYPE NONE  TITLE D1+D2+0.75L1+0.75W(-Z)
REFERENCE LOAD
R1 1.0 R9 1.0 R2 0.75 R5 0.75
LOAD 111 LOADTYPE NONE  TITLE D1+D2+0.75L1+0.75W(+Z)
REFERENCE LOAD
R1 1.0 R9 1.0 R2 0.75 R6 0.75
*0.6D+W
LOAD 112 0.6(D2)+W(+X)+D1
REFERENCE LOAD
R1 1.0 R9 0.6 R3 1.0
LOAD 113 0.6(D2)+W(-X)+D1
REFERENCE LOAD
R1 1.0 R9 0.6 R4 1.0
LOAD 114 LOADTYPE NONE  TITLE 0.6(D2)+W(-Z)+D1
REFERENCE LOAD
R1 1.0 R9 0.6 R5 1.0
LOAD 115 LOADTYPE NONE  TITLE 0.6(D2)+W(+Z)+D1
REFERENCE LOAD
R1 1.0 R9 0.6 R6 1.0
*D+0.7E
LOAD 116 D1+D2+0.7E_MAX
REFERENCE LOAD
R1 1.0 R9 1.0 R10 0.7
LOAD 117 D1+D2+0.7E_MIN
REFERENCE LOAD
R1 1.0 R9 1.0 R11 0.7
*D+0.75*0.7E+0.75L
LOAD 118 D1+D2+0.75L1+0.525E_MAX
REFERENCE LOAD
R1 1.0 R9 1.0 R2 0.75 R10 0.525
LOAD 119 D1+D2+0.75L1+0.525E_MIN
REFERENCE LOAD
R1 1.0 R9 1.0 R2 0.75 R11 0.525
*0.6D+0.7W
LOAD 120 0.6(D2)+0.7E_MAX+D1
REFERENCE LOAD
R1 1.0 R9 0.6 R10 0.7
LOAD 121 0.6(D2)+0.7E_MIN+D1
REFERENCE LOAD
R1 1.0 R9 0.6 R11 0.7
*FACTORED LOAD COMBINATIONS
**1.4D
LOAD 201 1.4(D1+D2)
REFERENCE LOAD
R1 1.4 R9 1.4
**1.4(D+T)+1.7L
LOAD 202 1.4(D1+D2)+1.4T(+VE)+1.7L1
REFERENCE LOAD
R1 1.4 R9 1.4 R7 1.4 R2 1.7
LOAD 203 1.4(D1+D2)+1.4T(-VE)+1.7L1
REFERENCE LOAD
R1 1.4 R9 1.4 R8 1.4 R2 1.7
**1.2D+0.8W
LOAD 204 1.2(D1+D2)+0.8W(+X)
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382. REFERENCE LOAD
383. R1 1.2 R9 1.2 R3 0.8
384. LOAD 205 1.2(D1+D2)+0.8W(-X)
385. REFERENCE LOAD
386. R1 1.2 R9 1.2 R4 0.8
387. LOAD 206 LOADTYPE NONE  TITLE 1.2(D1+D2)+0.8W(-Z)
388. REFERENCE LOAD
389. R1 1.2 R9 1.2 R5 0.8
390. LOAD 207 LOADTYPE NONE  TITLE 1.2(D1+D2)+0.8W(+Z)
391. REFERENCE LOAD
392. R1 1.2 R9 1.2 R6 0.8
393. ***1.2D+1.6W+1.0L
394. LOAD 208 1.2(D1+D2)+1.0L1+1.6W(+X)
395. REFERENCE LOAD
396. R1 1.2 R9 1.2 R2 1.0 R3 1.6
397. LOAD 209 1.2(D1+D2)+1.0L1+1.6W(-X)
398. REFERENCE LOAD
399. R1 1.2 R9 1.2 R2 1.0 R4 1.6
400. LOAD 210 LOADTYPE NONE  TITLE 1.2(D1+D2)+1.0L1+1.6W(-Z)
401. REFERENCE LOAD
402. R1 1.2 R9 1.2 R2 1.0 R5 1.6
403. LOAD 211 LOADTYPE NONE  TITLE 1.2(D1+D2)+1.0L1+1.6W(+Z)
404. REFERENCE LOAD
405. R1 1.2 R9 1.2 R2 1.0 R6 1.6
406. *
407. **0.9D+1.6W
408. LOAD 212 0.9(D2)+W(+X)+D1
409. REFERENCE LOAD
410. R9 0.9 R3 1.6 R1 1.0
411. LOAD 213 0.9(D2)+W(-X)+D1
412. REFERENCE LOAD
413. R9 0.9 R4 1.6 R1 1.0
414. LOAD 214 LOADTYPE NONE  TITLE 0.9(D2)+W(-Z)+D1
415. REFERENCE LOAD
416. R9 0.9 R5 1.6 R1 1.0
417. LOAD 215 LOADTYPE NONE  TITLE 0.9(D2)+W(+Z)+D1
418. REFERENCE LOAD
419. R9 0.9 R6 1.6 R1 1.0
421. *1.2D+1.0E+1.0L
422. LOAD 216 1.2(D1+D2)+1.0L1+1.0E_MAX
423. REFERENCE LOAD
424. R1 1.2 R9 1.2 R2 1 R10 1
425. LOAD 217 1.2(D1+D2)+1.0L1+1.0E_MIN
426. REFERENCE LOAD
427. R1 1.2 R9 1.2 R2 1 R11 1
428. *0.9D+1.0E
429. LOAD 218 0.9(D2)+1.0E_MAX+D1
430. REFERENCE LOAD
431. R1 1.0 R9 0.9 R10 1.0
432. LOAD 219 0.9(D2)+1.0E_MIN+D1
433. REFERENCE LOAD
434. R1 1.0 R9 0.9 R11 1.0
435. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	382	NUMBER OF MEMBERS	4
NUMBER OF PLATES	320	NUMBER OF SOLIDS	0
NUMBER OF SURFACES	0	NUMBER OF SUPPORTS	378

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 270/ 26/ 81 DOF
TOTAL PRIMARY LOAD CASES = 40, TOTAL DEGREES OF FREEDOM = 1158
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 94 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 15.3/ 193125.2 MB

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

*** LOAD CASE 102 -- START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**NOTE-Tension/Compression converged after 1 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 111

**NOTE-Tension/Compression converged after 1 iterations, Case= 112

**NOTE-Tension/Compression converged after 1 iterations, Case= 113

**NOTE-Tension/Compression converged after 1 iterations, Case= 114

**NOTE-Tension/Compression converged after 1 iterations, Case= 115

**NOTE-Tension/Compression converged after 1 iterations, Case= 116

**NOTE-Tension/Compression converged after 1 iterations, Case= 117

**NOTE-Tension/Compression converged after 1 iterations, Case= 118

**NOTE-Tension/Compression converged after 1 iterations, Case= 119

**NOTE-Tension/Compression converged after 1 iterations, Case= 120

**NOTE-Tension/Compression converged after 1 iterations, Case= 121

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**NOTE-Tension/Compression converged after 1 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

**NOTE-Tension/Compression converged after 1 iterations, Case= 215

**NOTE-Tension/Compression converged after 1 iterations, Case= 217

**NOTE-Tension/Compression converged after 1 iterations, Case= 218

**NOTE-Tension/Compression converged after 1 iterations, Case= 219

436. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 30,2020 TIME= 16:26: 0 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****

ATTACHMENT 4 STAAD OUTPUT : MD-216-3510-EG-CV-CAL-1281_A01_PIT IN CCD AREA

PROPRIETARY & CONFIDENTIAL

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```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    MAY 30, 2020                *
*           Time=    16:49:11                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: D:\samrat dey 14.10.15\7. Maaden\1.CCD _Neutralization area\Pipe rack\CCD area\design\A02.. .STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 20-MAR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 -0.05 -1.65 -0.05; 2 1.55 -1.65 -0.05; 3 -0.05 -1.65 1.55; 4 1.55 -1.65 1.55
     9. 5 0.05 -1.65 1.55; 6 0.05 -1.65 1.45; 7 -0.05 -1.65 1.45; 8 0.4 -1.65 1.55
    10. 9 0.4 -1.65 1.45; 10 0.75 -1.65 1.55; 11 0.75 -1.65 1.45; 12 1.1 -1.65 1.55
    11. 13 1.1 -1.65 1.45; 14 1.45 -1.65 1.55; 15 1.45 -1.65 1.45; 16 1.55 -1.65 1.45
    12. 17 0.05 -1.65 1.1; 18 -0.05 -1.65 1.1; 19 0.4 -1.65 1.1; 20 0.75 -1.65 1.1
    13. 21 1.1 -1.65 1.1; 22 1.45 -1.65 1.1; 23 1.55 -1.65 1.1; 24 0.05 -1.65 0.75
    14. 25 -0.05 -1.65 0.75; 26 0.4 -1.65 0.75; 27 0.75 -1.65 0.75; 28 1.1 -1.65 0.75
    15. 29 1.45 -1.65 0.75; 30 1.55 -1.65 0.75; 31 0.05 -1.65 0.4; 32 -0.05 -1.65 0.4
    16. 33 0.4 -1.65 0.4; 34 0.75 -1.65 0.4; 35 1.1 -1.65 0.4; 36 1.45 -1.65 0.4
    17. 37 1.55 -1.65 0.4; 38 0.05 -1.65 0.05; 39 -0.05 -1.65 0.05; 40 0.4 -1.65 0.05
    18. 41 0.75 -1.65 0.05; 42 1.1 -1.65 0.05; 43 1.45 -1.65 0.05; 44 1.55 -1.65 0.05
    19. 45 0.05 -1.65 -0.05; 46 0.4 -1.65 -0.05; 47 0.75 -1.65 -0.05
    20. 48 1.1 -1.65 -0.05; 49 1.45 -1.65 -0.05; 50 -0.05 0 -0.05; 51 1.55 0 -0.05
    21. 52 -0.05 0 1.55; 53 1.55 0 1.55; 54 0.05 -1.4 -0.05; 55 -0.05 -1.4 -0.05
    22. 56 0.4 -1.4 -0.05; 57 0.75 -1.4 -0.05; 58 1.1 -1.4 -0.05; 59 1.45 -1.4 -0.05
    23. 60 1.55 -1.4 -0.05; 61 0.05 -1 -0.05; 62 -0.05 -1 -0.05; 63 0.4 -1 -0.05
    24. 64 0.75 -1 -0.05; 65 1.1 -1 -0.05; 66 1.45 -1 -0.05; 67 1.55 -1 -0.05
    25. 68 0.05 -0.75 -0.05; 69 -0.05 -0.75 -0.05; 70 0.4 -0.75 -0.05
    26. 71 0.75 -0.75 -0.05; 72 1.1 -0.75 -0.05; 73 1.45 -0.75 -0.05
    27. 74 1.55 -0.75 -0.05; 75 0.05 -0.5 -0.05; 76 -0.05 -0.5 -0.05
    28. 77 0.4 -0.5 -0.05; 78 0.75 -0.5 -0.05; 79 1.1 -0.5 -0.05; 80 1.45 -0.5 -0.05
    29. 81 1.55 -0.5 -0.05; 82 0.05 -0.25 -0.05; 83 -0.05 -0.25 -0.05
    30. 84 0.4 -0.25 -0.05; 85 0.75 -0.25 -0.05; 86 1.1 -0.25 -0.05
    31. 87 1.45 -0.25 -0.05; 88 1.55 -0.25 -0.05; 89 0.05 0 -0.05; 90 0.4 0 -0.05
    32. 91 0.75 0 -0.05; 92 1.1 0 -0.05; 93 1.45 0 -0.05; 94 1.55 -1.4 0.05
    33. 95 1.55 -1.4 0.4; 96 1.55 -1.4 0.75; 97 1.55 -1.4 1.1; 98 1.55 -1.4 1.45
    34. 99 1.55 -1.4 1.55; 100 1.55 -1 0.05; 101 1.55 -1 0.4; 102 1.55 -1 0.75
    35. 103 1.55 -1 1.1; 104 1.55 -1 1.45; 105 1.55 -1 1.55; 106 1.55 -0.75 0.05
    36. 107 1.55 -0.75 0.4; 108 1.55 -0.75 0.75; 109 1.55 -0.75 1.1
    37. 110 1.55 -0.75 1.45; 111 1.55 -0.75 1.55; 112 1.55 -0.5 0.05
    38. 113 1.55 -0.5 0.4; 114 1.55 -0.5 0.75; 115 1.55 -0.5 1.1; 116 1.55 -0.5 1.45
```

39. 117 1.55 -0.5 1.55; 118 1.55 -0.25 0.05; 119 1.55 -0.25 0.4
40. 120 1.55 -0.25 0.75; 121 1.55 -0.25 1.1; 122 1.55 -0.25 1.45
41. 123 1.55 -0.25 1.55; 124 1.55 0 0.05; 125 1.55 0 0.4; 126 1.55 0 0.75
42. 127 1.55 0 1.1; 128 1.55 0 1.45; 129 1.45 -1.4 1.55; 130 1.1 -1.4 1.55
43. 131 0.75 -1.4 1.55; 132 0.4 -1.4 1.55; 133 0.05 -1.4 1.55; 134 -0.05 -1.4 1.55
44. 135 1.45 -1 1.55; 136 1.1 -1 1.55; 137 0.75 -1 1.55; 138 0.4 -1 1.55
45. 139 0.05 -1 1.55; 140 -0.05 -1 1.55; 141 1.45 -0.75 1.55; 142 1.1 -0.75 1.55
46. 143 0.75 -0.75 1.55; 144 0.4 -0.75 1.55; 145 0.05 -0.75 1.55
47. 146 -0.05 -0.75 1.55; 147 1.45 -0.5 1.55; 148 1.1 -0.5 1.55
48. 149 0.75 -0.5 1.55; 150 0.4 -0.5 1.55; 151 0.05 -0.5 1.55; 152 -0.05 -0.5 1.55
49. 153 1.45 -0.25 1.55; 154 1.1 -0.25 1.55; 155 0.75 -0.25 1.55
50. 156 0.4 -0.25 1.55; 157 0.05 -0.25 1.55; 158 -0.05 -0.25 1.55; 159 1.45 0 1.55
51. 160 1.1 0 1.55; 161 0.75 0 1.55; 162 0.4 0 1.55; 163 0.05 0 1.55
52. 164 -0.05 -1.4 1.45; 165 -0.05 -1.4 1.1; 166 -0.05 -1.4 0.75
53. 167 -0.05 -1.4 0.4; 168 -0.05 -1.4 0.05; 169 -0.05 -1 1.45; 170 -0.05 -1 1.1
54. 171 -0.05 -1 0.75; 172 -0.05 -1 0.4; 173 -0.05 -1 0.05; 174 -0.05 -0.75 1.45
55. 175 -0.05 -0.75 1.1; 176 -0.05 -0.75 0.75; 177 -0.05 -0.75 0.4
56. 178 -0.05 -0.75 0.05; 179 -0.05 -0.5 1.45; 180 -0.05 -0.5 1.1
57. 181 -0.05 -0.5 0.75; 182 -0.05 -0.5 0.4; 183 -0.05 -0.5 0.05
58. 184 -0.05 -0.25 1.45; 185 -0.05 -0.25 1.1; 186 -0.05 -0.25 0.75
59. 187 -0.05 -0.25 0.4; 188 -0.05 -0.25 0.05; 189 -0.05 0 1.45; 190 -0.05 0 1.1
60. 191 -0.05 0 0.75; 192 -0.05 0 0.4; 193 -0.05 0 0.05; 194 -0.05 0.35 -0.05
61. 195 1.55 0.35 -0.05; 196 -0.05 0.35 1.55; 197 1.55 0.35 1.55
62. 198 0.05 0.35 -0.05; 199 0.4 0.35 -0.05; 200 0.75 0.35 -0.05
63. 201 1.1 0.35 -0.05; 202 1.45 0.35 -0.05; 203 1.55 0.35 0.05; 204 1.55 0.35 0.4
64. 205 1.55 0.35 0.75; 206 1.55 0.35 1.1; 207 1.55 0.35 1.45; 208 1.45 0.35 1.55
65. 209 1.1 0.35 1.55; 210 0.75 0.35 1.55; 211 0.4 0.35 1.55; 212 0.05 0.35 1.55
66. 213 -0.05 0.35 1.45; 214 -0.05 0.35 1.1; 215 -0.05 0.35 0.75
67. 216 -0.05 0.35 0.4; 217 -0.05 0.35 0.05; 218 -0.05 0.55 -0.05
68. 219 1.55 0.55 -0.05; 220 -0.05 0.55 1.55; 221 1.55 0.55 1.55
69. 222 0.05 0.55 -0.05; 223 0.4 0.55 -0.05; 224 0.75 0.55 -0.05
70. 225 1.1 0.55 -0.05; 226 1.45 0.55 -0.05; 227 1.55 0.55 0.05; 228 1.55 0.55 0.4
71. 229 1.55 0.55 0.75; 230 1.55 0.55 1.1; 231 1.55 0.55 1.45; 232 1.45 0.55 1.55
72. 233 1.1 0.55 1.55; 234 0.75 0.55 1.55; 235 0.4 0.55 1.55; 236 0.05 0.55 1.55
73. 237 -0.05 0.55 1.45; 238 -0.05 0.55 1.1; 239 -0.05 0.55 0.75
74. 240 -0.05 0.55 0.4; 241 -0.05 0.55 0.05; 242 -0.05 0.85 -0.05
75. 243 1.55 0.85 -0.05; 244 -0.05 0.85 1.55; 245 1.55 0.85 1.55
76. 246 0.05 0.85 -0.05; 247 0.4 0.85 -0.05; 248 0.75 0.85 -0.05
77. 249 1.1 0.85 -0.05; 250 1.45 0.85 -0.05; 251 1.55 0.85 0.05; 252 1.55 0.85 0.4
78. 253 1.55 0.85 0.75; 254 1.55 0.85 1.1; 255 1.55 0.85 1.45; 256 1.45 0.85 1.55
79. 257 1.1 0.85 1.55; 258 0.75 0.85 1.55; 259 0.4 0.85 1.55; 260 0.05 0.85 1.55
80. 261 -0.05 0.85 1.45; 262 -0.05 0.85 1.1; 263 -0.05 0.85 0.75
81. 264 -0.05 0.85 0.4; 265 -0.05 0.85 0.05
82. ELEMENT INCIDENCES SHELL
83. 1 3 5 6 7; 2 5 8 9 6; 3 8 10 11 9; 4 10 12 13 11; 5 12 14 15 13; 6 14 4 16 15
84. 7 7 6 17 18; 8 6 9 19 17; 9 9 11 20 19; 10 11 13 21 20; 11 13 15 22 21
85. 12 15 16 23 22; 13 18 17 24 25; 14 17 19 26 24; 15 19 20 27 26; 16 20 21 28 27
86. 17 21 22 29 28; 18 22 23 30 29; 19 25 24 31 32; 20 24 26 33 31; 21 26 27 34 33
87. 22 27 28 35 34; 23 28 29 36 35; 24 29 30 37 36; 25 32 31 38 39; 26 31 33 40 38
88. 27 33 34 41 40; 28 34 35 42 41; 29 35 36 43 42; 30 36 37 44 43; 31 39 38 45 1
89. 32 38 40 46 45; 33 40 41 47 46; 34 41 42 48 47; 35 42 43 49 48; 36 43 44 2 49
90. 37 1 45 54 55; 38 45 46 56 54; 39 46 47 57 56; 40 47 48 58 57; 41 48 49 59 58
91. 42 49 2 60 59; 43 55 54 61 62; 44 54 56 63 61; 45 56 57 64 63; 46 57 58 65 64
92. 47 58 59 66 65; 48 59 60 67 66; 49 62 61 68 69; 50 61 63 70 68; 51 63 64 71 70
93. 52 64 65 72 71; 53 65 66 73 72; 54 66 67 74 73; 55 69 68 75 76; 56 68 70 77 75
94. 57 70 71 78 77; 58 71 72 79 78; 59 72 73 80 79; 60 73 74 81 80; 61 76 75 82 83

95. 62 75 77 84 82; 63 77 78 85 84; 64 78 79 86 85; 65 79 80 87 86; 66 80 81 88 87
96. 67 83 82 89 50; 68 82 84 90 89; 69 84 85 91 90; 70 85 86 92 91; 71 86 87 93 92
97. 72 87 88 51 93; 73 2 44 94 60; 74 44 37 95 94; 75 37 30 96 95; 76 30 23 97 96
98. 77 23 16 98 97; 78 16 4 99 98; 79 60 94 100 67; 80 94 95 101 100
99. 81 95 96 102 101; 82 96 97 103 102; 83 97 98 104 103; 84 98 99 105 104
100. 85 67 100 106 74; 86 100 101 107 106; 87 101 102 108 107; 88 102 103 109 108
101. 89 103 104 110 109; 90 104 105 111 110; 91 74 106 112 81; 92 106 107 113 112
102. 93 107 108 114 113; 94 108 109 115 114; 95 109 110 116 115; 96 110 111 117 116
103. 97 81 112 118 88; 98 112 113 119 118; 99 113 114 120 119; 100 114 115 121 120
104. 101 115 116 122 121; 102 116 117 123 122; 103 88 118 124 51
105. 104 118 119 125 124; 105 119 120 126 125; 106 120 121 127 126
106. 107 121 122 128 127; 108 122 123 53 128; 109 4 14 129 99; 110 14 12 130 129
107. 111 12 10 131 130; 112 10 8 132 131; 113 8 5 133 132; 114 5 3 134 133
108. 115 99 129 135 105; 116 129 130 136 135; 117 130 131 137 136
109. 118 131 132 138 137; 119 132 133 139 138; 120 133 134 140 139
110. 121 105 135 141 111; 122 135 136 142 141; 123 136 137 143 142
111. 124 137 138 144 143; 125 138 139 145 144; 126 139 140 146 145
112. 127 111 141 147 117; 128 141 142 148 147; 129 142 143 149 148
113. 130 143 144 150 149; 131 144 145 151 150; 132 145 146 152 151
114. 133 117 147 153 123; 134 147 148 154 153; 135 148 149 155 154
115. 136 149 150 156 155; 137 150 151 157 156; 138 151 152 158 157
116. 139 123 153 159 53; 140 153 154 160 159; 141 154 155 161 160
117. 142 155 156 162 161; 143 156 157 163 162; 144 157 158 52 163; 145 3 7 164 134
118. 146 7 18 165 164; 147 18 25 166 165; 148 25 32 167 166; 149 32 39 168 167
119. 150 39 1 55 168; 151 134 164 169 140; 152 164 165 170 169; 153 165 166 171 170
120. 154 166 167 172 171; 155 167 168 173 172; 156 168 55 62 173
121. 157 140 169 174 146; 158 169 170 175 174; 159 170 171 176 175
122. 160 171 172 177 176; 161 172 173 178 177; 162 173 62 69 178
123. 163 146 174 179 152; 164 174 175 180 179; 165 175 176 181 180
124. 166 176 177 182 181; 167 177 178 183 182; 168 178 69 76 183
125. 169 152 179 184 158; 170 179 180 185 184; 171 180 181 186 185
126. 172 181 182 187 186; 173 182 183 188 187; 174 183 76 83 188
127. 175 158 184 189 52; 176 184 185 190 189; 177 185 186 191 190
128. 178 186 187 192 191; 179 187 188 193 192; 180 188 83 50 193; 181 50 89 198 194
129. 182 89 90 199 198; 183 90 91 200 199; 184 91 92 201 200; 185 92 93 202 201
130. 186 93 51 195 202; 187 51 124 203 195; 188 124 125 204 203
131. 189 125 126 205 204; 190 126 127 206 205; 191 127 128 207 206
132. 192 128 53 197 207; 193 53 159 208 197; 194 159 160 209 208
133. 195 160 161 210 209; 196 161 162 211 210; 197 162 163 212 211
134. 198 163 52 196 212; 199 52 189 213 196; 200 189 190 214 213
135. 201 190 191 215 214; 202 191 192 216 215; 203 192 193 217 216
136. 204 193 50 194 217; 205 194 198 222 218; 206 198 199 223 222
137. 207 199 200 224 223; 208 200 201 225 224; 209 201 202 226 225
138. 210 202 195 219 226; 211 195 203 227 219; 212 203 204 228 227
139. 213 204 205 229 228; 214 205 206 230 229; 215 206 207 231 230
140. 216 207 197 221 231; 217 197 208 232 221; 218 208 209 233 232
141. 219 209 210 234 233; 220 210 211 235 234; 221 211 212 236 235
142. 222 212 196 220 236; 223 196 213 237 220; 224 213 214 238 237
143. 225 214 215 239 238; 226 215 216 240 239; 227 216 217 241 240
144. 228 217 194 218 241; 229 218 222 246 242; 230 222 223 247 246
145. 231 223 224 248 247; 232 224 225 249 248; 233 225 226 250 249
146. 234 226 219 243 250; 235 219 227 251 243; 236 227 228 252 251
147. 237 228 229 253 252; 238 229 230 254 253; 239 230 231 255 254
148. 240 231 221 245 255; 241 221 232 256 245; 242 232 233 257 256
149. 243 233 234 258 257; 244 234 235 259 258; 245 235 236 260 259
150. 246 236 220 244 260; 247 220 237 261 244; 248 237 238 262 261

```
151. 249 238 239 263 262; 250 239 240 264 263; 251 240 241 265 264
152. 252 241 218 242 265
153. START GROUP DEFINITION
154. MEMBER
155. FLOOR
156. ELEMENT
157. _FDN 8 TO 11 14 TO 17 20 TO 23 26 TO 29
158. _WALL 38 TO 41 44 TO 47 50 TO 53 56 TO 59 62 TO 65 68 TO 71 74 TO 77 -
159. 80 TO 83 86 TO 89 92 TO 95 98 TO 101 104 TO 107 110 TO 113 116 TO 119 122 -
160. 123 TO 125 128 TO 131 134 TO 137 140 TO 143 146 TO 149 152 TO 155 158 TO 161 -
161. 164 TO 167 170 TO 173 176 TO 179 182 TO 185 188 TO 191 194 TO 197 -
162. 200 TO 203 206 TO 209 212 TO 215 218 TO 221 224 TO 227 230 TO 233 -
163. 236 TO 239 242 TO 245 248 TO 251
164. END GROUP DEFINITION
165. ELEMENT PROPERTY
166. 1 TO 36 THICKNESS 0.2
167. 37 TO 252 THICKNESS 0.2
168. DEFINE MATERIAL START
169. ISOTROPIC CONCRETE
170. E 2.95804E+007
171. POISSON 0.17
172. DENSITY 25
173. ALPHA 5.5E-006
174. DAMP 0.05
175. TYPE CONCRETE
176. STRENGTH FCU 35000
177. END DEFINE MATERIAL
178. CONSTANTS
179. MATERIAL CONCRETE ALL
180. SUPPORTS
181. 1 TO 36 PLATE MAT DIRECT Y SUBGRADE 117600 PRINT
```

JOINT	INFLUENCE AREA
1	0.002500
2	0.002500
3	0.002500
4	0.002500
5	0.011250
6	0.050625
7	0.011250
8	0.017500
9	0.078750
10	0.017500
11	0.078750
12	0.017500
13	0.078750
14	0.011250
15	0.050625
16	0.011250
17	0.078750
18	0.017500
19	0.122500
20	0.122500
21	0.122500

22	0.078750
23	0.017500
24	0.078750
25	0.017500
26	0.122500
27	0.122500
28	0.122500
29	0.078750
30	0.017500
31	0.078750
32	0.017500
33	0.122500
34	0.122500
35	0.122500
36	0.078750
37	0.017500
38	0.050625
39	0.011250
40	0.078750
41	0.078750
42	0.078750
43	0.050625
44	0.011250
45	0.011250
46	0.017500
47	0.017500
48	0.017500
49	0.011250
TOTAL	2.559999

```
182. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD(DL)
183. SELFWEIGHT Y -1
184. LOAD 2 LOADTYPE NONE  TITLE SOIL PR (DRY COND)(SDC)
185. ELEMENT LOAD
186. 37 TO 42 73 TO 78 109 TO 114 145 TO 150 TRAP JT 21 21 18.6136 18.6136
187. 43 TO 48 79 TO 84 115 TO 120 151 TO 156 TRAP JT 18.6136 18.6136 14.7955 14.7955
188. 49 TO 54 85 TO 90 121 TO 126 157 TO 162 TRAP JT 14.7955 14.7955 12.4091 12.4091
189. 55 TO 60 91 TO 96 127 TO 132 163 TO 168 TRAP JT 12.4091 12.4091 10.0227 10.0227
190. 61 TO 66 97 TO 102 133 TO 138 169 TO 173 -
191. 174 TRAP JT 10.0227 10.0227 7.63636 7.63636
192. 67 TO 72 103 TO 108 139 TO 144 175 TO 180 TRAP JT 7.63636 7.63636 5.25 5.25
193. 181 TO 204 TRAP JT 5.25 5.25 1.90909 1.90909
194. 205 TO 228 TRAP JT 1.90909 1.90909 5.68953E-007 5.68953E-007
195. LOAD 3 LOADTYPE NONE  TITLE SURCHARGE PR(SP)
196. ELEMENT LOAD
197. 37 TO 228 PR 5
198. LOAD 4 LOADTYPE NONE  TITLE WATER PRESSURE(WP)
199. ELEMENT LOAD
200. 1 TO 36 PR GY -25
201. 37 TO 42 73 TO 78 109 TO 114 145 TO 150 TRAP JT -25 -25 -22.5 -22.5
202. 229 TO 252 TRAP JT -2.5 -2.5 0 0
203. 205 TO 228 TRAP JT -5 -5 -2.5 -2.5
204. 181 TO 204 TRAP JT -8.5 -8.5 -5 -5
205. 67 TO 72 103 TO 108 139 TO 144 175 TO 180 TRAP JT -11 -11 -8.5 -8.5
206. 61 TO 66 97 TO 102 133 TO 138 169 TO 174 TRAP JT -13.5 -13.5 -11 -11
207. 55 TO 60 91 TO 96 127 TO 132 163 TO 168 TRAP JT -16 -16 -13.5 -13.5
208. 49 TO 54 85 TO 90 121 TO 126 157 TO 162 TRAP JT -18.5 -18.5 -16 -16
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209. 43 TO 48 79 TO 84 115 TO 120 151 TO 156 TRAP JT -22.5 -22.5 -18.5 -18.5
210. **
211. LOAD COMB 101 DL
212. 1 1.0
213. LOAD COMB 102 DL+SDC+SP
214. 1 1.0 2 1.0 3 1.0
215. LOAD COMB 103 DL+WP+SDC+SP
216. 1 1.0 2 1.0 3 1.0 4 1.0
217. LOAD COMB 201 1.4DL
218. 1 1.4
219. LOAD COMB 202 1.4DL + 1.7(SDC+SP)
220. 1 1.4 2 1.7 3 1.7
221. LOAD COMB 203 1.4(DL+WP) + 1.7(SDC+SP)
222. 1 1.4 2 1.7 3 1.7 4 1.4
223. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**265**	**NUMBER OF MEMBERS**	**0**
NUMBER OF PLATES	**252**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**49**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 167/ 28/ 174 DOF
TOTAL PRIMARY LOAD CASES = 4, TOTAL DEGREES OF FREEDOM = 1443
TOTAL LOAD COMBINATION CASES = 6 SO FAR.
SIZE OF STIFFNESS MATRIX = 252 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 16.1/ 193119.2 MB

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE NONE TITLE DEAD LOAD(DL)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.749999953E+00
Y = -0.572413733E+00
Z = 0.749999953E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -92.80
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 69.60 MY= 0.00 MZ= -69.60

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 92.80
SUMMATION FORCE-Z = -0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -69.60 MY= -0.00 MZ= 69.60

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X	-3.58775E-04	166
Y	-3.19770E-02	263
Z	-3.58775E-04	57
RX	-3.69409E-05	41
RY	-4.45572E-06	65
RZ	3.69409E-05	24

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE NONE TITLE SOIL PR (DRY COND)(SDC)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 3.1126168E-15
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 5.4304408E-16

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 1.2788833E-15 MY= -4.6052360E-08 MZ= 4.0003076E-16

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 8.8492072E-14
SUMMATION FORCE-Y = -2.9275912E-13
SUMMATION FORCE-Z = 3.2717683E-14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 2.9683733E-13 MY= 4.6052356E-08 MZ= -1.2644970E-13

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	1.04639E-03	176
Y =	3.00159E-04	1
Z =	1.04639E-03	71
RX=	1.71383E-05	57
RY=	-1.44058E-05	175
RZ=	-1.71383E-05	166

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE NONE TITLE SURCHARGE PR(SP)

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 4.7070251E-16
SUMMATION FORCE-Y = 0.0000000E+00
SUMMATION FORCE-Z = 5.5558329E-16

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 8.0907741E-16 MY= -1.2418397E-15 MZ= -4.5471405E-17

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 5.7232795E-14
SUMMATION FORCE-Y = -1.0001006E-13
SUMMATION FORCE-Z = 3.3458460E-14

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 7.7244941E-14 MY= 3.9638521E-15 MZ= -3.0995818E-14

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	4.83875E-04	181
Y =	1.05992E-04	263
Z =	4.83875E-04	78
RX=	6.21883E-06	57
RY=	6.88766E-06	121
RZ=	-6.21883E-06	166

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE NONE TITLE WATER PRESSURE(WP)

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.749999940E+00
Y = -0.165000004E+01
Z = 0.749999940E+00

***TOTAL APPLIED LOAD (KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -64.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 48.00 MY= 0.00 MZ= -48.00

***TOTAL REACTION LOAD(KN METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = -0.00
SUMMATION FORCE-Y = 64.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -48.00 MY= -0.00 MZ= 48.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	-1.36730E-03	176
Y =	-2.16331E-02	1
Z =	-1.36730E-03	71
RX=	-2.16751E-05	57
RY=	-1.89092E-05	72
RZ=	2.16751E-05	166

********** END OF DATA FROM INTERNAL STORAGE **********

224. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= MAY 30,2020 TIME= 16:49:13 ****

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* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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